# Beyond the Label: Do Ecodesign Regulations Reduce Real-World Energy Use?

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**Abstract**

This paper studies how ecodesign regulation shapes product design and consumer welfare when compliance is tied to a single operating mode rather than overall use. I examine the German washing machine market following the 2021 EU reform, which rescaled energy labels and capped the duration of the regulated Eco program. Combining detailed product and wash-program data with an original household survey, I find that rescaling shifts demand toward higher-rated products, although Eco accounts for only about 10 percent of reported wash cycles. This response is consistent with households interpreting the label as measuring whole-machine efficiency rather than Eco mode alone. Meanwhile, firms offer more energy-efficient Eco programs but more energy-intensive Cotton programs. I develop and estimate an equilibrium model of product entry, pricing, and program design incorporating household purchase and program choices. The estimates reveal a mechanism for regulatory leakage: constraints on Eco design increase incentives to improve substitute Cotton programs, raising their energy consumption. Removing the Eco-duration cap raises consumer surplus by € 8.4 million and reduces electricity consumption by 3.14 percent. Broadening the efficiency index to include Cotton programs can further increase consumer surplus and reduce electricity expenditure. These findings highlight the importance of substitution across operating modes in designing energy-efficiency regulation.

*Keywords:* energy labels, product portfolio choice, information design
*JEL codes:* D83, L15, L68, Q48, Q58

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## 1 Introduction

Many product regulations rely on standardized tests, even though their benefits depend on how products perform in actual use. Automobile emissions provide a prominent example: Volkswagen's use of defeat-device software allowed vehicles to satisfy laboratory tests while producing substantially higher emissions on the road (Barrett et al., 2015). A related test–use distinction arises in medicine, where drug efficacy in controlled clinical trials can differ from effectiveness in routine practice (Eichler et al., 2011; Rothwell, 2005). These examples raise a broader question: when regulation rewards outcomes measured under standardized conditions, how do responses along both measured and unmeasured dimensions shape its realized benefits?[1]

This paper studies mechanisms through which measured and realized performance can diverge, using the German washing-machine market as a setting. European energy-efficiency regulation evaluates washing machines using a standardized Eco program, while households choose among multiple programs in use, including Cotton and Easy-Care. The regulated efficiency metric is prominently displayed at purchase, whereas energy use in other programs is less visible. Regulation may therefore strengthen incentives to reduce energy use in Eco while encouraging firms to enhance washing quality in unregulated programs through more energy-intensive designs. These design responses may also alter subsequent program choice by changing the relative attributes of different wash programs. Realized energy use can therefore depend not only on improvements in the regulated mode, but also on how firms design unregulated programs and how consumers use the product after purchase.

I exploit a major reform of European washing-machine regulation implemented in 2021 (European Commission, 2019a). The reform changed both how energy efficiency was measured and the design constraints firms faced. First, the energy label was rescaled from the old classes $A^{+++}$ through $D$ to classes $A$ through $G$, and the formula used to measure energy efficiency was revised. Because most products had accumulated in the highest classes under the old system, the rescaling restored differentiation in the label and increased the market reward to lowering measured Eco energy use. Second, the reform imposed a maximum duration on the Eco cycle to make the program more convenient to use. This constraint may, however, reduce washing quality and restrict an important energy-saving margin: lowering wash temperature and compensating for the resulting loss in washing quality with a longer cycle.[2] Together, these changes increase the return to improving measured Eco efficiency while restricting how firms can achieve those improvements, providing useful variation for studying firms' design responses.

[1] A related incentive problem arises in education, where test-based accountability can shift effort toward tested outcomes and away from dimensions of learning that are not measured (Jacob, 2005).

[2] European washing machines heat wash water electrically within the machine. Water heating is by far the largest component of energy use during a wash cycle, making lower wash temperatures an important source of energy savings.

To trace these responses across product design and household use, I assemble a novel dataset that links product-market outcomes to program-level design and household behavior. The main data combine detailed model-level sales with hand-collected attributes from washing-machine manuals, including electricity use, water use, duration, and temperature for both regulated and unregulated wash programs. I complement these data with an original survey that I designed and commissioned in Germany, which measures households' program choices and their interpretation of the energy label. Together, these data allow me to study firms' design responses across programs, consumers' purchase and usage decisions, and the implications for realized energy consumption.

The data reveal three main findings. First, consumers respond strongly to the energy class but often misunderstand its scope. The 2021 label rescaling increases sales of models rated $C$ or higher by 39.4 percent relative to those rated $D$ or lower, while survey respondents often interpret the energy class as measuring the efficiency of the whole machine rather than the Eco program alone. Eco nevertheless accounts for only about 10 percent of reported wash cycles. Second, improvements in regulated Eco efficiency do not extend to unregulated programs. Within the same machine, Eco and Cotton energy use per cycle are negatively correlated, particularly for Cotton 60°C. Event-study estimates show that by the end of 2023, unweighted average energy use per cycle fell by 8.6 percent for Eco but rose by 11.3 percent for Cotton 60°C. Third, this divergence is consistent with firms shifting design toward unregulated programs. The duration cap makes Eco less attractive to consumers who value longer cycles, while the stricter energy-efficiency requirement limits compensation through more energy-intensive Eco inputs. These consumers instead turn toward Cotton, which can raise the marginal return to providing longer, more intensive Cotton cycles. This sorting creates incentives for energy-intensive design changes outside the program evaluated by the label. Consistent with this mechanism, a 10-percentage-point increase in exposure to the duration cap is associated with 5.9 percent higher energy use and 2.4 percent longer duration in Cotton 60°C.[3]

To explain these findings, I develop an equilibrium model of the washing machine market. Demand consists of two stages: product purchase and program use. At the purchase stage, consumers value the new energy class once the label becomes observable but do not directly value the energy efficiency of unregulated programs. They also anticipate the utility from the bundle of wash programs they will use. At the use stage, households choose among programs based on expected washing quality and convenience, which depend on program-specific inputs such as temperature, duration, and water use. The expected utility from program use enters product demand through a household-specific inclusive value, linking firms' program-design decisions to consumers' willingness to pay.

[3]Water use rises broadly across wash programs after the reform. The stricter rinsing-effectiveness requirement, which limits residual detergent after washing, may contribute to this increase.

On the supply side, firms make discrete product-entry and continuous program-design decisions. Firms choose which products to introduce and their energy-efficiency classes, then select program-specific inputs to achieve washing-quality targets, subject to class-specific energy-efficiency requirements and duration constraints for Eco. Firms trade off the demand benefits of higher energy ratings and improved program quality against design costs and regulatory constraints. By limiting firms' ability to maintain Eco washing quality, tighter regulatory constraints increase the marginal benefit of quality-enhancing inputs in unregulated programs.

I estimate demand in two steps. First, I estimate program-use preferences by simulated maximum likelihood, combining survey respondents' reported program-use frequencies with program inputs from product manuals matched to the washing machines they own. Variation in program inputs across machines and differences in household usage patterns identify preferences and their heterogeneity. Second, I use the estimated preferences to construct household-specific expected utility from each machine's bundle of wash programs and incorporate it into product demand. I identify product-demand preferences using variation in product characteristics and program designs generated by the entry of new models, which firms choose before observing demand shocks.

On the supply side, I first recover program-design cost parameters conditional on observed product portfolios. I use a minimum-distance estimator that matches observed program inputs to firms' optimal choices under engineering and regulatory constraints. Taking these costs as given, I estimate entry costs in a static discrete-choice game of incomplete information, in the spirit of Seim (2006). Estimation presents two computational challenges. First, the number of possible portfolios grows exponentially with the number of product lines. Rather than enumerate firm-level portfolio probabilities, I maximize a composite conditional likelihood based on product-line choices, conditioning on observed choices in the same firm's other lines and integrating over beliefs about rivals' portfolios. I use Gibbs sampling (Casella and George, 1992) to simulate rival portfolios from product-line conditional choice probabilities, avoiding explicit construction of their high-dimensional joint distributions. I iterate between estimating entry parameters and updating portfolio beliefs until the conditional choice probabilities converge (Aguirregabiria and Mira, 2007). Second, evaluating each candidate portfolio requires solving for optimal continuous program inputs. I train a neural-network surrogate to predict optimal inputs and profits net of design costs, avoiding repeated constrained optimization during entry estimation.

The estimated model reveals substantial heterogeneity in preferences for cycle duration. Relative to $A^{+++}$ durations, a 30-minute reduction lowers Eco use by 5.5 p.p. among consumers who prefer longer cycles but raises it by 0.3 p.p. among those who prefer shorter cycles. On average, Eco's share falls by 2.2 p.p. and the combined Cotton share rises by 2.0 p.p. The product-demand estimates further show that consumers value both higher efficiency ratings and higher expected wash utility. These preferences give firms incentives to balance measured efficiency against the attractiveness

of the full menu of wash programs. Consistent with the sorting mechanism, shortening Eco also raises the marginal return to extending Cotton programs. Under the same duration comparison, the average marginal operating profit from extending Cotton programs rises by about 19 percent. These findings reveal a channel through which restrictions on the regulated program strengthen incentives to improve unregulated alternatives.

**Policy Implications** I use the estimated model to evaluate three policy changes: removing existing regulatory restrictions, broadening the efficiency index to include Cotton programs, and introducing an energy-based producer tax. Firms adjust product introductions and program designs, while households adjust purchases and program use. I compare electricity consumption and welfare across policies for purchase cohorts from 2020Q1 through 2023Q4, accounting for continued machine use by nonbuyers.

First, removing the Eco-duration cap while retaining energy labels increases consumer welfare by € 8.4 million and reduces electricity consumption by 3.14 percent. Relaxing the cap allows firms to offer longer Eco cycles at lower temperatures while continuing to meet efficiency requirements. Longer Eco cycles also appeal to households that value additional washing time, reducing firms' incentives to serve these preferences through longer Cotton programs. Consistent with this mechanism, firms offer longer, cooler Eco cycles and shorter Cotton 40°C and Cotton 60°C cycles. Lifetime wash benefits increase by € 1.3 million and electricity expenditure falls by € 7.4 million, while purchase-stage surplus changes little. Operating profits rise by 6.51 percent, bringing the aggregate welfare gain to € 61.2 million. These results suggest that greater flexibility in program duration can improve consumer and producer outcomes while preserving the efficiency incentives provided by energy labels.

Second, extending the efficiency index to Cotton programs improves outcomes relative to an Eco-only index, with the duration cap removed in both cases. Assigning 52 percent of the weight to Eco and 24 percent to each Cotton program reduces electricity consumption by 2.51 percent and increases consumer welfare by € 25.6 million. Broadening the test makes Cotton energy consumption relevant to firms' ratings, changing incentives to allocate performance-related inputs across Eco and Cotton programs. The consumer-welfare gain comes primarily from lower electricity expenditure, which more than offsets a small decline in lifetime wash benefits. Purchases remain nearly unchanged, indicating that the electricity savings primarily reflect changes in the machines purchased and how households use them, rather than fewer purchases.

Third, an energy-based producer tax reduces electricity consumption but generates substantial producer losses. A tax of € 50 per machine for each kWh of average per-cycle electricity consumption across Eco and the two Cotton programs reduces electricity consumption by 0.72%. Consumer welfare rises by € 53.7 million after rebates, but operating profits fall by € 123.8 million, yielding

an aggregate welfare loss of € 70.1 million. Taxing Cotton alone achieves nearly the same electricity reduction with a smaller aggregate welfare loss. Thus, targeting energy use outside Eco can improve the effectiveness of taxation, but consumer gains do not necessarily outweigh producer losses.

**Related Literature** This paper contributes to three strands of literature. First, it relates to research on multitask incentives and performance measurement. Holmstrom and Milgrom (1991) and Baker (1992) show that incentives based on incomplete performance measures can distort the allocation of effort across tasks. Empirical studies examine how performance-based incentives and quality disclosure affect the allocation of effort and investment across quality dimensions (Jacob, 2005; Lu, 2012; Gupta, 2021; Vatter, 2022). I extend this literature to environmental product regulation and identify a demand-mediated channel for multitasking distortions. When regulation constrains one operating mode, consumers may shift toward unregulated alternatives or sort across products, changing firms' incentives to improve unregulated programs. These responses can induce more energy-intensive designs even as measured efficiency improves.

Second, the paper contributes to research on the equilibrium effects of product regulation and endogenous attributes. Existing studies examine labeling and certification policies (Barahona et al., 2023; Houde, 2018, 2026; Davis and Metcalf, 2016) and environmental standards and attribute-based regulations (Reynaert, 2021; Reynaert and Sallee, 2021; Ito and Sallee, 2018; Barwick et al., 2024). My setting allows me to observe both regulated and unregulated operating modes within the same product, together with household program-use information. These data allow me to examine how regulation targeting one operating mode affects the design of other modes and how consumers allocate use across them. By incorporating both margins, I evaluate whether improvements in measured efficiency translate into lower actual energy consumption and higher consumer welfare.

Third, the paper contributes to empirical entry games and endogenous product variety. Seim (2006) estimates an incomplete-information entry game using equilibrium choice probabilities, while Jia (2008) exploits strategic structure to solve a game with interdependent entry decisions. These approaches become computationally demanding when many firms choose among large portfolios of products. An alternative literature uses moment inequalities to estimate fixed costs without solving the full portfolio game, typically under complete information (Holmes, 2011; Houde et al., 2023; Eizenberg, 2014; Wollmann, 2018; Montag, 2023; Fan and Yang, 2025). While these approaches accommodate large action spaces, they can yield set identification and complicate counterfactual analysis, which may require additional equilibrium computations and assumptions about equilibrium selection. I develop an alternative approach that retains likelihood-based estimation in an incomplete-information game with large product portfolios. I use product-line conditional choice probabilities to reduce the dimensionality of firms' decisions and Gibbs sampling to construct the implied joint distribution of rival portfolios. A neural-network surrogate approximates optimal pro-

gram inputs and profits, avoiding repeated constrained optimization. This approach permits point estimation and extends naturally to counterfactual analysis with endogenous portfolio choices and continuous product design.

The rest of the paper proceeds as follows. Section 2 describes the institutional setting and the 2021 label reform and presents the data and documents the main empirical patterns. Section 5 develops the equilibrium model. Section 6 describes the estimation strategy, and Section 7 presents the estimation results. Section 8 presents the counterfactual policy analysis. Section 9 concludes.

## 2 Institutional Background and Data

### 2.1 The 2021 EU Energy Label Reform

The EU energy-labeling framework covers a wide range of household appliances, including refrigerators, washing machines, dishwashers, and air conditioners. Its purpose is to provide consumers with standardized information on product performance, particularly energy efficiency, and to strengthen firms' incentives to improve product design. Retailers are required to display energy labels both in physical stores and in online listings. This paper focuses on household washing machines.

The EU has implemented three major energy-labeling regimes for household washing machines, introduced in 1995, 2011, and 2021. This paper studies the transition from the 2011 regime to the 2021 regime. The reform was announced in March 2019 and took effect on March 1, 2021. Transitional provisions effective in December 2019 allowed manufacturers to update control panels in anticipation of the new requirements, enabling firms to begin adapting their products in early 2020. Consumers, however, continued to see the old energy labels until March 2021, when the new labels were introduced. From March 2021, newly placed products had to comply with the revised ecodesign requirements, while existing inventory could remain on sale under transitional rules.[4]

The 2021 reform introduced two major changes. First, it rescaled energy-efficiency ratings and revised both the standardized testing procedure and the formula used to determine them. Second, it introduced additional restrictions on the design of the regulated program, particularly limits on cycle duration. Together, these changes altered both the information consumers observe when purchasing washing machines and the constraints firms face when designing wash programs. Figure 3 compares the labels under the two regimes.

**Energy-Label Rescaling** Washing machines offer multiple wash programs, including Cotton, Easy Care, and Quick Wash. The EU energy label, however, measures performance under stan-

[4]Retailers had 14 working days to replace existing labels. Certain discontinued models could continue to be sold with their old labels until November 30, 2021.

dardized test cycles rather than across all available programs.[5] Throughout the paper, I refer to these regulated test cycles as the Eco program, whose definition differs across regimes. Under the 2011 regime, the Eco program refers to the Standard Cotton 60°C cycle at full and half load and the Standard Cotton 40°C cycle at half load.[6] Under the 2021 regime, the Eco program refers to Eco 40–60, designed to clean normally soiled cotton laundry washable at either 40°C or 60°C in a single cycle. Other wash programs, including regular Cotton, are available to consumers but do not enter the calculation of the energy-efficiency rating.

Energy-efficiency classes are determined by an Energy Efficiency Index (EEI), which compares energy consumption measured under the Eco program with a capacity-dependent reference level. By the late 2010s, most washing machines had clustered in the highest efficiency classes, reducing the label's ability to differentiate among products. To restore differentiation and allow room for future efficiency improvements, the 2021 reform revised the EEI formula and replaced the previous efficiency scale, ranging from $A^{+++}$ to $D$, with a new scale ranging from $A$ to $G$. Appendix A.1 provides the detailed formulas and classification thresholds under both regimes.

Two distinctions matter. First, neither the differences across wash programs nor the change in the Eco program across regimes imply fundamentally different engineering technologies. Wash programs generally share the same hardware but use different combinations of temperature, duration, water use, and mechanical action to serve different washing needs.[7] Temperature designations such as 40°C and 60°C are nominal program labels rather than mandated wash temperatures, and firms retain discretion over program inputs subject to regulatory constraints.[8] Second, energy classes are not directly comparable across regimes because both the test-load conditions and the EEI formula changed. A change in the displayed class therefore does not necessarily reflect a change in actual energy consumption. As the European Commission notes, a machine previously rated $A^{+++}$ could receive a $C$ rating under the new label without any change to the machine (European Commission, 2021).

**Attribute Restrictions** Beyond energy labeling, EU regulations impose attribute restrictions on the Eco program that do not apply to other wash programs. These restrictions constrain how firms balance cleaning performance against energy consumption. According to Sinner's Circle (Sinner, 1960), wash performance depends on four interacting factors: temperature, mechanical action,

[5] Using designated test cycles facilitates comparability across products and limits certification costs, but may not reflect consumers' actual program choices.

[6] The Standard Cotton programs are distinct from the regular Cotton programs, and both are available on the machine's control panel.

[7] For example, Cotton 60°C is intended for more intensive cleaning, Easy Care prioritizes gentler fabric treatment, and Quick Wash is designed for time convenience.

[8] In practice, Eco programs typically operate at wash temperatures between 30°C and 45°C, below their nominal temperature designations.

duration, and detergent chemistry.[9] Higher temperatures, longer cycles, and stronger mechanical action can improve stain and odor removal but may also increase fabric wear. Water serves as an intermediate input: more water facilitates rinsing but dilutes detergent and reduces friction between garments, while less water increases mechanical action but may hinder stain removal. Program inputs also affect electricity consumption, although their effects differ. Water heating accounts for more than 70 percent of electricity consumption in a typical wash cycle, making temperature reduction a primary means of reducing energy consumption. Lower temperatures, however, may compromise cleaning performance unless offset by adjustments in other inputs. Regulatory restrictions on program attributes can alter this tradeoff.

The main change in attribute restrictions under the 2021 reform was the introduction of duration limits on the Eco program. Under the earlier regime, manufacturers could improve measured energy efficiency by lowering wash temperatures and compensating with longer cycles while maintaining wash performance (Alborzi et al., 2017). However, excessively long cycles may discourage consumers from using the Eco program, limiting energy savings in actual use. To improve time convenience and encourage Eco use, the 2021 reform imposed capacity-dependent duration caps, with separate limits at full, half, and quarter loads. The effect on consumer demand, however, is ambiguous. Consumers who value time convenience may prefer shorter Eco cycles, while those who associate longer cycles with better cleaning performance may find them less attractive and switch to other programs. The resulting effect on actual energy consumption depends on both firms' program-design responses and consumers' program choices.

The EU also imposes water-consumption and minimum wash-performance requirements under both regimes. The water constraint primarily aims to conserve water. The 2021 reform revised its measurement and introduced a separate rinsing-effectiveness requirement to limit residual detergent in clothing, which may cause skin irritation and allergic reactions (European Commission, 2019b). As an intermediate input, water serves multiple purposes, and its use may largely reflect the need to satisfy rinsing requirements rather than considerations of wash quality and energy efficiency alone. Beyond water constraints, both regimes require a minimum level of stain removal, measured by a washing-efficiency index using standardized stain strips relative to a reference machine. The minimum threshold remained unchanged across regimes, although testing conditions changed. Advances in washing technology have made it possible to meet this requirement even at low temperatures (Josephy et al., 2015), suggesting that compliance need not be a major technological obstacle. The index is not displayed on the energy label but is reported in the product information sheet. The reported washing-efficiency index clusters near the regulatory threshold, consistent with limited incentives to report performance above the minimum requirement. I therefore treat wash

[9] Because detergent chemistry is standardized in regulatory tests, I treat it as exogenous to manufacturers' program-design decisions.

performance as a pass-or-fail certification constraint rather than a continuous attribute valued by consumers. Appendix A.2 reports the detailed regulatory constraints.

### 2.2 Data

I combine three main data sources: a product-level sales panel from GfK, program-level consumption information collected from product manuals, and a household laundry-habits survey conducted in Germany by Norstat on my behalf.

**Product-Level Sales and Characteristics** My primary data are a monthly panel of washing-machine sales in Germany from 2018 to 2023, provided by GfK. The data cover more than 90 percent of sales through electrical retailers across six geographic regions. For each region-month-model observation, the data report unit sales, revenues, and product characteristics, including energy-efficiency class, capacity, maximum spin speed, spin-drying efficiency, and noise level. To construct monthly market shares, I divide model-level unit sales by the estimated number of potential purchasers in each region-month. I define market size as the number of households in each region multiplied by the national washing-machine ownership rate and a monthly replacement probability implied by an average product lifespan of 12 years. To account for income heterogeneity in product demand, I supplement the sales data with the Panel on Household Finances (PHF), conducted by the Deutsche Bundesbank. I use the PHF to construct the household income distribution for demand estimation.

I focus on seven firms operating eight brands: BSH (Bosch and Siemens), Electrolux (AEG), Samsung, Miele, LG, Whirlpool (Bauknecht), and Arçelik (Beko). These firms account for approximately 93 percent of sales. I restrict the sample to washing machines with commonly purchased capacities of 7, 8, 9, and 10 kilograms. I define a product line as a combination of firm, load capacity, and maximum spin speed (e.g., Miele, 8 kg, 1,400 rpm). I define a product as a combination of firm, load capacity, maximum spin speed, energy-efficiency class, and quarter of initial market appearance (e.g., Miele, 8 kg, 1,400 rpm, class $A$, 2022Q1). The average transaction price is calculated as total revenue divided by total unit sales, aggregated across individual models within each product and region-month. The final sample contains 42 product lines and 349 products.

The composition of product introductions changes substantially over the sample period. Figure B14 in the appendix shows that early introductions are concentrated in lower energy-efficiency classes under the new label, while class $A$ becomes increasingly prevalent over time. Exits are relatively infrequent, suggesting that the shift toward higher-rated products occurs primarily through entry rather than the withdrawal of lower-rated products.[10]

[10]Exit dates are not explicitly recorded. I identify an exit when a product disappears from the dataset or records zero sales for at least one quarter. Exit measurement is therefore right-censored at the end of the sample.

**Program-Level Attribute Data** My second data source consists of model-program-level information collected from 2,087 washing-machine user manuals. These manuals report energy consumption, load size, water consumption, duration, and temperature for individual wash programs. Unlike the energy label, which bases efficiency ratings on Eco performance, the manuals also report attributes of non-Eco programs, allowing me to compare regulated and unregulated programs within the same machine. The completeness of reporting differs across regulatory regimes. Under the 2021 regime, manufacturers are required to report Eco attributes at full, half, and quarter loads, including energy consumption, water use, duration, and temperature. Manuals also report commonly used non-Eco programs, including Cotton 20°C, Cotton 60°C, and Easy Care, while Cotton 40°C is reported by some brands. Under the earlier regime, reporting was less standardized, with temperature and other attributes frequently missing.

To improve coverage, I supplement the manuals with the European Product Registry for Energy Labelling (EPREL), using overlapping observations to validate reported attributes and fill missing values.[11] I impute the remaining missing attributes using the procedure described in Appendix B, obtaining complete program-level attributes for all models in the demand sample. I then aggregate these attributes to the product level by taking the unweighted average of each program's attributes across models belonging to the same product, as defined above.

**Household Laundry-Habits Survey** My third data source is a household laundry-habits survey that I commissioned in Germany in December 2025 and fielded through Norstat. The survey contains 508 completed responses from households that purchased washing machines under either the old or new labeling regime. I collect household demographics, appliance purchase history, and the model identifier and purchase date of each respondent's washing machine. To measure consumer information, I ask whether respondents considered the energy-efficiency class when purchasing their machine and whether they interpret the rating as reflecting the efficiency of specific wash programs or the entire machine. I also collect self-reported washing behavior, including the frequency of use of each available wash program and the frequency of washing lightly and heavily soiled laundry.[12]

Using the reported model identifiers, I match each respondent to the corresponding product manual and recover the attributes of the wash programs available on their machine. The matched data combine household characteristics, beliefs about energy labels, program-use frequencies, and program-specific energy consumption, temperature, water use, and duration.

Table 1 reports summary statistics for product characteristics, demand, and wash-program at-

[11] EPREL is the EU product database containing technical information submitted by manufacturers for energy-labelled appliances. Under the 2021 regime, its technical documentation includes Eco energy consumption, water use, duration, and temperature at full, half, and quarter loads.

[12] The English version of the questionnaire is provided in Appendix C.

tributes. Panel A summarizes product characteristics and sales by energy-efficiency class. Higher-rated products tend to be more expensive and have larger capacities. Under the new label, average prices increase from €409 for class $D$ to €601 for class $A$, while average capacity rises from 7.73 to 8.51 kg. Panel B reports program attributes and self-reported usage. Eco consumes less energy than Cotton 40°C and Cotton 60°C, averaging 0.87 kWh per cycle compared with 1.04 and 1.49 kWh, respectively. Eco also operates at lower temperatures and uses less water but runs substantially longer, averaging 225 minutes compared with approximately 184 minutes for the two Cotton programs.

Table 1: Summary Statistics

| **Panel A. Products** | | | | | |
|---|---|---|---|---|---|
| Energy class | $A^{+++}$ | $D$ | $C$ | $B$ | $A$ |
| Average price (€) | 616.41 | 408.94 | 512.27 | 522.63 | 600.95 |
| Average monthly unit sales | 839.84 | 562.75 | 673.09 | 640.14 | 809.89 |
| Average capacity (kg) | 7.99 | 7.73 | 8.36 | 8.05 | 8.51 |
| Number of products | 139 | 44 | 40 | 41 | 85 |

| **Panel B. Program attributes and usage** | | | | | | |
|---|---|---|---|---|---|---|
| Program | Energy (kWh) | Water (L) | Duration (min) | Load size (kg) | Temperature (°C) | Reported cycles |
| Eco | 0.87 | 56.38 | 224.53 | 8.12 | 37.98 | 7,304 |
| Cotton 20°C | 0.34 | 81.47 | 171.31 | 8.12 | 21.62 | 3,247 |
| Cotton 40°C | 1.04 | 83.00 | 184.30 | 8.12 | 41.26 | 17,317 |
| Cotton 60°C | 1.49 | 81.94 | 183.70 | 8.12 | 56.12 | 10,580 |
| Easy Care | 0.66 | 57.82 | 132.57 | 3.77 | 39.67 | 14,732 |

*Notes:* Product statistics use national-market observations for the products retained in demand estimation, from 2018m1 to 2023m12. Panel A reports unweighted product-month means and distinct aggregated product counts by energy class. $A^{+++}$ includes only old-label products; $D$–$A$ use the new label. Prices are CPI-deflated euros. Panel B is pooled across energy classes and reports unweighted product means of program attributes, averaging within products over time first and excluding missing values separately for each attribute. Eco combines old-label Standard 60 and new-label Eco at full load. Attributes include existing imputations. Reported cycles are totals from both survey waves, with old and new Eco combined; survey counts are not restricted to full-load cycles or to products in the demand sample.

## 3 Regulation-Induced Energy Shifting

This section develops a stylized model, in the spirit of Holmstrom and Milgrom (1991) and Baker (1992), to illustrate how regulation targeting Eco can shift energy consumption toward unregulated

programs, potentially increase total electricity use, and generate socially inefficient choices.

Consider a firm offering a washing machine with two programs, Eco and Cotton, with durations $d_E$ and $d_C$. Their energy consumption is given by $e_E(d_E)$ and $e_C(d_C)$, respectively. Both functions are increasing in duration, while Eco consumes less energy at a given duration. Consumers observe Eco energy consumption at purchase but mistakenly perceive it as representative of both programs. Because perceived energy costs are identical across programs, they enter product-purchase demand but cancel from program-choice comparisons. Program utility is therefore $v_r = \beta d_r + \varepsilon_r$, where $\beta$ captures the average preference for duration and $\varepsilon_r$ follows a standard type-I extreme-value distribution. Expected program utility is

$$\mathcal{I}(d_E, d_C) = \log\left(e^{\beta d_E} + e^{\beta d_C}\right).$$

Taking product price $P$ as given and normalizing the firm's markup to one, the firm chooses program durations to maximize demand net of separable design costs:

$$\max_{d_E, d_C} \quad Q\left(P + \kappa e_E(d_E) - \theta \mathcal{I}(d_E, d_C)\right) - C_E(d_E) - C_C(d_C) \qquad \text{s.t.} \qquad d_E \leq \bar{d}_E,$$

where $Q' < 0$, $\kappa > 0$ captures consumers' valuation of perceived energy consumption, $\theta > 0$ measures the contribution of expected program utility to product demand, and $C_E$ and $C_C$ are separable design costs.[13]

Program designs are substitutes in expected utility, since $\frac{\partial^2 \mathcal{I}}{\partial d_E \partial d_C} = -\beta^2 s_E s_C \leq 0$, where $s_r$ denotes the program-choice probability. Shortening one program increases the marginal value of extending the other, regardless of the sign of $\beta$. When this substitution carries over to product demand, tightening the Eco-duration cap increases the marginal return to extending Cotton.

**Proposition 1** (Regulation-Induced Energy Shifting)**.** *Suppose the Eco-duration cap binds and Cotton duration has an interior local optimum satisfying the second-order condition. If* $\frac{\partial^2 Q}{\partial d_C \partial d_E} < 0$*, tightening the Eco-duration cap increases optimal Cotton duration and energy consumption:*

$$\frac{\partial d_C^*}{\partial \bar{d}_E} < 0, \qquad \frac{\partial e_C(d_C^*)}{\partial \bar{d}_E} < 0.$$

*Proof.* See Appendix D.1. □

The proposition establishes that restricting Eco duration can induce firms to increase energy consumption in the unregulated Cotton program. Whether this shifting raises electricity consumption also depends on consumer program choices. Conditional on machine purchase, average electricity

[13] The reduced-form demand specification follows from logit product choice when perceived energy consumption and expected program utility enter purchase utility additively, with coefficients expressed in price-equivalent units.

consumption per cycle is $\mathcal{E} = s_E e_E(d_E^*) + s_C e_C(d_C^*)$, where $s_E + s_C = 1$. Differentiating with respect to the Eco-duration cap yields

$$\frac{\partial \mathcal{E}}{\partial \bar{d}_E} = \underbrace{s_E \frac{\partial e_E}{\partial d_E}}_{\text{Direct Eco effect}} + \underbrace{s_C \frac{\partial e_C}{\partial d_C} \frac{\partial d_C^*}{\partial \bar{d}_E}}_{\text{Firm design response}} + \underbrace{\left[e_C(d_C^*) - e_E(d_E^*)\right] \frac{\partial s_C}{\partial \bar{d}_E}}_{\text{Consumer substitution}}.$$

where the Eco-duration cap binds. Tightening the cap reduces Eco energy consumption but increases Cotton energy consumption. When Cotton consumes more energy than Eco, tightening the cap shifts program use toward Cotton if the average preference for duration is positive, reinforcing the increase in energy consumption. Average electricity consumption can therefore increase if the Cotton energy and program-substitution effects outweigh the Eco savings.

The interaction between energy-efficiency and duration constraints can further affect firms' design responses. When firms can adjust other Eco inputs, such as temperature, a tighter duration cap creates room under a binding Eco energy target to increase these inputs, potentially offsetting the shift toward Cotton. A stricter energy target limits this adjustment and can strengthen incentives to improve Cotton instead. Appendix D.2 formalizes this interaction.

These design responses also have welfare implications when consumers have incomplete energy information. Consumers perceive Eco energy consumption as representative of both programs, overlooking differences in their actual electricity costs. A social planner instead accounts for program-specific energy consumption when choosing program designs and allocating washing cycles. Firms may therefore choose excessively long and energy-intensive Cotton programs relative to the social optimum. Details are shown in Appendix D.3.

## 4 Descriptive Evidence

This section presents three empirical patterns that motivate the structural model in Section 5 and relate to the mechanisms developed in Section 3. First, following the reform, Eco energy consumption per cycle initially rises before declining, while Cotton energy consumption increases and remains elevated, documenting divergent responses across programs. Second, products more exposed to the Eco-duration cap exhibit longer and more energy-intensive Cotton cycles relative to old-label benchmarks, consistent with the proposed mechanism of regulation-induced energy shifting. Third, consumers respond to energy ratings at purchase but often interpret Eco-based ratings as measuring whole-machine efficiency, even though non-Eco programs account for most reported washing cycles.

### 4.1 Changes in Program-Level Energy Consumption

Because program characteristics are fixed over a product's lifetime, I examine changes in mean energy consumption across products offered within each product line, weighting products equally. These changes reflect product turnover rather than shifts in sales shares. I restrict the sample to product lines observed between 2018Q1 and 2023Q4. For comparability across labeling regimes, I measure energy consumption per kilogram and apply common load weights to Eco programs.[14] I estimate

$$\log \bar{e}_{\ell rt} = \sum_{r' \in \mathcal{R}} \sum_{k} \beta_{kr'} \mathbf{1}\{t = k\} \mathbf{1}\{r = r'\} + \mu_{\ell r} + \mu_t + \varepsilon_{\ell rt}, \tag{1}$$

where $\bar{e}_{\ell rt}$ is mean energy consumption per kilogram in product line $\ell$, program $r$, and quarter $t$. The specification includes product-line-by-program and quarter fixed effects. The set $\mathcal{R}$ contains Eco, Cotton 20°C, Cotton 40°C, Cotton 60°C, and Easy Care. Wool serves as the reference program because it requires specialized care for delicate fabrics, making its design less likely to respond to policy-induced changes in program inputs. The coefficients capture changes relative to Wool.

Figure 1 shows divergent trends relative to Wool and the pre-policy reference quarter. Eco energy consumption initially rises before declining after 2021, whereas Cotton 60°C energy consumption rises and remains elevated. By 2023Q4, the estimated changes are −2.2 percent for Eco and 14.9 percent for Cotton 60°C. Cotton 40°C and Easy Care also increase by 8.4 and 7.9 percent, respectively.

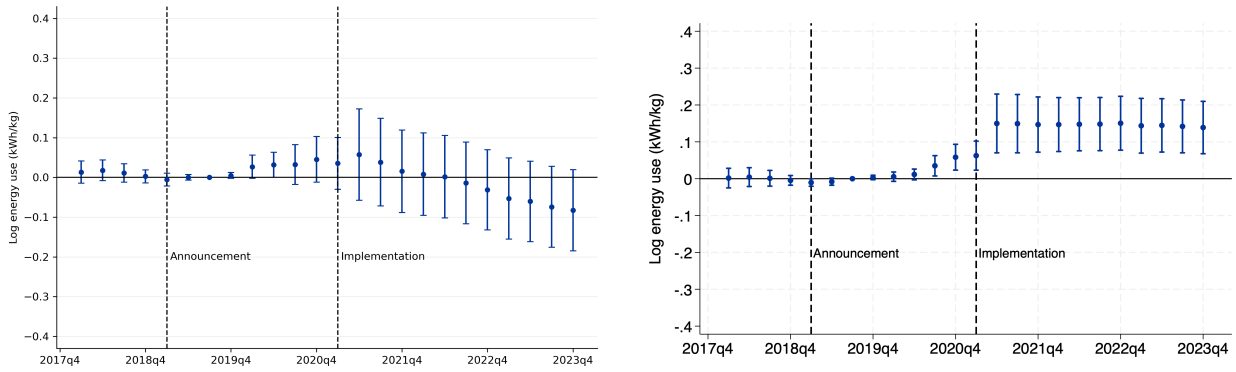
To examine the relationship between energy consumption per cycle across programs within a machine, Figure 4 plots Eco against Cotton energy consumption by machine capacity. It suggests that greater efficiency in the regulated Eco program does not necessarily translate into lower energy consumption in unregulated programs. Across all three capacities, machines with lower Eco energy consumption tend to have higher Cotton 60°C energy consumption, while the relationship for Cotton 40°C is weaker.

### 4.2 Accounting for Changes in Energy Consumption

To examine the inputs underlying changes in energy consumption, I estimate a relationship between electricity use, program characteristics, and product-specific energy efficiency. The approach follows the distinction between attribute choices and technological progress in Knittel (2011). Whereas time effects capture shifts in the technological frontier in that setting, I exploit observations on multiple programs and load sizes within a washing machine to allow conditional energy efficiency to differ

[14] I apply the new-label weights, normalized across full and half load, to Eco 40–60 under the new regime and to Standard Cotton 60°C at full load and Standard Cotton 40°C at half load under the old regime.

Figure 1: Changes in Program-Level Energy Consumption




(a) Eco

(b) Cotton 60°C

*Notes:* Panels (a) and (b) report changes in log energy consumption per kilogram for Eco and Cotton 60°C, respectively, relative to Wool. Dots denote the estimated coefficients $\beta_{kr'}$ from equation (1), normalized to zero in 2019Q3. The sample contains 31 product lines (firm × capacity × spin speed), observed in every quarter from 2018Q1 to 2023Q4, with products weighted equally within each line. Bars denote 95 percent confidence intervals based on standard errors clustered by product line. Dashed vertical lines mark the reform's announcement in March 2019 and implementation in March 2021.

across products.

Product-specific efficiency can reflect differences in motors, drum design, and control systems. These features may affect electricity consumption directly or permit a given level of cleaning performance at different combinations of temperature, duration, and water use.[15] I therefore interpret the product effect as conditional energy efficiency rather than as a direct measure of a particular hardware investment.

Specifically, I estimate

$$\log e_{jro} = z_j + \delta_{g(r)} + \theta_T \log T_{jro} + \theta_L \log L_{jro} + \theta_d \log d_{jro} + \theta_w \log w_{jro} + \varepsilon_{jro}. \tag{2}$$

Here, $e_{jro}$ is electricity consumption in kWh per cycle for product $j$, program $r$, and load condition $o$. For Eco, the observed load conditions are full, half, and quarter load; the available conditions differ across other programs and the earlier labeling regime. The variables $T_{jro}$, $L_{jro}$, $d_{jro}$, and $w_{jro}$ denote temperature in degrees Celsius, laundry load in kilograms, duration in hours, and water

[15]For example, Samsung describes Ecobubble as mixing detergent, water, and air before the main wash to improve detergent penetration and support washing at lower temperatures. AEG describes ProSense as using load sensing to adjust cycle duration, water, and energy use. Availability and operation vary across models and programs. These examples illustrate possible mechanisms rather than measured adoption in the estimation sample. Effects operating through observed temperature, duration, or water use enter those input terms rather than necessarily appearing in the product fixed effect. See Samsung, "What is the Ecobubble Feature on my Washing Machine?" and AEG, "Saving Money on Laundry."

use in liters. The product fixed effect $z_j$ is shared across programs and loads, with lower values indicating lower energy consumption conditional on observed inputs. The program-group effect $\delta_{g(r)}$ distinguishes Cotton from Eco and the legacy Standard programs. Cotton 20°C, Cotton 40°C, and Cotton 60°C share one effect, and Eco/Standard is the omitted group.

I estimate equation (2) by ordinary least squares with product fixed effects. Input coefficients are identified by variation across programs and load conditions within products, net of program-group effects. The sample retains eligible Eco/Standard observations, including imputed values, and Cotton observations with nonimputed energy consumption and inputs. Easy Care and Wool are excluded. The disturbance captures measurement error and remaining program- and load-specific operating differences. Estimation assumes that this disturbance has conditional mean zero given the inputs and fixed effects; conventional standard errors additionally assume independent, homoskedastic disturbances.

Appendix Table A1 reports the estimates. The equation provides an accounting framework for separating observed input changes from changes in conditional efficiency. This distinction also helps discipline the margins of adjustment included in the structural model, following the emphasis on alternative abatement strategies in Reynaert (2021). Because the product effect can absorb omitted engineering characteristics and systematic measurement differences, the accounting exercise does not separately identify technological adoption.

Table A2 reports energy consumption per cycle and per kilogram, their percentage changes relative to 2020Q2, and the associated input levels for Eco and Cotton 60°C. Both programs use the same product portfolio in each quarter, with equal weights across products. Eco outcomes use the regulatory full-, half-, and quarter-load weights. Products may enter and exit, so changes reflect portfolio turnover rather than changes within a continuing product. These comparisons differ from the preceding estimates relative to Wool and use a different load-weighting convention.

By 2023Q4, mean Eco energy consumption is 12.8 percent lower per cycle and 15.4 percent lower per kilogram than in 2020Q2. Over the same period, Cotton 60°C consumption rises by 9.3 percent per cycle and 7.5 percent per kilogram. Eco temperature falls from 34.0 to 31.3°C, while Cotton duration rises from 3.15 to 3.32 hours and its water use increases from 76.2 to 84.2 liters. The mean product effect becomes more negative, consistent with improved conditional efficiency in the offered portfolio. These patterns motivate the distinction between program-input choices and product-level efficiency in the model.

### 4.3 Eco Duration-Cap Exposure and Non-Eco Program Design

Proposition 1 in Section 3 predicts that a binding Eco duration cap shifts performance-related inputs toward unregulated Cotton programs. To investigate this mechanism, I examine whether

products more exposed to the cap have longer and more energy-intensive Cotton programs relative to pre-policy benchmarks.

The sample consists of dual-label products with observed old- and new-label energy classes. Because their non-Eco outcomes are not observed under both regimes, I compare them with matched old-label-only $A^{+++}$ products. For each program $r$ and outcome $q \in \{e, d, w\}$, denoting energy consumption, duration, and water use, respectively, I estimate

$$\log q_{jr}^{\text{new}} - \log \widehat{q}_{jr}^{\text{old}} = \beta_r^q \, \text{Exposure}_j + \mu_{k(j)}^{q,r} + \eta_{b(j)}^{q,r} + \varepsilon_{jr}^q, \tag{3}$$

where $\widehat{q}_{jr}^{\text{old}}$ is the mean outcome among matched old-label products, and $\mu_{k(j)}^{q,r}$ and $\eta_{b(j)}^{q,r}$ denote capacity and brand fixed effects. Exposure measures how much pre-policy regulated-program duration exceeds the new cap:

$$\text{Exposure}_j = \max\left\{0, \frac{\overline{d}_{f(j)k(j),\text{reg}}^{\text{pre}} - ub_{k(j)}}{ub_{k(j)}}\right\}, \tag{4}$$

where $\overline{d}_{f(j)k(j),\text{reg}}^{\text{pre}}$ is the mean pre-policy duration within a firm–capacity group and $ub_{k(j)}$ is the new duration cap. Identification comes from variation in exposure across firm–capacity groups, conditional on brand and capacity fixed effects.

Table 4 reports the estimates. A 10-percentage-point increase in exposure is associated with 4.5 percent higher energy consumption and 1.5 percent longer duration in Cotton 60°C relative to old-label benchmarks. Cotton 40°C energy consumption also increases by 1.9 percent, although its duration response is not statistically significant. Neither program exhibits a significant change in water use. These findings are consistent with Proposition 1, suggesting that a binding Eco duration cap shifts performance-related inputs toward Cotton programs, particularly Cotton 60°C.

### 4.4 Consumer Program Use, Label Beliefs, and Purchase Responses

The household survey reveals a substantial gap between how consumers use washing machines and what they believe energy ratings measure. Figure 5 shows that Eco accounts for only 9% of reported washing cycles among new-label households, compared with 25% for Cotton 40°C, 17% for Cotton 60°C, and 22% for Easy Care. Eco usage is also limited among old-label households, accounting for 13% of cycles. Despite this limited use, Figure 6 shows that 71% of new-label households mistakenly believe the energy class reflects energy consumption across all programs, even though it is determined by Eco performance alone. Moreover, 67% rely on the energy class rather than reported kWh consumption to assess energy costs. These findings suggest that consumers may value energy ratings as indicators of whole-machine efficiency, even though the ratings do not reflect energy consumption in their most frequently used programs.

To investigate purchase responses, I exploit the 2021 label rescaling from $A^{+++}$–$D$ to $A$–$G$. Before the reform, most products were rated $A^{+++}$, providing little differentiation across models. The rescaling redistributed these products across the new $A$–$D$ classes, creating variation in the information conveyed by energy ratings. I examine whether products receiving more favorable ratings experienced larger sales gains when the new labels became visible at retail.

Importantly, the transition in product design preceded the retail label switch. Beginning in 2020, manufacturers started updating control panels and Eco programs to accommodate the new regulatory requirements, following permission to display the Eco designation from late December 2019.[16] The new energy labels, however, became visible at retail only on March 1, 2021. Thus, for products whose designs remained unchanged around the rollout, the label switch altered the information presented to consumers without changing their underlying energy consumption. This timing allows me to examine the demand response to the newly displayed energy ratings separately from the earlier adjustments in program design.

Using a pre-implementation snapshot, I predict each product's energy class under the new system and define $E_j = 1$ for products predicted to receive $A$, $B$, or $C$, and $E_j = 0$ for those predicted to receive $D$. I estimate an event study centered on the March 2021 retail rollout, using February 2021 as the reference month:

$$\log(q_{jmt}) = \sum_{k \neq -1} \beta_k 1\{t = k\} E_j + \beta_p \log(p_{jmt}) + \mu_{jm} + \mu_t + \epsilon_{jmt}, \tag{5}$$

where $q_{jmt}$ and $p_{jmt}$ denote product sales and prices, $\mu_{jm}$ are product–market fixed effects, and $\mu_t$ are month fixed effects. I censor each product's panel at its first observed post-rollout rating change to limit contamination from subsequent redesign. The coefficients $\beta_k$ capture differential sales changes between products predicted to receive $A$–$C$ and those predicted to receive $D$, holding product identity fixed.

Figure 7 reports the estimates. Pre-reform coefficients are close to zero, indicating little evidence of differential pre-trends. Following the retail rollout, products predicted to receive $A$–$C$ experience an immediate and persistent increase in sales relative to $D$ products. The average post-reform coefficient of 0.44 implies approximately 55% higher relative sales. Because product-design adjustments began before the retail rollout, the sharp increase in relative sales around March 2021 is consistent with a demand response to the newly salient energy ratings rather than contemporaneous changes in energy performance. Together with the survey evidence, these results suggest that consumers place substantial value on energy ratings, which they often mistakenly interpret as measuring whole-machine efficiency.

[16] See Commission Regulation (EU) 2019/2023, Article 11, and Czyzewski (2022).

## 5 Model

I develop a static equilibrium model of washing-machine demand and supply to examine how energy labels and ecodesign requirements affect product design, consumer choices, and energy consumption. Within each period, the model follows four stages:

**S1. Product Entry.** Private portfolio-level entry shocks are realized. Firms simultaneously choose which product configurations to introduce, taking the distribution of rivals' entry decisions as given.

**S2. Program Design.** Market entry outcomes and design costs are realized and observed by all firms. Firms choose the wash-program inputs of entering products relative to a predetermined design frontier, subject to the energy-efficiency target and attribute constraints. Program inputs remain fixed over the product's lifetime.

**S3. Price Setting and Product Choice.** Product-level demand and marginal-cost shocks are realized. Firms set prices in Bertrand–Nash equilibrium, and households choose among the available washing machines and the outside option. Product demand internalizes the expected utility from subsequent program use.

**S4. Program Use.** Program-use shocks are realized before each wash occasion. Households then choose among the available wash programs of the purchased washing machine.

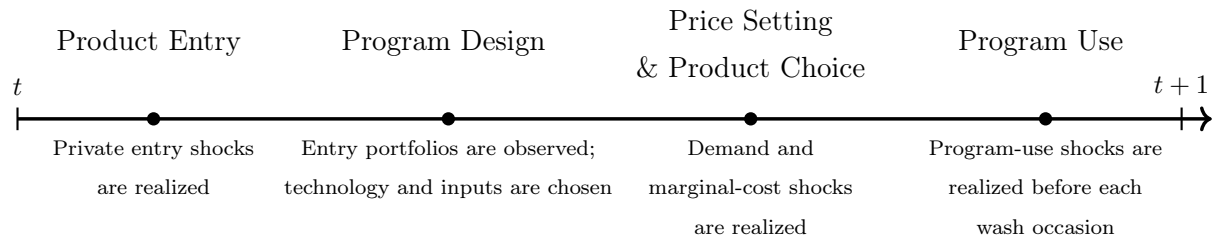



Figure 2: Timing within period $t$.

### 5.1 Demand

A market is indexed by $(m, t)$, where $m$ denotes region and $t$ quarter. I model household demand as a two-stage decision problem, combining a differentiated-products demand system in the spirit of Berry et al. (1995) with a subsequent usage decision, as in the appliance ownership and utilization framework of Dubin and McFadden (1984).

There is a continuum of households indexed by $i$. Household heterogeneity relevant for demand is summarized by income $y_i$ and the household's share of high-stain laundry $\tau_i$. Income shifts price sensitivity, while the high-stain share shifts preferences over wash-program temperature and duration. I begin with the program-use decision because it determines the continuation value associated with owning each product.

**Program Use** Let $r \in \mathcal{R}$ index a common set of inside wash programs: Cotton 20°C, Cotton 40°C, Cotton 60°C, Easy Care, and Eco. Let $r = 0$ denote an outside option, interpreted as the use of any other program, and let $s$ index wash occasions. Conditional on owning product $j$, household $i$'s utility from program $r \in \mathcal{R}$ on occasion $s$ is

$$v_{ijrs} = \delta_{c(r)} + \rho_i^T T_{jr} + \rho_i^d d_{jr} + \rho_w w_{jr} + \rho_L L_{jr} + \varepsilon_{ijrs}^v, \tag{6}$$

where $T_{jr}$, $d_{jr}$, $w_{jr}$, and $L_{jr}$ denote temperature, duration, water use, and load size;[17] $\varepsilon_{ijrs}^v$ is i.i.d. type-I extreme value and is redrawn across wash occasions; and the outside option is normalized to $v_{ij0s} = \varepsilon_{ij0s}^v$. Preferences for temperature and duration vary with the household's share of high-stain laundry according to $\rho_i^q = \rho_0^q + \rho_1^q \tau_i + \sigma_q \nu_i^q$, for $q \in \{T, d\}$ and $\nu_i^q \sim N(0,1)$.

I assume that households do not observe the product-specific energy cost of an individual wash cycle when choosing programs. Instead, they have persistent perceptions of the relative operating costs of different program categories. The program-category fixed effect $\delta_{c(r)}$ therefore captures both baseline preferences and washing habits associated with category $c(r)$, as well as perceived differences in relative operating costs across program categories.[18]

Let $\bar{v}_{ijr} \equiv v_{ijrs} - \varepsilon_{ijrs}^v$ denote the deterministic component of utility from program $r$. Under the type-I extreme value assumption, the conditional probability that household $i$ chooses program $r \in \mathcal{R}$ on product $j$ is

$$P_{ijr} = \frac{\exp\left(\bar{v}_{ijr}\right)}{1 + \sum_{r' \in \mathcal{R}} \exp\left(\bar{v}_{ijr'}\right)}. \tag{7}$$

The expected utility from product $j$'s program bundle is

$$I_{ij} \equiv \mathbb{E}_{\varepsilon^v}\left[\max_{r \in \{0\} \cup \mathcal{R}} v_{ijrs}\right] = \log\left(1 + \sum_{r \in \mathcal{R}} \exp\left(\bar{v}_{ijr}\right)\right), \tag{8}$$

[17] Households may not observe individual program inputs directly, but they perceive the resulting washing quality. I represent perceived quality as a function of the underlying program inputs without requiring households to evaluate each input separately. Technology may also affect washing quality, but I assume that it is common across programs within a product and incorporate its effect in the product-choice stage.

[18] Program categories are Cotton, Easy Care, and Eco. In particular, Cotton 20°C, Cotton 40°C, and Cotton 60°C share the same category fixed effect.

up to the Euler–Mascheroni constant, which is common across products and normalized away.

**Product Choice** In each market $(m,t)$, household $i$ chooses among the set of available washing machines $\mathcal{J}_{mt}$ and an outside option. For product $j \in \mathcal{J}_{mt}$, indirect utility is

$$u_{ijmt} = -\alpha_i p_{jmt} + \sum_{c \in \mathcal{C}} \beta_c \mathbf{1}\{\text{ec21}(j) = c\}\mathbf{1}_{\text{new},t} + \beta^I \mathcal{I}_{ij} + x_j'\beta^x + \delta_{b(j)} + \delta_m + \delta_t + \xi_{jmt} + \epsilon_{ijmt}, \quad (9)$$

where $\epsilon_{ijmt}$ is i.i.d. type-I extreme value and the outside-option mean utility is normalized to zero. Price sensitivity varies with household income according to $\alpha_i = \exp(\bar{\alpha} + \alpha^y y_i)$.

The indicator $\mathbf{1}_{\text{new},t}$ equals one under the new $A$–$G$ labeling regime, while $\mathbf{1}\{\text{ec21}(j) = c\}$ indicates whether product $j$ belongs to energy class $c \in \mathcal{C} = \{A, B, C, D\}$, with class $D$ normalized to zero. I assume that consumers do not know the regulatory formula underlying the energy class and do not translate the label into product-specific energy costs. Instead, the displayed class enters utility directly as a coarse signal of energy efficiency, consistent with evidence that consumers may respond to energy certifications without fully processing the underlying energy-cost information (Houde, 2018). The coefficients $\beta_c$ therefore capture consumers' valuation of each displayed energy class under the new label. I omit the pre-reform energy-class effects because almost all products were rated $A^{+++}$ before the reform.

The inclusive value $\mathcal{I}_{ij}$ links program use to product choice by summarizing household $i$'s expected value from product $j$'s program bundle. Although I do not explicitly model reputation formation, this provides a parsimonious way to capture a reputation-like channel: consumers may learn about downstream product performance through reviews and other pre-purchase information (Reimers and Waldfogel, 2021), giving firms an incentive to design programs that generate higher expected use value. The coefficient $\beta^I$ determines how strongly this expected program-use value enters the product-purchase decision.

The vector $x_j$ contains other exogenous, time-invariant product characteristics, including capacity, spin speed, and technology. The terms $\delta_{b(j)}$, $\delta_m$, and $\delta_t$ denote brand, region, and quarter fixed effects, while $\xi_{jmt}$ captures product-market-time attributes observed by households but not by the econometrician.

Let $\bar{u}_{ijmt} = u_{ijmt} - \epsilon_{ijmt}$ denote the deterministic component of product utility. Conditional on household characteristics, the probability that household $i$ purchases product $j$ is

$$s_{ijmt} = \frac{\exp\left(\bar{u}_{ijmt}\right)}{1 + \sum_{k \in \mathcal{J}_{mt}} \exp\left(\bar{u}_{ikmt}\right)}. \quad (10)$$

The market share of product $j$ integrates this choice probability over the distribution of household

heterogeneity:

$$s_{jmt} = \int s_{ijmt}\, dF_{mt}(y_i, \tau_i, \nu_i^T, \nu_i^d). \tag{11}$$

### 5.2 Supply

**Actions.** Let $\mathcal{L}_f$ denote firm $f$'s set of product lines, where a line $\ell$ is defined by brand $\times$ capacity $\times$ spin speed. On each line, the firm can introduce at most one new product in quarter $t$. The line-level action is $a_{f\ell t} \in \mathcal{A}_{f\ell t} \subseteq \{0, A, B, C, D\}$, where $a_{f\ell t} = 0$ denotes no entry and $a_{f\ell t} \in \{A, B, C, D\}$ denotes entry of a product with the corresponding new-label energy class, subject to the line's feasibility restrictions. The firm's portfolio action is $a_{ft} = (a_{f\ell t})_{\ell \in \mathcal{L}_f}$, and $a_t = (a_{ft}, a_{-f,t})$ denotes the joint entry action of all firms. Let $\mathcal{J}_t(a_t)$ denote the resulting product portfolio.

For each entering product $j$, let $q_{fjrt}$ denote the vector of inputs chosen for wash program $r$, including temperature, duration, water use, and other design inputs. I collect these program-design choices in $q_{ft}$ and denote the industry-wide profile by $q_t = (q_{ft}, q_{-f,t})$. Program inputs of incumbent products remain fixed. Given the resulting product portfolio and program designs, firms then compete in prices across markets.

**Price Setting.** Given the realized product portfolio $\mathcal{J}_t(a_t)$ and program designs $q_t$, firms compete in prices across markets. Firm $f$ chooses $\{p_{jmt}\}_{j \in \mathcal{J}_{ft}(a_t)}$ to maximize

$$\pi_{ft}\big(\mathcal{J}_t(a_t), q_t\big) = \sum_m \sum_{j \in \mathcal{J}_{ft}(a_t)} \big(p_{jmt} - mc_{jt}\big) s_{jmt}\big(p; \mathcal{J}_t(a_t), q_t\big) M_{mt}, \tag{12}$$

where $s_{jmt}$ is product $j$'s market share and $M_{mt}$ is market size. Prices are determined in Bertrand–Nash equilibrium. Marginal cost is

$$\log mc_{jt} = x_j^{mc\prime}\gamma + \gamma_{b(j)} + \gamma_t + \omega_{jt}, \tag{13}$$

where $x_j^{mc}$ contains capacity, spin speed, and technology, $\gamma_{b(j)}$ and $\gamma_t$ are brand and quarter fixed effects, and $\omega_{jt}$ is an idiosyncratic marginal-cost shock observed by firms. Program inputs other than technology are control-algorithm parameters and therefore do not directly enter marginal production cost.

**Program Design.** After entry outcomes are realized and observed, firms simultaneously choose the wash-program inputs of entering products. For product $j$ entering on line $\ell$, let $q_{f\ell r}^{\text{ref}}$ denote its predetermined reference design. Duration references remain anchored to the matched $A^{+++}$ benchmark; technology and temperature use the latest strictly earlier first-class introduction on the

line, with an $A^{+++}$ fallback. A repeated introduction of the same class copies that class's first design rather than resetting the duration reference. Define $\Delta q_{fjrt} = q_{fjrt} - q^{\text{ref}}_{f\ell r}$. The design cost of product $j$ is

$$DC_{fjt} = \sum_{r \in \mathcal{R}} \left[ \kappa_r' \Delta q_{fjrt} + \Delta q_{fjrt}' C_r \Delta q_{fjrt} \right], \tag{14}$$

where $\kappa_r$ and $C_r \succeq 0$ govern the linear and quadratic costs of departing from the reference design.

Conditional on the realized portfolio $a_t$, firm $f$ solves

$$\begin{aligned} W_{ft}(a_t) = \max_{q_{ft}} \mathbb{E}_{\xi,\omega} \left[ \pi_{ft}\big(\mathcal{J}_t(a_t), q_t; \xi, \omega\big) \right] - \sum_{j \in \mathcal{J}^{\text{new}}_{ft}(a_{ft})} DC_{fjt} \\ \text{s.t.} \quad \text{EEI}_j(q_{j,\text{Eco}}) \in \mathcal{E}(e_j), \\ d_{jo}(q_{j,\text{Eco}}) \le \overline{d}_o(k_j), \qquad o \in \{F, H, Q\}. \end{aligned} \tag{15}$$

Here $k_j$ denotes product $j$'s rated capacity, $\mathcal{E}(e_j)$ is the EEI range associated with the energy class chosen at entry, and $\overline{d}_o(\cdot)$ denotes the applicable load-specific duration limits. The exact regulatory formulas are provided in the Appendix. Non-Eco programs are unconstrained by these requirements but affect program use, product demand, and operating profits.

**Product Entry.** Firms simultaneously choose their product introductions, taking the distribution of rivals' entry decisions, $\mathcal{P}_{-f,t}$, as given. Each firm receives private portfolio-specific payoff shocks, which capture unobserved factors affecting its introduction decisions. Firm $f$ solves

$$V_{ft} = \max_{a_{ft} \in \mathcal{F}_f(J_{f,t-1})} \left\{ \mathbb{E}_{\mathcal{P}_{-f,t}} \left[ W_{ft}(a_{ft}, a_{-f,t}) \right] - C^{\text{entry}}_{ft}(a_{ft}) + \sigma \eta_{ft}(a_{ft}) \right\}, \tag{16}$$

where $\eta_{ft}(a_{ft})$ follows a type-I extreme-value distribution and $\sigma$ determines the scale of payoff shocks.

I specify entry costs to capture both the fixed cost of introducing a product and the additional cost of introducing a different energy class:

$$\begin{aligned} C^{\text{entry}}_{ft}(a_{ft}) = \sum_{\ell : a_{f\ell t} \neq 0} \big[ \kappa_L + \gamma_{q(t)} + \beta_N \log(1 + N_{f\ell,t-1}) + \beta_G \log(1 + G_{f\ell t}) \\ + \kappa^{h(t)}_{a_{f\ell t}} \mathbf{1}\{a_{f\ell t} \notin \mathcal{C}^{\text{last}}_{f\ell,t}\} \big]. \end{aligned} \tag{17}$$

The common launch cost $\kappa_L$ captures fixed expenses associated with introducing a new product, while $\gamma_{q(t)}$ allows these costs to vary seasonally, with $\gamma_1 = 0$. I allow launch costs to depend on the number of existing products on a line, $N_{f\ell,t-1}$, to capture differences in the cost of expanding an established product line, and on the number of quarters since its previous introduction, $G_{f\ell t}$, to account for variation in product replacement cycles. The switching premium $\kappa^{h(t)}_{a_{f\ell t}}$ captures

additional costs of introducing an energy class not offered in the line's most recent introduction quarter, such as class-specific development and preparation costs beyond the continuous design adjustments. These premiums differ between the transition and post-policy periods, with classes $C$ and $D$ sharing a coefficient. The specification allows firms to incur switching costs when returning to an older class.

### 5.3 Discussion

The model makes two main simplifying assumptions. First, I treat consumer purchase and firm entry decisions as static. I assume that consumers replace washing machines only upon failure, rather than optimally timing replacement as in Goettler and Gordon (2011). Unlike microprocessors, where consumers may delay upgrades in anticipation of technological improvements and price declines, washing-machine replacement is more likely driven by necessity. Firms likewise maximize current-period profits, even though products remain on the market for approximately 2.5 years on average. Following Wollmann (2018), static entry decisions can approximate dynamic investment decisions when current profitability predicts future returns, with estimated entry costs reflecting an implicit hurdle rate. Although these assumptions abstract from dynamic replacement and entry incentives, such considerations are less central to the paper's main mechanism of regulation-induced changes in program design and household substitution across wash programs.

Second, I simplify how firms adjust program designs across product introductions. The data show minimal changes in program inputs across cohorts within the same product line and energy class. I therefore allow firms to redesign all programs when introducing a new energy class on a line, while subsequent introductions of the same class inherit the existing design. The model includes a fixed cost of product entry but abstracts from program-level fixed costs of modifying existing designs. In practice, such costs may discourage firms from adjusting unregulated programs and lead them to concentrate redesign efforts on Eco. This simplification captures the observed stability of designs within line-class combinations but does not explicitly model firms' decisions about which programs to redesign.

## 6 Estimation

I estimate program preferences, product demand, and the engineering relationship first; recover marginal costs from pricing conditions; then estimate design and entry costs. Two neural-network approximations make repeated portfolio and input evaluations feasible. The first predicts operating profits and own-firm input derivatives. The second predicts optimized inputs and operating and net profits after solving the design game. Appendix F.1 details their training. This use of profit

approximation is related to Bodéré (2026); the static entry estimator below differs from his dynamic model and fixed-rival-beliefs estimation.

### 6.1 Program Use, Demand, and Marginal Cost

For survey household $i$, let $n_{ir}$ count reported uses of program $r$. Simulated maximum likelihood integrates household-specific tastes jointly across its reported washes:

$$\max_{\rho} \sum_{i} \log \left[ \frac{1}{R} \sum_{b=1}^{R} \prod_{r} P_{ijr}(\nu_i^b; \rho)^{n_{ir}} \right]. \tag{18}$$

The count multinomial constant can be omitted. Averaging inside the household likelihood preserves persistent preference heterogeneity. I then compute household-specific inclusive values for product demand.

Product demand is estimated by aggregate-share GMM using the inversion of Berry et al. (1995) and the implementation of Conlon and Gortmaker (2020). With demand residual $\xi(\theta)$ and instruments $Z$, the criterion is

$$g(\theta) = \frac{1}{n} Z'\xi(\theta), \qquad \widehat{\theta} = \arg\min_{\theta} g(\theta)' W g(\theta). \tag{19}$$

The current specification uses a one-step weighting matrix and aggregate moments, without additional micro moments. Excluded instruments include same-firm and rival characteristic sums, their interactions with market-quarter mean income, and mean program inclusive value. Characteristic sums use capacity, spin speed, and displayed electricity consumption. Technology is an included demand characteristic, not an additional BLP characteristic-sum instrument. Income, stain intensity, and temperature and duration tastes are integrated with 625 weighted agents per market. The validity of the instruments and the exogeneity of included technology are maintained identifying assumptions.

Given demand estimates and product ownership, the multiproduct Bertrand first-order conditions recover markups and marginal costs. I regress log recovered marginal cost on the variables in equation (13). Its residual and the demand residual determine the joint shock distribution used to integrate design-stage profits. All profit labels and entry costs are converted to euros, consistently with the price normalization in demand.

**Instruments.** The instruments follow the timing of the model. Product characteristics and program designs are chosen before the product-market demand shock $\xi_{jmt}$ and marginal-cost shock $\omega_{jt}$ are realized, whereas prices are set after both shocks are observed. Price is therefore endogenous,

while predetermined product and program characteristics are orthogonal to the contemporaneous demand shock.

I instrument for price using standard BLP instruments constructed from the capacity and spin speed of products offered by the same firm and by rival firms. Because price sensitivity varies with household income, I additionally interact these instruments with mean household income in market $(m, t)$. These interactions provide excluded variation for the price–income interaction and therefore identify $\alpha^y$ separately from average price sensitivity.

The coefficient $\beta^I$ is also nonlinear because the inclusive value $\mathcal{I}_{ij}$ varies across households. I use the market-average inclusive value

$$Z^I_{jmt} = \int \mathcal{I}_{ij}\, dF_{mt}(i) \tag{20}$$

as an additional instrument. Under the model timing, this object is determined by program designs chosen before $\xi_{jmt}$ is realized and therefore provides exogenous variation in expected program-use value. This use of excluded variation to identify nonlinear preference heterogeneity follows the general identification logic of Berry and Haile (2014).

Collecting the excluded instruments and exogenous product characteristics in $Z_{jmt}$, the aggregate moments satisfy

$$\mathbb{E}\left[Z_{jmt}\xi_{jmt}(\theta)\right] = 0. \tag{21}$$

### 6.2 Supply Estimation

I estimate supply parameters in two steps. First, I estimate the design costs governing program-level input choices conditional on the observed product portfolio. Second, taking the estimated design costs as given, I estimate the entry costs governing energy-class choices within each product line.

**Design costs.** I estimate the parameters $(\kappa, C)$ in equation (14) for five program-input groups: Cotton 40°C duration, Cotton 60°C duration, Eco duration, Eco temperature, and technology. There are ten parameters to estimate. I use a minimum distance (MD) estimator that minimizes a weighted distance between observed moments and their model-implied counterparts, following the moment-matching approach in Hall and Rust (2003), Goettler and Gordon (2011), Chen and Xu (2022), and Su et al. (2026).[19] For each candidate parameter vector, I solve firms' constrained input-choice problems in equation (15) using sequential least squares programming (SLSQP) (Kraft, 1988). Details of the numerical implementation are provided in Appendix F.

[19] An alternative is to recover design costs from firms' first-order conditions (Fan, 2013; Barwick et al., 2024; Ito et al., 2026). In my setting, binding regulatory constraints introduce product-specific shadow values. Solving the constrained design problem incorporates these shadow values implicitly, avoiding the need to estimate them separately.

The MD estimator solves

$$(\widehat{\kappa}, \widehat{C}) = \arg \min_{\substack{\kappa \in \mathbb{R}^5 \\ C \in \mathbb{R}^5_{++}}} \sum_{n=1}^{N} w_n \left[ \widehat{m}_n(\kappa, C) - m_n \right]^2 . \tag{22}$$

Here, $N$ is the number of moments, $m_n$ denotes observed moment $n$, and $\widehat{m}_n(\kappa, C)$ is its model-implied counterpart. The fixed weights $w_n$ normalize discrepancies by inverse squared standard deviations and assign equal total weight to each input group. I use three sets of moments:

- *Baseline design restrictions.* I match average marginal operating profits at the $A^{+++}$ reference designs to the linear cost coefficients. The corresponding discrepancy is $\kappa_{rq} - \frac{1}{|\mathcal{J}_{rq}|} \sum_{j \in \mathcal{J}_{rq}} \frac{\partial \Pi^{\text{op}}_{f(j)}(q^{\text{ref}})}{\partial q_{jr}}$, where $\Pi^{\text{op}}_{f(j)}$ denotes operating profit before design costs and $\mathcal{J}_{rq}$ contains eligible baseline products.
- *Input changes at product introduction.* I match observed input changes, $q_{jr} - q^{\text{ref}}_{jr}$, to predicted changes for each eligible product introduction.
- *Class-specific mean inputs and energy consumption.* I match observed mean inputs, $\frac{1}{|\mathcal{J}_{rq,ec}|} \sum_{j \in \mathcal{J}_{rq,ec}} q_{jr}$, and mean energy consumption to their model-predicted counterparts. Here, $\mathcal{J}_{rq,ec}$ denotes the corresponding eligible products in energy class $ec$.

The parameters are jointly estimated from the full set of moments, although different moment sets provide the main identifying information for different parameters. The baseline moments primarily inform the linear coefficients $\kappa_{rq}$. I treat the market in 2020Q1 as stationary and use each matched $A^{+++}$ product's own design as its reference. At the reference, the quadratic cost term has zero derivative, leaving $\kappa_{rq}$ as the marginal design cost. The EEI distribution in Figure A11 supports treating the old-label EEI constraint as slack for these products. For interior choices without other binding constraints, firms' first-order conditions therefore link the linear cost coefficients to marginal operating profits.

The input-change moments primarily inform the curvature parameters $C_{rq}$, which govern the magnitude of marginal design costs associated with deviations from the reference design. Conditional on the linear coefficients and changes in marginal operating profits, larger observed adjustments imply lower curvature.

The class-specific mean-input and energy moments discipline the model's fit to product design and energy consumption across energy classes. Mean-input moments place additional weight on systematic differences between observed and predicted designs, while energy moments discipline the combinations of inputs through the energy equation. These moments complement the baseline and input-change moments in the joint estimation, rather than primarily identifying a particular cost parameter.

**Entry costs.** I estimate the fixed entry cost $C^{\text{fix}}$ for repeated energy classes, the cost $C^{\text{trans}}$ of introducing a new energy class, the coefficient $\phi$ governing how entry costs vary with the number of products already on a line, and the scale $\sigma$ of private entry shocks. These parameters are jointly disciplined by variation in entry decisions, existing product counts, and the relative profitability of repeated and new energy classes. Estimation requires solving for equilibrium beliefs about rivals' entry decisions, which determine the expected profitability of each portfolio.

Building on the incomplete-information entry framework of Seim (2006), I model firms' portfolio choices given equilibrium beliefs about rival portfolios. With independent type-I extreme-value shocks across feasible firm portfolios, these choices follow a portfolio logit probability.[20] Expected profit net of entry costs is

$$\overline{W}_{ft}(a_{ft};\theta,\mathcal{P}_{-f,t}) = \mathbb{E}_{\mathcal{P}_{-f,t}}\left[W_{ft}(a_{ft},a_{-f,t})\right] - C^{\text{entry}}_{ft}(a_{ft};\theta),$$

where $\theta$ collects the entry-cost parameters and $\sigma$, and $\mathcal{P}_{-f,t}$ is the distribution of rival portfolios. Here, $W_{ft}$ is the payoff net of design costs obtained by solving the program-design problem in equation (15).

Direct estimation based on the full portfolio probabilities faces two computational challenges. First, the action space grows exponentially with the number of active product lines. The estimation sample contains 469 product-line–quarters across seven firms and 16 quarters, averaging approximately 29 eligible lines per quarter. With four energy-class choices and a no-entry option on each line, a market with 29 eligible lines has up to $5^{29} \approx 1.86 \times 10^{20}$ joint portfolio configurations. Enumerating these configurations to calculate expected profits and equilibrium choice probabilities is therefore infeasible. Second, evaluating payoffs for each candidate market portfolio requires solving the constrained program-design game for the newly introduced products.

I address these challenges using conditional choice probabilities, simulated rival portfolios, and a neural-network approximation to optimized payoffs. First, conditioning on the firm's choices on its other lines gives

$$\mathcal{P}_{f\ell t}(a \mid a_{f,-\ell,t};\theta,\mathcal{P}_{-f,t}) = \frac{\exp\left\{\overline{W}_{ft}((a,a_{f,-\ell,t});\theta,\mathcal{P}_{-f,t})/\sigma\right\}}{\sum_{a'\in\mathcal{A}_{f\ell t}(a_{f,-\ell,t})}\exp\left\{\overline{W}_{ft}((a',a_{f,-\ell,t});\theta,\mathcal{P}_{-f,t})/\sigma\right\}}. \tag{24}$$

Here, $a_{f,-\ell,t}$ denotes the firm's choices on its other lines, and $\mathcal{A}_{f\ell t}(a_{f,-\ell,t})$ contains at most five feasible actions on line $\ell$. Conditioning eliminates the sum over full firm portfolios while preserv-

[20] The full portfolio choice probability is

$$\mathcal{P}_{ft}(a_{ft};\theta,\mathcal{P}_{-f,t}) = \frac{\exp\{\overline{W}_{ft}(a_{ft};\theta,\mathcal{P}_{-f,t})/\sigma\}}{\sum_{a'_{ft}\in\mathcal{F}_f(J_{f,t-1})}\exp\{\overline{W}_{ft}(a'_{ft};\theta,\mathcal{P}_{-f,t})/\sigma\}}. \tag{23}$$

Conditioning on the firm's choices on its other lines cancels the full portfolio normalizer and yields equation (24).

ing interdependence across product lines. For given rival beliefs, I estimate the parameters by maximizing the conditional composite log likelihood (Varin et al., 2011):

$$\mathcal{L}(\theta; \mathcal{P}) = \sum_{(f,\ell,t)\in\mathcal{S}^{\text{entry}}} \log \mathcal{P}_{f\ell t}\left(a^{\text{obs}}_{f\ell t} \mid a^{\text{obs}}_{f,-\ell,t}; \theta, \mathcal{P}_{-f,t}\right), \tag{25}$$

where $\mathcal{S}^{\text{entry}}$ is the estimation sample.

Although conditioning simplifies the firm's choice probability, expected profits still depend on the joint distribution of rivals' portfolios, which must be consistent with equilibrium behavior. I approximate this distribution using Gibbs sampling, sequentially updating line-level actions according to their conditional probabilities to generate complete portfolio draws (Casella and George, 1992). Following the logic of nested pseudo-likelihood estimation (Aguirregabiria and Mira, 2007), I alternate between estimating entry-cost parameters given rival beliefs and updating those beliefs using simulated portfolios until the parameters and conditional choice probabilities stabilize.

Evaluating profits for these simulated portfolios would still require repeatedly solving the program-design game. To reduce this computational burden, I train a neural-network surrogate on payoffs obtained from solutions of that game. The surrogate approximates firms' payoffs after design optimization across market portfolios. Appendix F provides details of the surrogate, belief updates, and convergence checks.

## 7 Estimation Results

### 7.1 Demand Estimates

Panel A of Table 2 reports estimates of the program-use utility function in equation (6). On average, households prefer higher temperatures, shorter cycles, and greater water use. Households with a larger share of heavily stained laundry place less value on temperature and exhibit a smaller duration penalty. Duration preferences vary substantially across households, with an estimated standard deviation of 2.248 per hour. This heterogeneity shapes substitution across programs: holding other attributes fixed, shortening Eco by 30 minutes reduces its predicted usage share by 1.65 percentage points and raises the combined Cotton share by 1.61 percentage points.[21] Although households prefer shorter cycles on average, the aggregate response depends on the preferences of households most responsive to changes in Eco duration.

Panel B of Table 2 reports estimates of the product-choice utility function in equation (9). Price sensitivity decreases with income: the log-income coefficient is $-0.578$ and significant at the

[21] The simulation uses the 117 households in the new-label survey, weighted by reported wash counts, and holds all other program attributes fixed.

5 percent level. Households also value more efficient technology. The coefficient on product-level technology, $z_j$, is $-1.147$ and significant at the 1 percent level. Since lower values of $z_j$ indicate lower electricity consumption conditional on program inputs, this coefficient associates greater conditional efficiency with higher demand, holding observed program attributes and energy class fixed.

The estimates also imply a substantial purchase premium for class $A$. Relative to class $D$, the utility coefficients are 0.956 for class $A$, 0.186 for $B$, and 0.172 for $C$, all significant at the 1 percent level. The class-$A$ coefficient implies a willingness to pay of approximately €145 relative to class $D$ for a household at the sample mean log income, holding other characteristics fixed. This valuation is higher for households with greater income because they are less price sensitive. Finally, the coefficient on the expected value of the wash-program menu is 0.147 and significant at the 1 percent level, indicating that households are more likely to purchase machines offering programs better suited to their preferences.

### 7.2 Supply Estimates

Panel A of Table 3 reports the marginal-cost estimates from equation (13). More efficient technology is associated with higher production costs, although the magnitude is modest: the coefficient on $z_j$ is $-0.116$ and significant at the 5 percent level, implying that a technology improvement that reduces electricity consumption by 10 percent at fixed program inputs raises predicted marginal cost by only 1.23 percent. Marginal costs also increase with capacity, with 7, 8, and 9 kg machines costing less to produce than the omitted 10 kg category. Including firm and quarter fixed effects, the regression explains 75.4 percent of the variation in recovered log marginal costs.

Equation (14) reports the estimated costs of departing from a reference design. For input $k$, a deviation of $h$ costs $(\kappa_r)_k h + (C_r)_{kk} h^2$ euros per introduction, with curvature reported as $2(C_r)_{kk}$. A ten-minute change costs € 2,045 for Cotton 40°C duration and € 4,057 for Cotton 60°C duration, compared with € 50 for one Eco-duration coordinate. Extending both full- and half-load Eco cycles by ten minutes costs € 100. These amounts are costs per introduction, not per machine sold, and exclude discrete entry costs and changes in operating profits. They measure deviations from the reference design rather than incremental costs from an already modified design.

Eco duration is thus a relatively inexpensive margin of adjustment. Although the quadratic cost rises with the size of the change, firms can lengthen Eco cycles at much lower design cost than Cotton cycles. This difference helps explain why removing the duration cap produces a large Eco response. Equilibrium responses also depend on regulatory constraints, household preferences, and other design choices.

Technological improvements are considerably more expensive. A change in the log energy-requirement index of $\log(0.9)$, equivalent to a 10 percent reduction in energy requirements at fixed

Table 2: Program-Use and Product-Demand Estimates

| Variable | Parameter | Estimate | Std. Error |
|---|---|---|---|
| *Panel A: Program-use estimates* | | | |
| *Program attributes* | | | |
| Temperature | $\rho_T$ | 0.046 | (0.004) |
| Duration (hours) | $\rho_d$ | 0.582 | (0.064) |
| Water consumption | $\rho_w$ | 0.016 | (0.001) |
| Representative load (kg) | $\rho_L$ | 0.373 | (0.027) |
| *Household preference variation* | | | |
| Temperature × high-stain share | $\gamma_T$ | −0.041 | (0.012) |
| Duration × high-stain share | $\gamma_d$ | 0.362 | (0.171) |
| Temperature taste SD | $\sigma_T$ | 0.064 | (0.002) |
| Duration taste SD | $\sigma_d$ | 2.248 | (0.068) |
| *Panel B: Product-demand estimates* | | | |
| *Linear demand coefficients* | | | |
| Shared technology | $\beta_{tech}$ | −1.147 | (0.227) |
| Capacity: 7 kg | $\beta_{7kg}$ | 0.309 | (0.251) |
| Capacity: 8 kg | $\beta_{8kg}$ | 0.220 | (0.185) |
| Capacity: 9 kg | $\beta_{9kg}$ | 0.412 | (0.102) |
| Spin speed: 1400 rpm | $\beta_{1400}$ | 0.140 | (0.125) |
| Spin speed: 1600 rpm | $\beta_{1600}$ | 0.392 | (0.126) |
| Energy class A $\times I_{new}$ | $\delta_A$ | 0.956 | (0.131) |
| Energy class B $\times I_{new}$ | $\delta_B$ | 0.186 | (0.064) |
| Energy class C $\times I_{new}$ | $\delta_C$ | 0.172 | (0.049) |
| *Household-specific coefficients* | | | |
| Log price-sensitivity intercept | $a$ | 1.619 | (1.057) |
| Log-income effect on price sensitivity | $b$ | −0.578 | (0.278) |
| Program value | $\beta^{\mathcal{I}}$ | 0.147 | (0.045) |

*Notes:* Panel A pools the old- and new-label surveys; parentheses report Hessian-based standard errors. The outside option is normalized to zero, and temperature and duration slopes allow household heterogeneity. Panel B uses prices and income in hundreds of euros; parentheses report robust standard errors. Price sensitivity is $-\exp(a + b \log y_i)$, with residual heterogeneity fixed at zero. Program value is the household-specific inclusive value from program choice. Class D is omitted; label effects apply from 2021Q1. Demand covers 2020Q1–2023Q4.

inputs, costs approximately € 3.05 million per introduction. This calculation uses the index in its original units rather than the table's $10^{-3}$ display scale. It holds other design coordinates fixed and excludes entry costs and changes in marginal production cost. The magnitude helps explain why firms may prefer to adjust program inputs instead of making large technological improvements.

The model fits Eco duration closely: observed and predicted levels are 190.1 and 190.0 minutes, and the corresponding reductions from the reference are 41.4 and 41.6 minutes. It is less successful at explaining increases in Cotton duration and improvements in technology. For the selected technology-improvement sample, the predicted change in the energy-requirement index is

approximately zero, compared with −0.078 in the data (Tables A4 and **??**; Figure F3).

Table 3: Supply Estimates

| Variable | Parameter | Estimate | Std. Error |
|---|---|---|---|
| *Panel A: Marginal costs* | | | |
| Shared technology | $\gamma_{tech}$ | -0.116 | (0.047) |
| Capacity: 7 kg | $\gamma_{7kg}$ | -0.552 | (0.016) |
| Capacity: 8 kg | $\gamma_{8kg}$ | -0.347 | (0.015) |
| Capacity: 9 kg | $\gamma_{9kg}$ | -0.146 | (0.015) |
| Spin speed: 1400 rpm | $\gamma_{1400}$ | -0.179 | (0.028) |
| Spin speed: 1600 rpm | $\gamma_{1600}$ | -0.009 | (0.029) |
| *Panel B: Design costs* | | | |
| *Linear coefficients* | | | |
| Cotton 40°C duration | $a_{40}$ | 25.19 | (9.23) |
| Cotton 60°C duration | $a_{60}$ | 56.03 | (22.83) |
| Eco duration | $a_d$ | -1.13 | (5.13) |
| Eco temperature | $a_T$ | 40.82 | (13.89) |
| *Quadratic coefficients* | | | |
| Cotton 40°C and Eco duration | $b_{40} = b_d$ | 11.82 | (3.64) |
| Cotton 60°C duration | $b_{60}$ | 46.20 | (56.23) |
| Eco temperature | $b_T$ | 101.02 | (415.97) |
| *Panel C: Entry costs* | | | |
| Launch intercept | $\kappa_L$ | 2.118 | (1.259) |
| Existing products: $\log(1+N)$ | $\beta_N$ | -0.759 | (0.391) |
| Time since introduction: $\log(1+G)$ | $\beta_G$ | 1.344 | (0.855) |
| Private-shock scale | $\sigma$ | 1.617 | (0.853) |
| *Switching premiums: Transition* | | | |
| Class A | $\kappa^{\text{tr}}_{NA}$ | 2.576 | (1.574) |
| Class B | $\kappa^{\text{tr}}_{NB}$ | 3.434 | (2.124) |
| Classes C and D | $\kappa^{\text{tr}}_{NCD}$ | 1.609 | (0.959) |
| *Switching premiums: Post-policy* | | | |
| Class A | $\kappa^{\text{post}}_{NA}$ | 1.892 | (0.768) |
| Class B | $\kappa^{\text{post}}_{NB}$ | 3.494 | (2.048) |
| Classes C and D | $\kappa^{\text{post}}_{NCD}$ | 4.735 | (2.602) |

*Notes:* Panel A uses log marginal cost, 7,496 observations, and firm and quarter fixed effects; omitted categories are 10 kg and 1200 rpm. Panel B uses $a_g(x-r)+b_g(x-r)^2/2$, in euros, with minutes and degrees Celsius as input units. Its S.E.s are local firm-clustered approximations (seven firms) at the manually adjusted curvature vector, not an objective minimizer. Panel C uses millions of euros and 469 line–quarters; quarter effects are unreported. Switching premiums apply to classes absent from the latest introduction. Panels A and C report conditional HC1 S.E.s excluding upstream uncertainty. Entry estimates predate the design-cost update.

### 7.3 Entry Costs

Table 3 reports a common launch-cost intercept of € 2.12 million, before adjustments for season, product stock, time since the previous introduction, and energy-class switching. The estimates imply lower launch costs for established lines and higher costs following longer gaps between introductions. Increasing the preceding product count from one to three reduces launch costs by € 0.53 million, while increasing the gap from one to three quarters raises them by € 0.93 million, holding other characteristics fixed. These patterns are consistent with economies from maintaining an active product line and costs of restarting introductions after a period of inactivity.

Switching premiums apply when a firm introduces a class absent from the line's most recent introduction quarter. During the transition, the estimated premiums are € 2.58 million for class $A$, € 3.43 million for $B$, and € 1.61 million for $C/D$. After the transition, they are € 1.89 million, € 3.49 million, and € 4.73 million, respectively. These premiums favor persistence in recently introduced classes, particularly $C/D$ after the transition. They are additional to the launch cost and do not apply when the chosen class is already in the reference set.

The size of these estimates reflects the definition of an introduction and the static nature of the model. A modeled product can cover several commercial variants across markets, so its launch may entail tooling, certification, marketing, and distribution expenses beyond program design. The data do not identify these expenditures separately. Moreover, the estimated costs may absorb omitted continuation values, the option to delay introduction, and errors in predicted portfolio profits. They should therefore be interpreted as effective costs that rationalize observed entry decisions, not direct measures of engineering expenditures.

The estimated payoff-shock scale is € 1.62 million. The model predicts introductions in 33.8 percent of eligible line–quarters, compared with 36.0 percent in the data. It underpredicts class-$A$ entry, slightly overpredicts $B$ and $C$, and closely matches the class-$D$ share (Figure F4; Table **??**). These fit measures condition on the observed decisions on a firm's other lines; the counterfactuals instead allow its portfolio history to evolve.

## 8 Counterfactuals

I use the estimated model to conduct three sets of counterfactual analyses to assess whether alternative policies can limit energy leakage from regulated Eco programs to unregulated Cotton programs. First, I evaluate the Eco duration cap and energy-label rescaling by removing them separately and jointly, isolating their contributions to this leakage. Second, I extend the current Eco-only energy efficiency index to include Cotton programs and vary their weights, bringing non-Eco energy consumption within the scope of regulation. Third, I introduce a carbon tax that adds a charge for

each program's per-cycle electricity consumption to the marginal cost of each machine, extending incentives to reduce energy use beyond Eco.

To evaluate these policies, I recompute equilibrium product entry, program design, prices, and household purchase and program choices, and compare the resulting electricity consumption, consumer welfare, and firms' profits across scenarios. In the no-policy scenario, the market remains at the stationary 2020Q1 $A^{+++}$ portfolio. Policy scenarios use their respective equilibrium portfolios.

### 8.1 Welfare Accounting

I evaluate policies using purchase-stage consumer surplus, lifetime use-stage welfare, operating profits, and electricity consumption. Consumer welfare excludes utility attributable solely to energy labels and accounts for electricity expenditure during use. Purchase and program-choice probabilities are generated by the estimated model and remain unchanged when applying the welfare corrections. For the carbon-tax policy, tax revenue is rebated to households through lump-sum transfers.

**Purchase-stage consumer surplus.** To separate purchase and use-stage welfare, I remove both the energy-label component and the program inclusive value from purchase utility for welfare accounting:

$$\bar{u}^P_{ijmt} = \bar{u}_{ijmt} - \sum_{c\in\mathcal{C}} \beta_c \mathbf{1}\{\text{ec21}(j) = c\}\mathbf{1}_{\text{new},t} - \beta^{\mathcal{I}}\mathcal{I}_{ij}. \tag{26}$$

Labels and the expected value of program use continue to affect purchase decisions, but label utility provides no intrinsic welfare benefit, and program benefits are accounted for separately at the use stage.

Purchase-stage consumer surplus evaluates these corrections at the predicted purchase probabilities:

$$\mathcal{CS}^P_{mt} = M_{mt}\int \frac{1}{\alpha_i}\left[\log\left(1+\sum_{j\in\mathcal{J}_{mt}}\exp(\bar{u}_{ijmt})\right) + \sum_{j\in\mathcal{J}_{mt}} s_{ijmt}\left(\bar{u}^P_{ijmt} - \bar{u}_{ijmt}\right)\right] dF_{mt}(i) + \mathcal{R}_{mt}, \tag{27}$$

where $\mathcal{R}_{mt}$ is total carbon-tax revenue rebated to households in market $(m,t)$ and is zero in the other policy experiments.

**Use-stage welfare.** I incorporate electricity expenditure into program utility for welfare accounting:

$$v^W_{ijrs} = v_{ijrs} - \beta^e_i p^e e_{jr}, \tag{28}$$

where $v_{ijrs}$ is program utility defined in equation (6), including its program-choice shock, $p^e$ is the electricity price in euros per kWh, and $e_{jr}$ is electricity consumption per cycle. I assume $\beta_i^e = \alpha_i$: one euro has the same utility value at the purchase and program-use stages. This assumption provides a common monetary scale across stages; it is not identified by the estimated program-choice model.

I evaluate corrected utility at the programs households choose under the estimated model. Expected program utility for welfare accounting is therefore

$$\mathcal{I}_{ij}^W = \mathcal{I}_{ij} - \alpha_i p^e \sum_{r \in \mathcal{R}} P_{ijr} e_{jr}. \tag{29}$$

I use the same expected-utility normalization across scenarios, omitting the common extreme-value constant. The program outside option has zero systematic utility and is assigned zero electricity consumption.

Assuming a 12-year product life, let

$$D_i = N_i \sum_{y=1}^{12} (1+\delta)^{-y}$$

denote the discounted number of lifetime wash cycles, where $N_i$ is annual wash frequency and $\delta$ is the annual discount rate.

Let $s_{i0mt} = 1 - \sum_{j \in \mathcal{J}_{mt}} s_{ijmt}$ denote the probability of not purchasing. Nonpurchasers retain a machine drawn from a fixed reference distribution over the 2021Q1 portfolio. Let $\mathcal{I}_{i0}$ and $e_{i0}$ denote its expected program inclusive value and electricity consumption per cycle, evaluated using household $i$'s preferences. Appendix G.1 describes their construction. I assign retained-machine services entirely to use-stage welfare and maintain the zero normalization for the residual purchase-stage outside option.

Holding preferences, program attributes, electricity prices, and annual wash frequency constant over the product's life, use-stage welfare is

$$\mathcal{CS}_{mt}^U = M_{mt} \int D_i \left[ \sum_{j \in \mathcal{J}_{mt}} s_{ijmt} \left( \frac{\mathcal{I}_{ij}}{\alpha_i} - p^e \sum_{r \in \mathcal{R}} P_{ijr} e_{jr} \right) + s_{i0mt} \left( \frac{\mathcal{I}_{i0}}{\alpha_i} - p^e e_{i0} \right) \right] dF_{mt}(i). \tag{30}$$

This expression values lifetime program benefits and deducts discounted electricity expenditure for purchasers and nonpurchasers, without changing their choice probabilities. Program benefits are counted only once because their purchase-utility component is removed from purchase-stage surplus. Both groups are evaluated over the same 12-year horizon; nonpurchasers are assumed to receive the reference machine's service throughout this period, through continued ownership or equivalent replacement. The retained-machine reference is distinct from the stationary 2020Q1 $A^{+++}$ no-policy scenario.

**Producer surplus.** I measure producer surplus using expected operating profits:

$$\mathcal{PS}_t = \sum_f \mathbb{E}_{\mathcal{P}_t}\left[\Pi^{\text{op}}_{ft}(a_t)\right], \tag{31}$$

where $\mathcal{P}_t$ is the equilibrium distribution of product portfolios and $\Pi^{\text{op}}_{ft}(a_t)$ is firm $f$'s revenue less variable production costs at the resulting designs, prices, and sales. Under the carbon-tax policy, variable costs include tax payments.

**Aggregate welfare benchmark.** I summarize consumer and producer outcomes using

$$\mathcal{W}_t = \sum_m \left(\mathcal{CS}^P_{mt} + \mathcal{CS}^U_{mt}\right) + \mathcal{PS}_t. \tag{32}$$

Consumer surplus, operating profits, and electricity consumption are evaluated at the same equilibrium allocation. Labels affect purchases and firms' profits but provide no intrinsic benefit in consumer welfare. Under the carbon-tax policy, household rebates equal tax revenue collected on equilibrium sales, offsetting the corresponding payments included in firms' costs. Electricity expenditure is incorporated in use-stage welfare and is not deducted separately. This benchmark excludes monetized environmental damages.

**Electricity consumption.** For each scenario, I compute expected electricity consumption using its product portfolio, purchase probabilities, and predicted program choices:

$$\mathcal{EC}_t = \sum_m M_{mt} \int N_i \left[\sum_{j\in\mathcal{J}_{mt}} s_{ijmt} \sum_{r\in\mathcal{R}} P_{ijr} e_{jr} + s_{i0mt} e_{i0}\right] dF_{mt}(i). \tag{33}$$

Here, $N_i$ is household $i$'s annual number of wash cycles. The expression measures annual electricity use by households represented in the purchase market, including those retaining existing machines. It does not measure consumption by the entire installed stock. The no-policy scenario uses the stationary 2020Q1 $A^{+++}$ purchase portfolio, while policy scenarios use their respective equilibrium portfolios; all scenarios use the same retained-machine reference for nonpurchasers. I compare consumption directly across scenarios without subtracting reference consumption. Electricity consumption is reported separately as an environmental outcome and does not enter $\mathcal{W}_t$ as an additional term.

### 8.2 Policy Effects on Product Design and Welfare

**Removing the duration cap or labeling requirements.** I consider two departures from current policy: removing the Eco duration cap while retaining the efficiency index and label incentives,

and removing efficiency-index restrictions and label utility while retaining the cap. Figure **??** reports percentage changes in unweighted mean inputs over 2020Q1–2023Q4. All comparisons use model-predicted outcomes and reflect both changes in product design and changes in entry composition.

Without the duration cap, mean Eco duration increases from 208.9 to 260.8 minutes, or 24.9%, while temperature falls from 31.7 to 29.9°C, or 5.7%. This combination is consistent with firms substituting time for temperature when Eco design is less constrained. Cotton 40°C duration falls from 186.4 to 180.8 minutes, a 3.0% decline, and Cotton 60°C duration falls from 168.7 to 164.7 minutes, a 2.4% decline. These patterns are consistent with reduced incentives to shift performance-related inputs toward Cotton programs. Longer Eco cycles do not themselves reduce electricity consumption in the estimated energy equation; any savings depend on accompanying changes in temperature, other inputs, and program use.

Removing efficiency-index restrictions and label utility produces a different adjustment. Eco duration changes little, increasing from 208.9 to 210.8 minutes, or 0.9%, while temperature rises from 31.7 to 36.2°C, or 14.2%. The temperature increase is consistent with weaker incentives to conserve energy when measured efficiency no longer determines regulatory compliance or earns a label premium. Cotton 40°C duration falls to 177.2 minutes, a 4.9% decline, whereas Cotton 60°C duration rises to 171.3 minutes, a 1.5% increase. Thus, removing labeling requirements does not produce a uniform response across Cotton programs.

Capacity changes little in either experiment. Mean capacity falls from 8.079 to 8.064 kg without the duration cap and rises to 8.124 kg without labels and efficiency-index restrictions. Because capacity is fixed within each product line, these changes arise entirely from the composition of introductions.

Figure 8 reports welfare changes relative to the model-predicted current-policy equilibrium, using the full-capitalization benchmark. Unlike the input comparisons, these comparisons use model predictions on both sides. Electricity expenditure and use-stage welfare include nonbuyers retaining their existing machines.

Removing the duration cap has little effect on purchase-stage surplus, which falls by € 0.3 million, or 0.06%. Lifetime wash benefits rise by € 1.3 million, or 0.17%, while discounted electricity expenditure falls by € 7.4 million, or 0.14%. Total consumer welfare therefore increases by € 8.4 million. Operating profits rise by € 52.7 million, or 6.5%, bringing the increase in the aggregate welfare benchmark to € 61.2 million. The small electricity response, despite substantial changes in selected entrants' designs, reflects the broader population covered by the accounting: incumbent machines, repeated-class introductions, and machines retained by nonbuyers also contribute to consumption.

Removing efficiency-index restrictions and label utility raises purchase-stage surplus by € 114.5 million, or 22.2%, but reduces lifetime wash benefits by € 5.8 million, or 0.78%. Discounted electricity expenditure falls by € 48.0 million, or 0.93%, yielding a consumer-welfare gain of € 156.6 million. Labels continue to affect purchase decisions in the model but provide no intrinsic welfare benefit, so eliminating their influence can improve purchase-stage welfare even as fewer households buy new machines. Purchases fall from 4.02 million to 3.46 million, a 14.0% decline. Much of the reduction in electricity use by purchased machines is offset by continued use of retained machines. Operating profits fall by € 106.1 million, or 13.1%, leaving an aggregate welfare gain of € 50.5 million. These aggregate comparisons combine consumer welfare and operating profits; they exclude design costs, entry costs, and monetized environmental damages.

**Broadening the EEI test.** I broaden the efficiency index by replacing its Eco-only numerator with $w_E e_{jE} + w_{40} e_{j40} + w_{60} e_{j60}$, retaining the capacity adjustment and existing class thresholds. The evaluated specifications assign 52–100% of the weight to Eco and up to 24% to each Cotton program. All remove the duration cap and are compared with an uncapped Eco-only benchmark. These weights determine regulatory performance rather than household program-use shares. Because class thresholds remain unchanged, the experiments alter both the coverage and the stringency of the efficiency requirement.

Under the full-capitalization benchmark, weights $(w_E, w_{40}, w_{60}) = (0.52, 0.24, 0.24)$ yield the largest consumer-welfare gain and the lowest electricity consumption among the 139 evaluated specifications. Relative to the uncapped Eco-only index, purchase-stage surplus increases by € 0.14 million, or 0.03%, while lifetime wash benefits decline by € 0.50 million, or 0.07%. Discounted electricity expenditure falls by € 25.95 million, or 0.51%, producing a consumer-welfare gain of € 25.59 million. Operating profits increase by € 31.61 million, or 3.78%, bringing the aggregate welfare gain to € 57.20 million.

These results suggest that extending the test to Cotton can reduce electricity expenditure with little change in the value households obtain from program use. Purchases decline by only 0.05%, so the electricity reduction is not primarily driven by fewer purchases. The preferred specification places the maximum tested weight on both Cotton programs; it therefore identifies the best outcome within the evaluated grid rather than a global optimum.

**An energy-based producer tax.** The producer-tax experiment removes the efficiency index, label utility, and duration cap and adds the following charge to each machine's marginal cost:

$$\mathcal{T}_j = \frac{\tau_E e_{jE} + \tau_C (e_{j40} + e_{j60})}{3}. \tag{34}$$

I evaluate 50 pairs of coefficients in $[0, 100]^2$, including zero and taxes on either program group alone. The coefficients translate assessed electricity use per cycle into euros per machine; they are not carbon prices per tonne. Household electricity prices remain unchanged. Tax revenue is rebated equally to households within each market, including nonbuyers. Each specification is compared with the zero-tax benchmark under the same regulatory regime.

The largest tested tax, $(\tau_E, \tau_C) = (100, 100)$, produces the lowest electricity consumption in the tax grid. Discounted electricity expenditure falls by € 36.96 million, or 0.72%, while lifetime wash benefits decline by € 4.09 million, or 0.55%. Purchase-stage surplus, including rebates, increases by € 20.80 million, or 3.31%, yielding a consumer-welfare gain of € 53.66 million. This purchase-stage gain includes € 254.81 million in rebates; before rebates, purchase-stage surplus falls by € 234.01 million. Operating profits decline by € 123.76 million, or 24.29%, so the aggregate welfare benchmark falls by € 70.10 million.

Purchases fall from 3.45 million to 2.49 million, a 27.81% decline, but nonbuyers continue using their retained machines. Consequently, the reduction in total electricity use is much smaller than the decline in purchases. A Cotton-only tax, $(0, 100)$, delivers nearly the same electricity reduction, 0.72%, and the largest consumer-welfare gain in the tax grid, € 56.57 million. Nevertheless, its operating-profit loss exceeds the consumer gain, reducing the aggregate welfare benchmark by € 32.15 million. Zero tax yields the highest aggregate welfare among the evaluated tax specifications. This ranking excludes monetized environmental damages.

## 9 Conclusion

This paper studies how energy labels and ecodesign constraints affect product design when regulatory tests capture only one of several operating modes. I use the 2021 EU washing-machine reform in Germany, which rescaled energy labels and imposed an Eco-duration cap. The evidence shows that energy consumption declined in the regulated Eco program but rose in hotter Cotton programs. Because Eco accounts for only a small share of household wash cycles, improvements in the tested mode need not translate into comparable savings in actual use.

The estimated model links these responses to firms' design incentives and households' program choices. Counterfactuals show that relaxing the Eco-duration cap can lower electricity use and improve consumer welfare while retaining labeling incentives. Including Cotton energy use in the efficiency index also improves outcomes under the tested specifications, as does an energy-based producer tax. Taken together, the results suggest that effective regulation should account for both firms' flexibility in meeting standards and consumers' use of programs outside the test environment.

## References


Aguirregabiria, V. and Mira, P. (2007). Sequential estimation of dynamic discrete games. *Econometrica*, 75(1):1–53.

Alborzi, F., Boyano, A., Cordella, M., Espinosa, N., Graulich, K., Hook, I., Rüdenauer, I., Stamminger, R., and Villanueva, A. (2017). Ecodesign and energy label for household washing machines and washer dryers: Preparatory study – final report. Technical Report EUR 28809 EN, Publications Office of the European Union, Luxembourg. JRC108604.

Baker, G. P. (1992). Incentive contracts and performance measurement. *Journal of political Economy*, 100(3):598–614.

Barahona, N., Otero, C., and Otero, S. (2023). Equilibrium effects of food labeling policies. *Econometrica*, 91(3):839–868.

Barrett, S. R. H., Speth, R. L., Eastham, S. D., Dedoussi, I. C., Ashok, A., Malina, R., and Keith, D. W. (2015). Impact of the volkswagen emissions control defeat device on us public health. *Environmental Research Letters*, 10(11):114005.

Barwick, P. J., Kwon, H.-S., and Li, S. (2024). Attribute-based subsidies and market power: an application to electric vehicles. Technical report, National Bureau of Economic Research.

Berry, S., Levinsohn, J., and Pakes, A. (1995). Automobile prices in market equilibrium. *Econometrica*, 63(4):841–890.

Berry, S. T. and Haile, P. A. (2014). Identification in differentiated products markets using market level data. *Econometrica*, 82(5):1749–1797.

Bodéré, P. (2026). Dynamic spatial competition in early education: An equilibrium analysis of the preschool market in pennsylvania. Working paper, July 22.

Casella, G. and George, E. I. (1992). Explaining the gibbs sampler. *The American Statistician*, 46(3):167–174.

Chen, Y. and Xu, D. Y. (2022). A structural empirical model of R&D investment, firm heterogeneity, and industry evolution. Working Paper 29733, National Bureau of Economic Research.

Conlon, C. and Gortmaker, J. (2020). Best practices for differentiated products demand estimation with pyblp. *The RAND Journal of Economics*, 51(4):1108–1161.

Czyzewski, G. (2022). Energy savings 2021: New european energy label and ecodesign requirements for washing machines. In Bertoldi, P., editor, *Energy Efficiency in Domestic Appliances and*

*Lighting: Proceedings of the 10th International Conference (EEDAL'19)*, pages 1–11. Springer, Cham.

Davis, L. W. and Metcalf, G. E. (2016). Does better information lead to better choices? evidence from energy-efficiency labels. *Journal of the Association of Environmental and Resource Economists*, 3(3):589–625.

Dubin, J. A. and McFadden, D. L. (1984). An econometric analysis of residential electric appliance holdings and consumption. *Econometrica*, 52(2):345–362.

Eichler, H.-G., Abadie, E., Breckenridge, A., Flamion, B., Gustafsson, L. L., Leufkens, H., Rowland, M., Schneider, C. K., and Bloechl-Daum, B. (2011). Bridging the efficacy–effectiveness gap: a regulator's perspective on addressing variability of drug response. *Nature Reviews Drug Discovery*, 10(7):495–506.

Eizenberg, A. (2014). Upstream innovation and product variety in the U.S. home PC market. *The Review of Economic Studies*, 81(3):1003–1045.

European Commission (2019a). Commission regulation (eu) 2019/2023 of 1 october 2019 laying down ecodesign requirements for household washing machines and household washer-dryers pursuant to directive 2009/125/ec of the european parliament and of the council, amending commission regulation (ec) no 1275/2008 and repealing commission regulation (eu) no 1015/2010 (text with eea relevance). Official Journal of the European Union, OJ L 315, 5.12.2019, p. 285. EUR-Lex.

European Commission (2019b). Impact assessment accompanying the ecodesign and energy labelling regulations for household washing machines and household washer-dryers. Commission Staff Working Document SWD(2019) 349 final, European Commission.

European Commission (2021). European energy labels: rescaling and transition periods. https://commission.europa.eu/system/files/2021-04/rescaled_eu_energy_labels_and_transition_period.pdf. Guidance document on rescaled EU energy labels.

Fan, Y. (2013). Ownership consolidation and product characteristics: A study of the us daily newspaper market. *American Economic Review*, 103(5):1598–1628.

Fan, Y. and Yang, C. (2025). Estimating discrete games with many firms and many decisions: An application to merger and product variety. *Journal of Political Economy*, 133(6):1886–1931.

Goettler, R. L. and Gordon, B. R. (2011). Does amd spur intel to innovate more? *Journal of Political Economy*, 119(6):1141–1200.

Gupta, A. (2021). Impacts of performance pay for hospitals: The readmissions reduction program. *American Economic Review*, 111(4):1241–1283.

Hall, G. and Rust, J. (2003). Simulated minimum distance estimation of a model of optimal commodity price speculation with endogenously sampled prices. *manuscript, Yale University*.

Holmes, T. J. (2011). The diffusion of Wal-Mart and economies of density. *Econometrica*, 79(1):253–302.

Holmstrom, B. and Milgrom, P. (1991). Multitask principal–agent analyses: Incentive contracts, asset ownership, and job design. *The Journal of Law, Economics, and Organization*, 7(special_issue):24–52.

Houde, J.-F., Newberry, P., and Seim, K. (2023). Nexus tax laws and economies of density in E-Commerce: A study of Amazon's fulfillment center network. *Econometrica*, 91(1):147–190.

Houde, S. (2018). How consumers respond to product certification and the value of energy information. *The RAND Journal of Economics*, 49(2):453–477.

Houde, S. (2026). The incidence of coarse certification: Evidence from the energy star program. *The RAND Journal of Economics*, 57(3):545–585.

Ito, K. and Sallee, J. M. (2018). The economics of attribute-based regulation: Theory and evidence from fuel economy standards. *Review of Economics and Statistics*, 100(2):319–336.

Ito, K., Sallee, J. M., and Smith, J. A. (2026). Global policy spillovers: How environmental policies propagate through product attributes. Technical report, National Bureau of Economic Research.

Jacob, B. A. (2005). Accountability, incentives and behavior: The impact of high-stakes testing in the chicago public schools. *Journal of Public Economics*, 89(5–6):761–796.

Jia, P. (2008). What happens when wal-mart comes to town: An empirical analysis of the discount retailing industry. *Econometrica*, 76(6):1263–1316.

Josephy, B., Bush, E., Attali, S., and Zuloaga, F. R. (2015). Cold wash: The cool and modern way to launder. In *Proceedings of the 2015 ECEEE Summer Study on Energy Efficiency: First Fuel Now*, pages 1527–1535. European Council for an Energy Efficient Economy.

Knittel, C. R. (2011). Automobiles on steroids: Product attribute trade-offs and technological progress in the automobile sector. *American Economic Review*, 101(7):3368–3399.

Kraft, D. (1988). A software package for sequential quadratic programming. Technical Report FB 88-28, DFVLR.

Lu, S. F. (2012). Multitasking, information disclosure, and product quality: Evidence from nursing homes. *Journal of Economics & Management Strategy*, 21(3):673–705.

Montag, F. (2023). Mergers, foreign competition, and jobs: Evidence from the us appliance industry. *George J. Stigler Center for the Study of the Economy & the State Working Paper*, (326).

Reimers, I. and Waldfogel, J. (2021). Digitization and pre-purchase information: The causal and welfare impacts of reviews and crowd ratings. *American Economic Review*, 111(6):1944–1971.

Reynaert, M. (2021). Abatement strategies and the cost of environmental regulation: Emission standards on the european car market. *The Review of Economic Studies*, 88(1):454–488.

Reynaert, M. and Sallee, J. M. (2021). Who benefits when firms game corrective policies? *American Economic Journal: Economic Policy*, 13(1):372–412.

Rothwell, P. M. (2005). External validity of randomised controlled trials: "to whom do the results of this trial apply?". *The Lancet*, 365(9453):82–93.

Seim, K. (2006). An empirical model of firm entry with endogenous product-type choices. *The RAND Journal of Economics*, 37(3):619–640.

Sinner, H. (1960). *Über das Waschen mit Haushaltswaschmaschinen*. Haus und Heim Verlag, Hamburg, Germany.

Su, R., Yang, C., and Sweeting, A. (2026). Competition, procurement and learning-by-doing in the space launch industry. Working Paper 34766, National Bureau of Economic Research.

Van Holsteijn en Kemna B.V., Vlaamse Instelling voor Technologisch Onderzoek NV, Viegand Maagøe A/S, and Wuppertal Institut für Klima, Umwelt, Energie GmbH (2014). "omnibus" review study on cold appliances, washing machines, dishwashers, washer-driers, lighting, set-top boxes and pumps: Final report. Technical report, Prepared for the European Commission, DG ENER-C3, Brussels/Delft. 12 March 2014; specific contract no. ENER/C3/2012-418 LOT2/03/SI2.654805.

Varin, C., Reid, N., and Firth, D. (2011). An overview of composite likelihood methods. *Statistica Sinica*, 21:5–42.

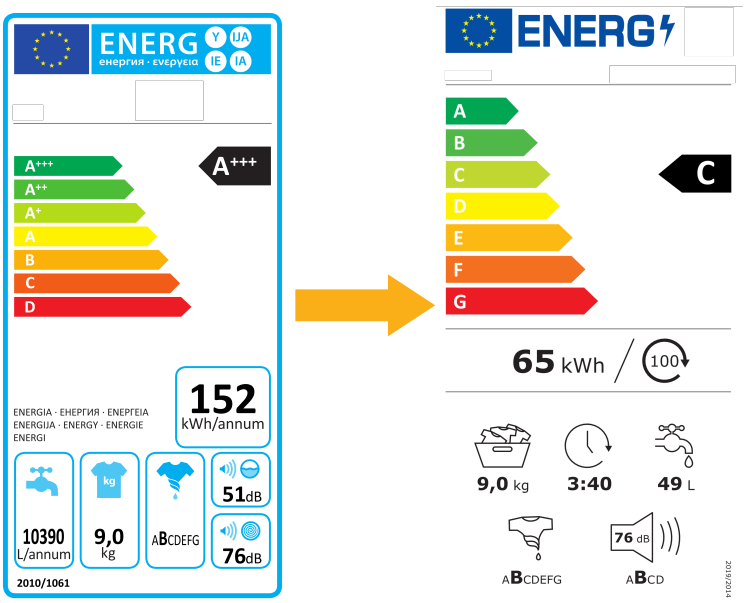
Vatter, B. (2022). Quality disclosure and regulation: Scoring design in medicare advantage. *Available at SSRN 4250361*.

Wollmann, T. G. (2018). Trucks without bailouts: Equilibrium product characteristics for commercial vehicles. *American Economic Review*, 108(6):1364–1406.

Zaheer, M., Kottur, S., Ravanbakhsh, S., Póczos, B., Salakhutdinov, R., and Smola, A. (2017). Deep sets. In *Advances in Neural Information Processing Systems*, volume 30.

## Figures

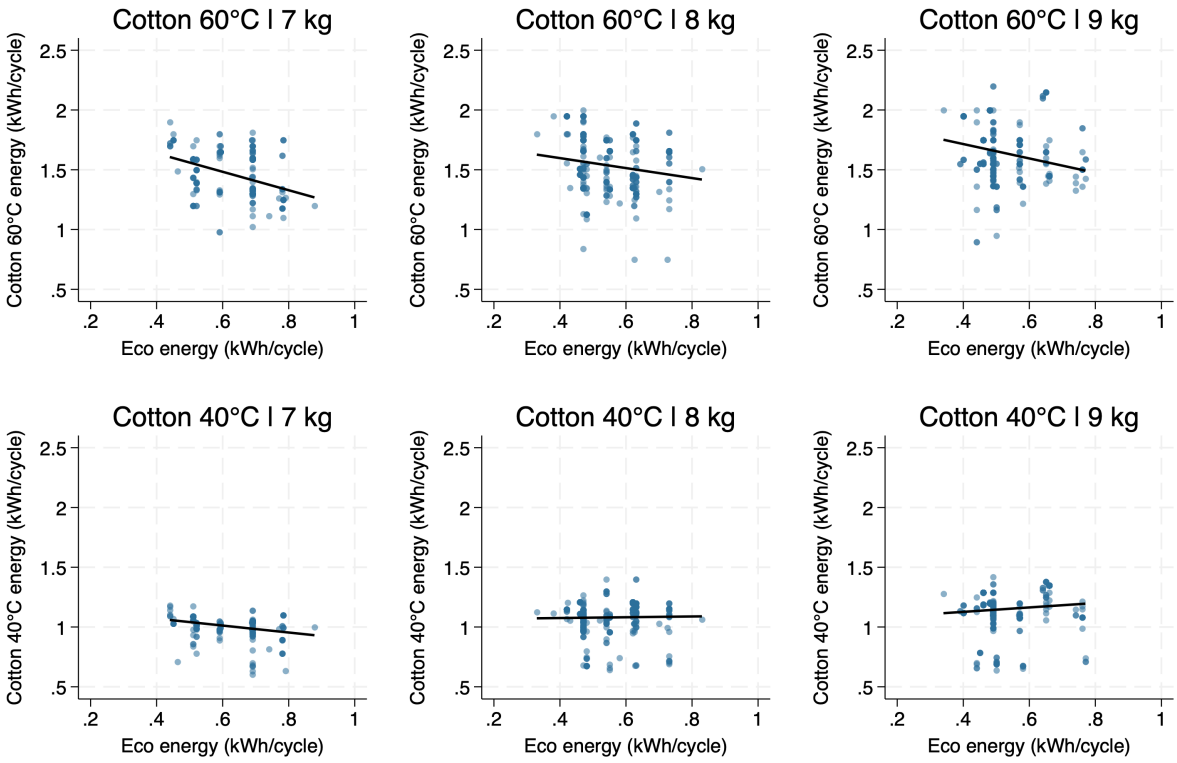
Figure 3: Energy Labels of Washing Machines in the EU




(a) 2011 Energy Label (b) 2021 Energy Label

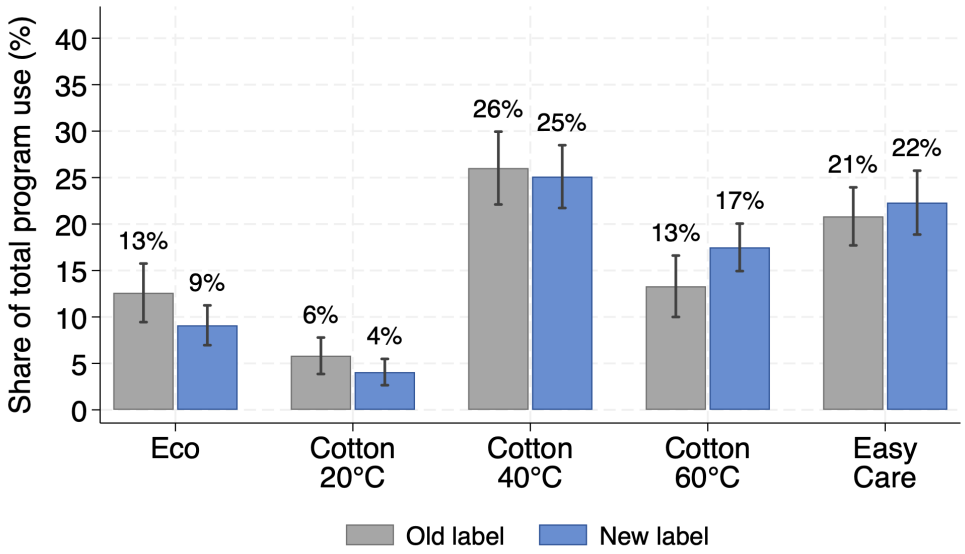
*Notes*: The figure compares two versions of the EU energy label for washing machines. Both labels are shown for the same washing machine model and report the attributes of its energy-saving wash program. To preserve anonymity, brand and model identifiers are removed. The labels report energy-efficiency class and selected program attributes, including energy use, capacity, water consumption, program duration, and noise.

Figure 4: Eco and Cotton Energy Consumption by Capacity



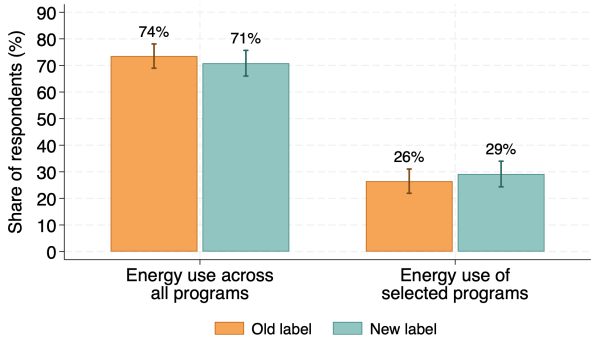

*Notes:* Each point represents a washing-machine model carrying the new energy label. The horizontal axis reports certified Eco energy consumption, and the vertical axis reports Cotton energy consumption, both in kWh per cycle. The top and bottom rows show Cotton 60°C and Cotton 40°C, respectively; columns distinguish capacities of 7, 8, and 9 kg. Solid lines show separate unadjusted linear fits within each panel. The sample includes models with nonmissing energy consumption for both programs, including imputed Cotton values.

Figure 5: Share of program use

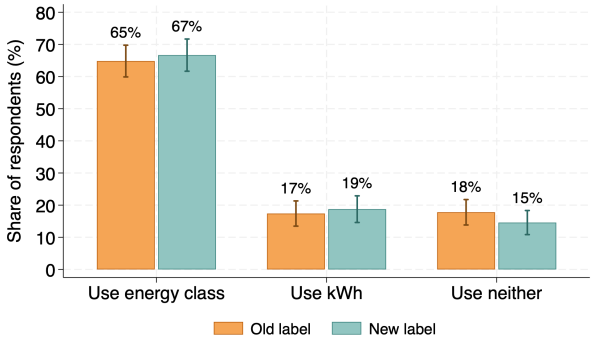



*Notes:* The figure reports the distribution of wash-program choices in the household survey, separately for owners of old- and new-label washing machines. Shares are calculated as the fraction of total reported wash cycles assigned to each program. Eco accounts for 13 percent of cycles among old-label households and 9 percent among new-label households. Cotton 40°C, Cotton 60°C, and Easy Care are the most frequently used programs. Remaining cycles belong to other programs not shown in the figure.

Figure 6: Consumer Beliefs About Energy Ratings and Energy Costs

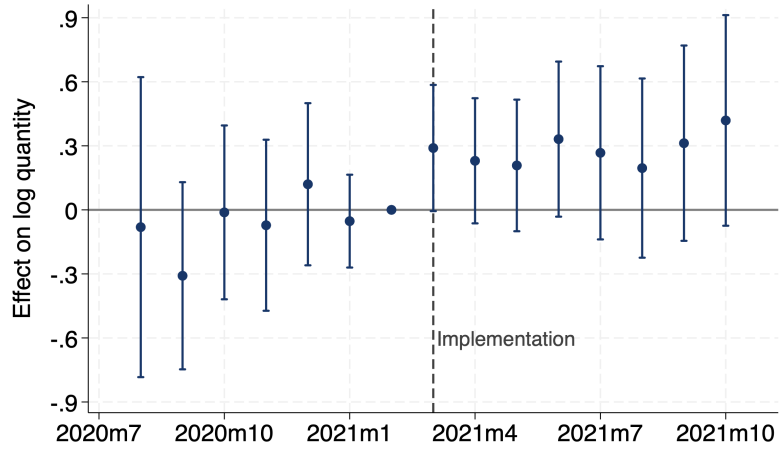



(a) Interpretation of Energy Ratings

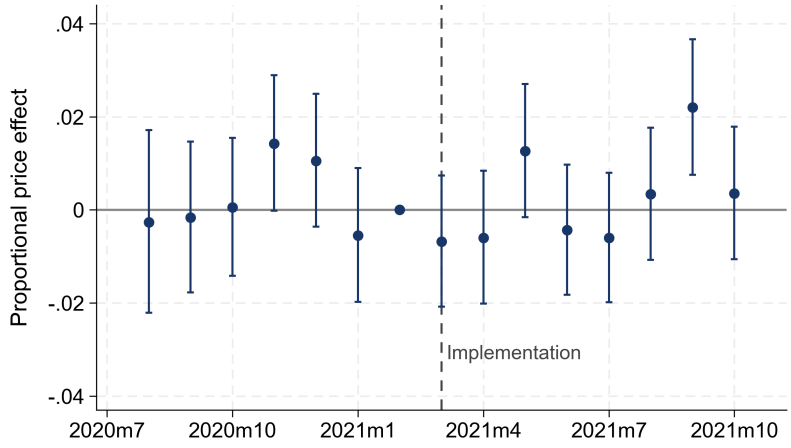



(b) Information Used to Assess Energy Costs

*Notes:* The figure reports household survey responses separately for owners of old- and new-label washing machines. Panel (a) shows whether respondents believe the energy rating reflects energy consumption across all programs or only selected programs. Panel (b) reports whether respondents use the energy class, reported kWh consumption, or neither to assess energy costs. Most respondents in both groups interpret the energy rating as measuring whole-machine efficiency, while approximately two-thirds rely on the energy class rather than reported kWh consumption.




(a) Demand effect




(b) Price effect

Figure 7: Demand and Price Effects

*Notes:* Event-study estimates around the March 2021 energy-label rescaling. Panel (a) reports coefficients for log product sales, controlling for log price. Panel (b) reports proportional price effects, $\exp(\widehat{\beta}_k) - 1$, from a log-price regression; confidence limits are transformed in the same way. Both specifications include product–market and calendar-month fixed effects and omit February 2021. Points are estimates and bars are 95 percent confidence intervals. The demand regression clusters standard errors by product; the price regression uses conventional standard errors. Each dashed line marks implementation. The two panels retain their respective estimation samples.

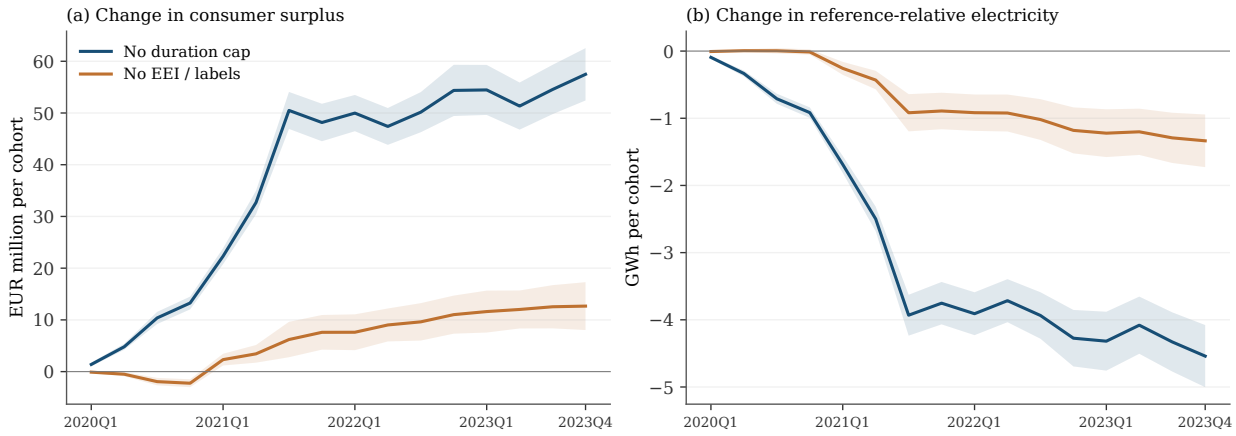



Figure 8: Quarterly effects relative to current policy

Means of 20 matched path differences. Shaded bands are plus or minus 1.96 Monte Carlo standard errors and exclude parameter and approximation uncertainty. Each point is a decision-cohort outcome: welfare includes discounted lifetime electricity costs, whereas electricity is quarterly use for that cohort. The frozen benchmark is omitted because it does not follow the same evolving market.

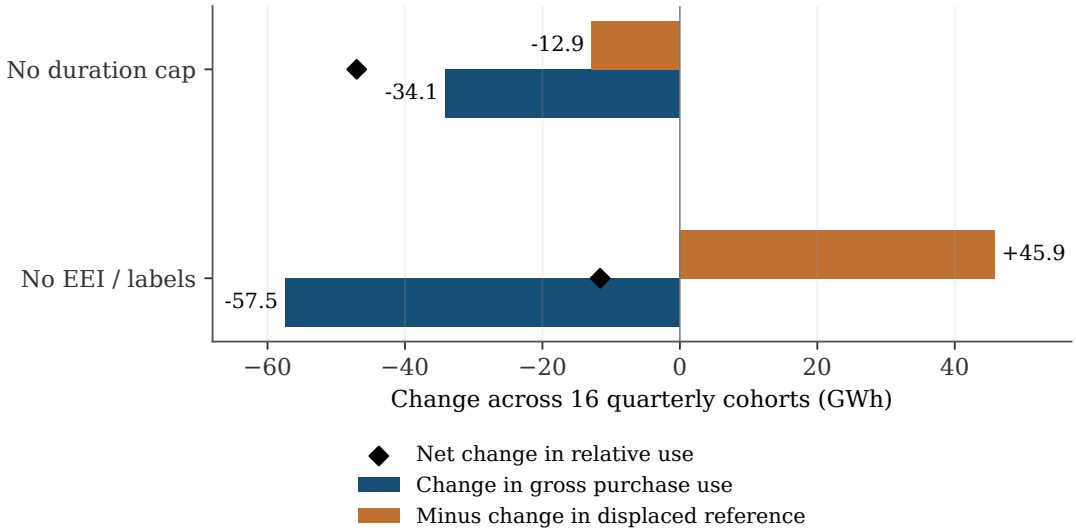



Figure 9: Accounting for the change in electricity

Bars show $\Delta G$ and $-\Delta D$; their sum is the net change $\Delta R$, shown by the diamond. Negative values reduce reference-relative electricity. Components are means of horizon totals across matched paths. This is an accounting identity, not a causal allocation to design, program switching, or purchase decisions.

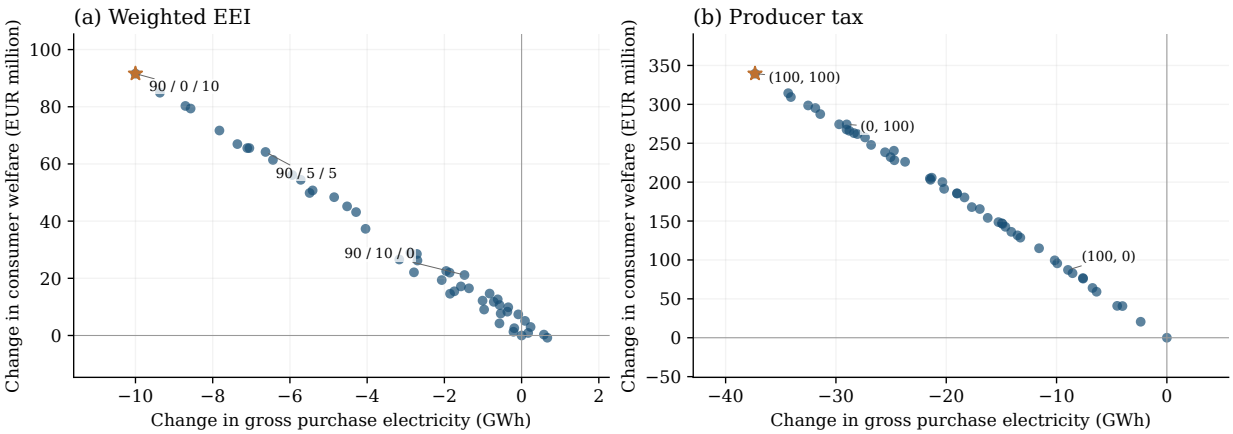



Figure 10: Consumer welfare and gross electricity across tested policies

Each panel shows 50 path-mean changes relative to its own baseline, using gross-use accounting. Stars identify the sole nondominated tested point in each family; upper-left outcomes have higher consumer welfare and lower gross purchase electricity. Labels give EEI percentages or tax-coefficient pairs. The panels do not share a policy baseline or a common Cotton engineering calibration.

## Tables

Table 4: Duration-Cap Exposure and Cotton and Easy Care Program Design

| | Cotton 60°C | | | Cotton 40°C | | | Easy Care | | |
|---|---|---|---|---|---|---|---|---|---|
| | Energy | Duration | Water | Energy | Duration | Water | Energy | Duration | Water |
| Exposure | 0.570*** | 0.211*** | -0.061 | 0.212*** | 0.096* | -0.051 | 0.136* | -0.072 | -0.153*** |
| | (0.055) | (0.058) | (0.045) | (0.061) | (0.050) | (0.055) | (0.081) | (0.058) | (0.053) |
| Capacity FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Brand FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Observations | 223 | 223 | 223 | 223 | 223 | 223 | 223 | 223 | 223 |
| R-squared | 0.554 | 0.307 | 0.460 | 0.373 | 0.487 | 0.408 | 0.111 | 0.574 | 0.419 |

*Notes:* The dependent variables are log differences between each dual-label product's energy, duration, or water consumption and the corresponding mean among old-label-only donors (EC11 observed, EC21 missing). Donors match the same brand and exact Standard 60 full-load duration and loading. Current completed inputs, including imputed values, are used. Exposure is max(0, Standard full-load duration minus the Eco full-load cap), divided by that cap. All regressions include capacity and brand fixed effects, with conventional standard errors in parentheses. Each column uses its own available sample. * p<0.10, ** p<0.05, *** p<0.01.

Table 5: Counterfactual policy definitions

| Policy | Regulatory treatment | Comparator and reported accounting |
|---|---|---|
| Current policy | Eco-only EEI, labels, duration cap | Common removal baseline; $R$, $CS$ |
| No duration cap | Eco-only EEI and labels | Current policy; $R$, $CS$ |
| No EEI / labels | Duration cap retained | Current policy; $R$, $CS$ |
| Frozen joint removal | No EEI, labels, or cap; initial state repeated | Descriptive comparison; $R$, $CS$ |
| Weighted EEI | 50 Eco/Cotton weight vectors; labels; no cap | Uncapped Eco-only; $G$, $CW^G$ |
| Producer tax | 50 Eco/Cotton coefficient pairs; no EEI, labels, or cap | Carbon-model zero tax; $G$, $CW^G$ |

*Notes:* Each evolving policy has 20 paths; the frozen benchmark is solved once. Both grids include their baseline among the 50 policies. "No EEI / labels" removes constraints and explicit label utility; it does not restore the old label scale. The grid and removal accounting conventions differ as described in Section **??**.

Table 6: Entry and product-design responses

| | Current policy | No duration cap | No EEI / labels |
|---|---|---|---|
| *Entry and availability* | | | |
| Product introductions, 2020–2023 | 89.45 | 87.10 | 66.10 |
| Active products, 2023Q4 | 89.45 | 87.10 | 66.10 |
| Introductions per quarter | 5.59 | 5.44 | 4.13 |
| *Final-quarter program design: minutes or degrees Celsius* | | | |
| Eco duration: full load | 211.7 | 381.2 | 212.2 |
| Eco duration: half load | 162.5 | 346.1 | 162.9 |
| Eco duration: quarter load (fixed) | 192.6 | 190.9 | 197.8 |
| Cotton 40°C duration | 183.4 | 171.4 | 174.6 |
| Cotton 60°C duration | 175.3 | 169.9 | 170.4 |
| Eco temperature: full load | 31.4 | 29.7 | 37.3 |
| *Final-quarter class composition: percent of products* | | | |
| Class *A* | 45.7 | 43.7 | — |
| Class *B* | 14.5 | 14.6 | — |
| Class *C* | 18.1 | 19.8 | — |
| Class *D* | 21.8 | 21.9 | — |

*Notes:* Means across 20 paths. Program attributes and class shares are equal-product averages within each path's active 2023Q4 portfolio, then averaged across paths; they are not sales weighted. Product introductions count simulated entry actions over 16 quarters. Final active-product counts coincide with cumulative introductions in these saved paths. Quarter-load Eco attributes are fixed at the product level, so changes in their mean reflect composition. Class shares are omitted for unlabeled entry: the *D* template is a physical implementation device, not a counterfactual rating.

Table 7: Product responses under selected grid policies

| | Weighted EEI | | Producer tax | |
|---|---|---|---|---|
| | 100 / 0 / 0 | 90 / 0 / 10 | (0, 0) | (100, 100) |
| Introductions, 2020–2023 | 87.10 | 80.75 | 62.55 | 59.40 |
| Active products, 2023Q4 | 87.10 | 80.75 | 62.55 | 59.40 |
| Eco duration: full load (min) | 381.2 | 357.2 | 345.1 | 344.0 |
| Eco duration: half load (min) | 346.1 | 319.9 | 278.3 | 272.7 |
| Cotton 40°C duration (min) | 171.4 | 174.4 | 166.2 | 165.2 |
| Cotton 60°C duration (min) | 169.9 | 163.3 | 168.4 | 168.3 |
| Eco temperature: full load (°C) | 29.7 | 26.2 | 30.4 | 30.1 |
| Eco temperature: half load (°C) | 28.7 | 25.4 | 28.3 | 27.9 |

*Notes:* Means across 20 paths. Design attributes are equal-product means in each active 2023Q4 portfolio, then averaged across paths. Entry counts sum introductions over 16 quarters. EEI weights are percentages in the order Eco, Cotton 40°C, Cotton 60°C. Tax columns use the carbon model; their levels are not a common-baseline comparison with the EEI family.

Table 8: Changes in consumer surplus and electricity

| | $\Delta CS$ (EUR m) | MCSE (EUR m) | $\Delta R$ (GWh) | MCSE (GWh) |
|---|---|---|---|---|
| Current policy | 0.00 | 0.00 | 0.00 | 0.00 |
| No duration cap | 603.22 | 20.55 | -47.02 | 1.86 |
| No EEI / labels | 100.87 | 18.93 | -11.60 | 1.64 |
| Frozen joint removal | 1,192.49 | 24.53 | -15.57 | 1.65 |

*Notes:* Changes relative to current policy, averaged across matched paths. $CS$ uses the purchase outside reference; $R$ sums reference-relative quarterly use across 16 cohorts. MCSE is the paired simulation standard error, not parameter uncertainty. Current-policy levels are EUR 588.35 million for $CS$ and 15.89 GWh for $R$. The frozen benchmark changes fundamentals and population. Grid outcomes use distinct accounting and appear in Table 10.

Table 9: Distribution of consumer-surplus changes

| | Baseline $CS$ (EUR m) | $\Delta CS$ (EUR m) | MCSE (EUR m) | $\Delta CS$/obs. (EUR) |
|---|---|---|---|---|
| *No duration cap* | | | | |
| All households | 588.35 | 603.22 | 20.55 | 62.44 |
| Positive duration preference | 176.52 | 596.53 | 20.56 | 264.65 |
| Nonpositive duration preference | 411.83 | 6.69 | 4.88 | 0.90 |
| *No EEI / labels* | | | | |
| All households | 588.35 | 100.87 | 18.93 | 10.44 |
| Positive duration preference | 176.52 | 95.14 | 16.45 | 42.21 |
| Nonpositive duration preference | 411.83 | 5.72 | 7.39 | 0.77 |

*Notes:* The groups are defined by the sign of the household's program-duration coefficient, not by observed cycle length. They contain 2.254 million positive-preference and 7.406 million nonpositive-preference weighted decision observations over the horizon; these are not counts of unique households. Per-observation gains divide each group's aggregate change by its own population sum. Baseline values are repeated to make the two policy comparisons self-contained.

Table 10: Selected policy-grid outcomes relative to matched baselines

| Policy | $\Delta G$ | $\Delta CS^G$ | Rebates | $\Delta CW^G$ |
|---|---|---|---|---|
| | GWh | EUR m | EUR m | EUR m |
| *Weighted EEI: Eco / Cotton 40°C / Cotton 60°C* | | | | |
| 90 / 5 / 5 | -6.63 | 64.19 | 0.00 | 64.19 |
| 90 / 10 / 0 | -1.48 | 21.15 | 0.00 | 21.15 |
| 90 / 0 / 10 | -10.00 | 91.59 | 0.00 | 91.59 |
| *Producer tax:* $(\tau_E, \tau_C)$ | | | | |
| (50, 50) | -19.02 | 80.48 | 104.81 | 185.29 |
| (100, 0) | -8.97 | 44.74 | 42.44 | 87.18 |
| (0, 100) | -29.02 | 122.74 | 151.51 | 274.25 |
| (100, 100) | -37.34 | 166.17 | 173.30 | 339.48 |

*Notes:* Means of paired differences over 20 paths and 16 cohorts. EEI compares with uncapped Eco-only; tax compares with zero tax in the carbon model. Baseline gross electricity is 165.89 and 128.87 GWh, respectively. $CS^G$ deducts gross purchase electricity expenditure; $CW^G$ adds rebates. Revenue equals aggregate rebates and is added once. These are distinct accounting columns from Table 8. Full grids and Monte Carlo errors appear in Tables A9 and A10.

Table 11: Distribution of selected grid-policy welfare changes

| Group | $\Delta CS^G$ | Rebates | $\Delta CW^G$ | $\Delta CW^G$/obs. |
|---|---|---|---|---|
| | EUR m | EUR m | EUR m | EUR |
| *EEI weights 90 / 0 / 10* | | | | |
| All households | 91.59 | 0.00 | 91.59 | 9.48 |
| Positive duration preference | 38.84 | 0.00 | 38.84 | 17.23 |
| Nonpositive duration preference | 52.75 | 0.00 | 52.75 | 7.12 |
| *Producer tax (100, 100)* | | | | |
| All households | 166.17 | 173.30 | 339.48 | 35.14 |
| Positive duration preference | 87.32 | 40.44 | 127.75 | 56.68 |
| Nonpositive duration preference | 78.86 | 132.86 | 211.72 | 28.59 |

*Notes:* Changes relative to each family's baseline under gross-use accounting. Groups and population weights are the same as in Table 9. Rebates are equal per household within each market. Aggregate subgroup changes sum to the all-household outcome; per-observation effects divide by each group's own horizon population.

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# A Regulatory Details

This appendix summarizes the energy-efficiency calculations and program-design constraints under the 2011 and 2021 EU regulations for household washing machines.

## A.1 Energy-Efficiency Index Calculations

### A.1.1 2011 Labeling Regime

Under the 2011 regime, the Energy Efficiency Index (EEI) measures weighted annual energy consumption relative to a capacity-based benchmark:

$$\mathrm{EEI}_{2011} = 100 \times \frac{\mathrm{AE}_c}{\mathrm{SAE}_c}, \qquad \mathrm{SAE}_c = 47c + 51.7, \tag{A.1}$$

where $\mathrm{AE}_c$ denotes annual energy consumption in kWh/year and $c$ is the rated capacity in kg, defined as the lower of the full-load capacities of the standard Cotton 60°C and Cotton 40°C programs.

Annual energy consumption is calculated as

$$\mathrm{AE}_c = 220E_t + \frac{(P_o + P_l)(525600 - 200T_t)}{2 \times 60 \times 1000}, \tag{A.2}$$

where $E_t$ is weighted energy consumption per cycle (kWh), $P_o$ and $P_l$ are weighted power consumption in off-mode and left-on mode (W), and $T_t$ is weighted program duration (minutes). The calculation assumes 220 standard washing cycles per year and includes energy consumed in low-power modes.[22]

Weighted energy consumption is calculated from three standardized Cotton test cycles:

$$E_t = \frac{3E_{60,\mathrm{full}} + 2E_{60,\mathrm{half}} + 2E_{40,\mathrm{half}}}{7}, \tag{A.3}$$

where $E_{60,\mathrm{full}}$ and $E_{60,\mathrm{half}}$ denote energy consumption of the standard Cotton 60°C program at full and half load, respectively, and $E_{40,\mathrm{half}}$ denotes energy consumption of the standard Cotton 40°C program at half load. The same weights apply to $P_o$, $P_l$, and $T_t$.

The resulting EEI determines the energy-efficiency class according to the following thresholds.

[22] The expression applies to the standard calculation without a separate adjustment for automatic power management. For machines equipped with a power-management system, the regulation provides an alternative formula incorporating the duration of left-on mode.

Table A12: Energy efficiency classes

| **Energy efficiency class** | **Energy Efficiency Index** |
|---|---|
| A+++ (most efficient) | EEI $< 46$ |
| A++ | $46 \leq$ EEI $< 52$ |
| A+ | $52 \leq$ EEI $< 59$ |
| A | $59 \leq$ EEI $< 68$ |

#### A.1.2 2021 Labeling Regime

Under the 2021 regime, the EEI is calculated from weighted energy consumption of the Eco 40–60 program:

$$\mathrm{EEI}_{2021} = 100 \times \frac{E_W}{\mathrm{SCE}_W}, \qquad \mathrm{SCE}_W = -0.0025c^2 + 0.0846c + 0.3920, \tag{A.4}$$

where $E_W$ is weighted energy consumption in kWh per cycle and $c$ is the rated capacity of the Eco 40–60 program in kg.

Weighted energy consumption combines measurements at full, half, and quarter load:

$$E_W = AE_{W,\mathrm{full}} + BE_{W,\mathrm{half}} + CE_{W,\mathrm{quarter}}, \tag{A.5}$$

where the capacity-dependent weights for machines with rated capacity above 3 kg are

$$\begin{aligned} A &= -0.0391c + 0.6918, \\ B &= -0.0109c + 0.3582, \\ C &= 1 - A - B. \end{aligned} \tag{A.6}$$

For machines with rated capacity at or below 3 kg, the regulation assigns $A = 1$ and $B = C = 0$.

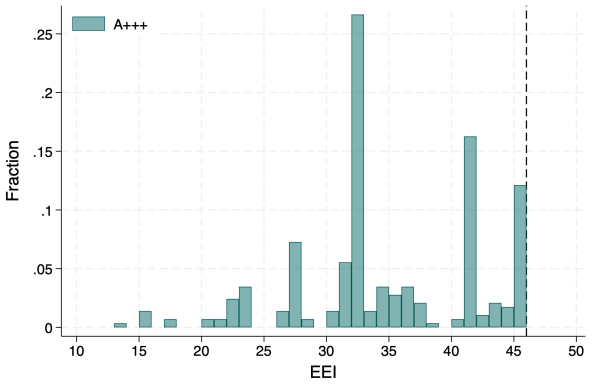
The resulting EEI determines the energy-efficiency class according to the following thresholds.

### A.2 Program-Design Constraints

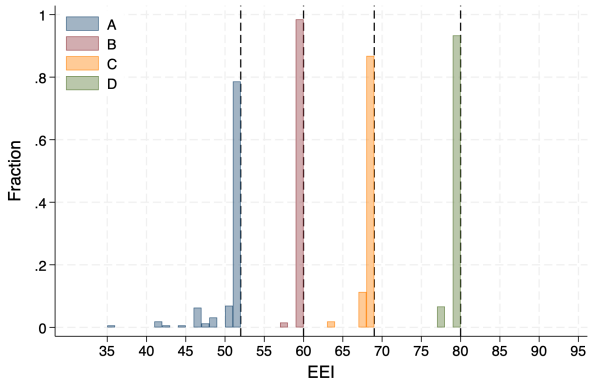
#### A.2.1 Cycle-Duration Limits under the 2021 Regime

The 2021 ecodesign regulation imposes capacity-dependent upper bounds on the duration of the Eco 40–60 program. The limits differ between full and partial loads:

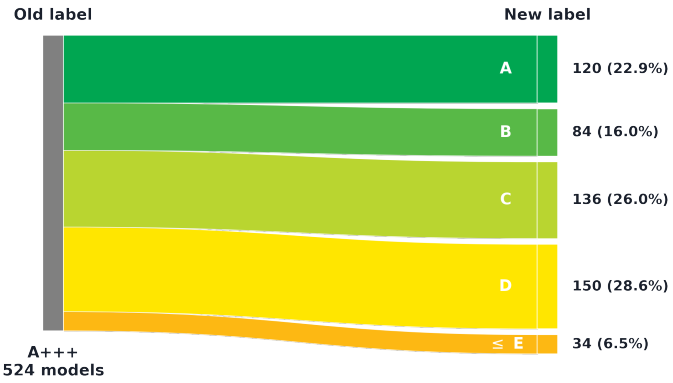
$$\begin{aligned} t_{\mathrm{cap}}^{\mathrm{full}}(c) &= \min\{137 + 10.2c,\ 240\}, \\ t_{\mathrm{cap}}^{\mathrm{half}}(c) = t_{\mathrm{cap}}^{\mathrm{quarter}}(c) &= \min\{120 + 6c,\ 180\}, \end{aligned} \tag{A.7}$$




(a) Old label




(b) New label




(c) Portfolio transformation

Figure A11: Energy Efficiency Indices and Portfolio Transformation

*Notes:* Panel (a) shows the distribution of EEI11 among $A^{+++}$ products sold between March 2019 and December 2020. Panel (b) shows the distribution of EEI21 among products classified $A$–$D$ under the new labeling regime. Panel (c) shows portfolio transformation across labeling regimes. Because the reform changed the underlying efficiency calculation and testing requirements, EEI11 and EEI21 are not directly comparable.

Table A13: Energy efficiency classes

| **Energy Efficiency Class** | **Energy Efficiency Index** |
|---|---|
| A | $\text{EEI} \leq 52$ |
| B | $52 < \text{EEI} \leq 60$ |
| C | $60 < \text{EEI} \leq 69$ |
| D | $69 < \text{EEI} \leq 80$ |
| E | $80 < \text{EEI} \leq 91$ |
| F | $91 < \text{EEI} \leq 102$ |
| G | $\text{EEI} > 102$ |

where $c$ is rated capacity in kg and duration is measured in minutes. Table A14 reports the corresponding limits for common washing-machine capacities.

Table A14: Eco 40–60 Duration Limits under the 2021 Regulation

| Rated capacity | Full load | Half/quarter load |
|---|---|---|
| 6 kg | 198.2 min | 156 min |
| 7 kg | 208.4 min | 162 min |
| 8 kg | 218.6 min | 168 min |
| 9 kg | 228.8 min | 174 min |
| 10 kg | 239.0 min | 180 min |
| $\geq$10.5 kg | 240 min | 180 min |

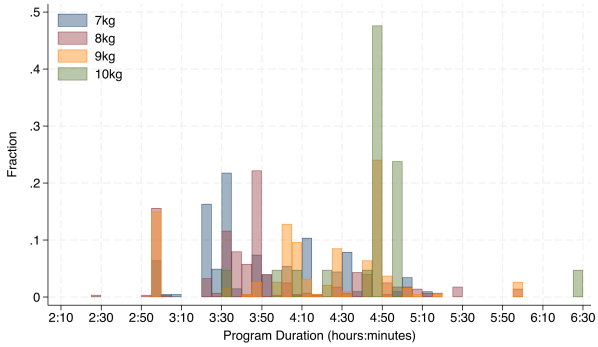
### A.2.2 Water-Consumption Limits under the 2011 and 2021 Regimes

Under the 2011 ecodesign regulation, water-consumption limits applied to the standard Cotton 60°C program at full load. From December 1, 2011, the requirement was

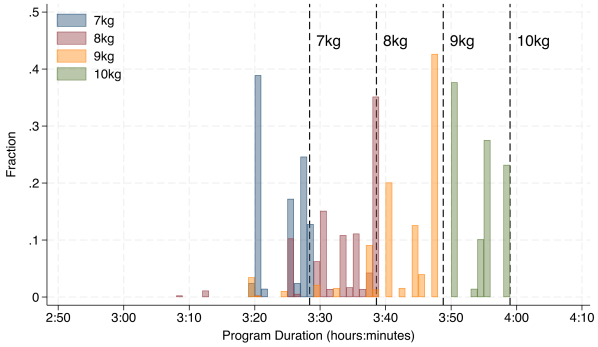
$$W_t \leq 5c + 35, \tag{A.8}$$

where $W_t$ is the water consumption of the standard Cotton 60°C program at full load (liters per cycle), and $c$ is the lower of the rated capacities of the standard Cotton 60°C and Cotton 40°C programs at full load.

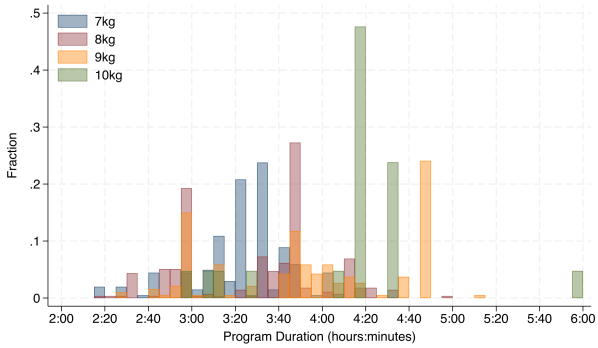
Figure A12: Eco program duration distributions under the old and new labels



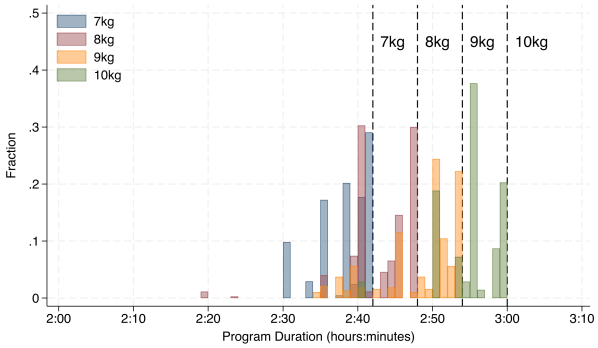

(a) Old label: Cotton 60°C, full load



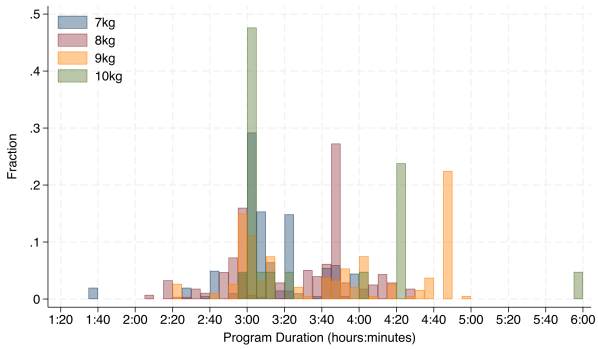

(b) New label: Eco, full load



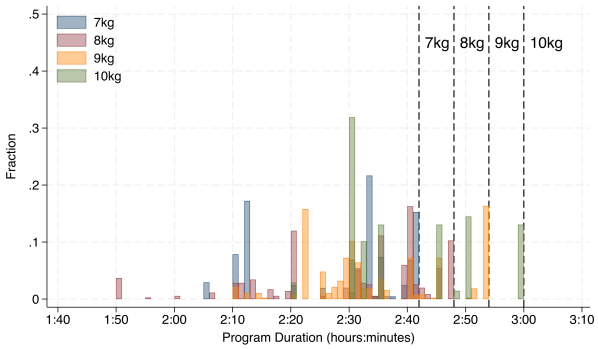

(c) Old label: Cotton 60°C, half load




(d) New label: Eco, half load




(e) Old label: Cotton 40°C, half load




(f) New label: Eco, quarter load

*Notes:* The figure shows program-duration distributions separately by rated capacity, with equal weight assigned to each model within each capacity group. The left column reports the three standard Cotton test conditions used under the old label; the right column reports Eco at full, half, and quarter load under the new label. Rows therefore do not always represent identical test conditions across regimes. Vertical dashed lines in the right column indicate capacity-specific regulatory duration limits. For plotting only, reported durations exceeding their applicable limits are adjusted to the largest whole minute strictly below the limit; the underlying data are unchanged.

From December 1, 2013, the requirement was tightened to

$$W_t \leq 5c_{1/2} + 35, \tag{A.9}$$

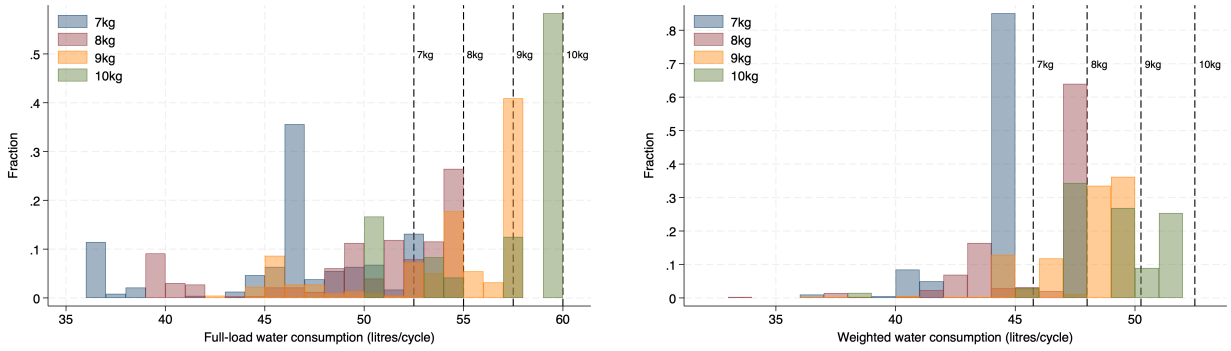
where $c_{1/2}$ is the lower of the rated capacities of the standard Cotton 60°C and Cotton 40°C programs at partial load.

The 2021 regulation instead constrains weighted water consumption of the Eco 40–60 program:

$$W_W \leq 2.25c + 30, \tag{A.10}$$

where $W_W$ is weighted water consumption in liters per cycle, calculated across full, half, and quarter loads using the same capacity-dependent weights $A$, $B$, and $C$ as in the energy-efficiency calculation.

Thus, the two regimes differ in both the regulated test programs and the measurement basis used to impose water-consumption limits.




(a) Old label: full-load water consumption

(b) New label: weighted water consumption

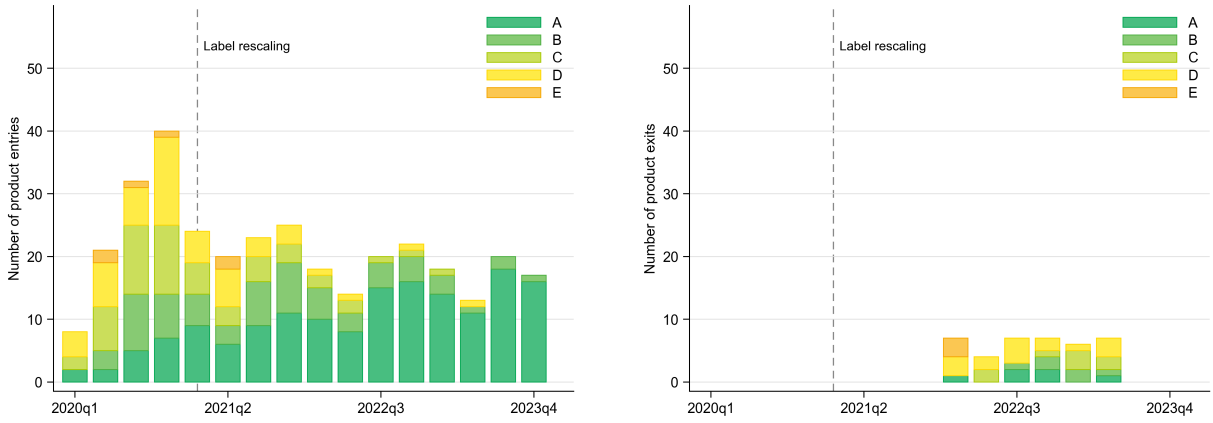
Figure A13: Distribution of Water Consumption

*Notes:* The figure shows water consumption in litres per cycle, separately by rated capacity, with equal weight assigned to each model within each capacity group. Panel (a) reports standard Cotton 60°C water consumption at full load under the old label. Panel (b) reports weighted Eco water consumption across full, half, and quarter loads under the new label. Vertical dashed lines indicate the capacity-specific regulatory limits. For plotting only, reported values exceeding their applicable limits are adjusted to the largest whole litre strictly below the limit. The two panels use different test conditions and aggregation rules and are therefore not directly comparable.

## B Data Imputation

**Standard-program temperature.** Under the pre-2021 labeling regime, product manuals identify the regulated test cycles as the standard Cotton 60°C program at full and half load and the standard Cotton 40°C program at half load, but they generally do not report the temperature

Figure B14: Product turnover by new energy class




(a) Entry

(b) Exit

*Notes:* Quarterly national product entries and recorded exits, 2020q1–2023q4, for new-label classes *A*–*E*. A product is the current demand-data group: firm × capacity/spin-speed line × entry-quarter cohort × energy-label class (new and legacy class identifiers). Each product is counted once across markets and wash programs, rather than once per physical model. Entry and exit use the group's recorded dates; a group remains active if any constituent model has no recorded exit. The sample endpoint is not treated as an exit. Bars retain the same *A*–*E* colors in both panels; the dashed line marks 2021q1 label rescaling.

actually reached during these cycles. I do not assign the nominal program temperature to these observations. This distinction is empirically important because the nominal program name need not correspond to the realized wash temperature. Industry and consumer testing has documented substantial departures from the nominal temperature. In particular, Czyzewski (2022), writing from the perspective of BSH and the CENELEC working group on laundry appliances, reports that measured sump temperatures in Cotton 60°C programs vary substantially across products and that most machines do not reach 60°C; temperatures experienced by the laundry are lower still. An earlier Stiftung Warentest market test provides a quantitative benchmark: among 14 tested washing machines, none reached 60°C in the tested 60°C cotton program, with reported temperatures generally around 45–50°C and the lowest observation approximately 44°C (Van Holsteijn en Kemna B.V. et al., 2014).

I therefore impute realized temperatures from the physical characteristics of the wash cycle rather than from its nominal program name. I estimate an XGBoost regression using Eco 40–60 cycles for which temperature is observed. The estimation sample pools full-, half-, and quarter-load Eco cycles and excludes observations whose temperature was itself previously model-imputed. The predictors are program load, energy consumption, water consumption, program duration, product capacity, and the ratio of program load to rated capacity. All predictors are required to be observed and strictly positive. Importantly, the specification does not include the nominal program temperature, program identity, energy-label class, manufacturer, or the certified aggregate energy measure. The identifying assumption for the transfer is that, conditional on these physical inputs, the relationship between temperature and the other dimensions of program design is common across the Eco and standard test programs.

I evaluate the prediction rule using a product-level holdout sample: 20 percent of brand–model combinations are excluded from estimation, so that cycles from the same product cannot appear in both the training and validation samples. Within the Eco programs, the holdout regression yields an $R^2$ of 0.988, an RMSE of 0.79°C, and a mean absolute error of 0.42°C. I then re-estimate the model on all eligible Eco observations and apply it to the three old-regime standard programs using their observed physical inputs. The procedure imputes 19,521 standard-program temperatures; observations with incomplete or nonpositive physical inputs remain missing.

The resulting temperatures are well below the nominal program settings. The median imputed temperature is 36.1°C for the standard Cotton 60°C full-load cycle, 34.7°C for the standard Cotton 60°C half-load cycle, and 32.1°C for the standard Cotton 40°C half-load cycle. These estimates are lower than the roughly 45–50°C range reported in the 2013 Stiftung Warentest test, but both the external tests and the imputation imply the same economically important feature of the old testing regime: the nominal 40°C or 60°C designation is a poor measure of the temperature actually delivered by an energy-efficient cycle. The lower imputed values should therefore not be

interpreted as reproducing the Stiftung Warentest distribution mechanically; rather, the external evidence supports the central feature of the imputation—substantial temperature reductions relative to the nominal setting.

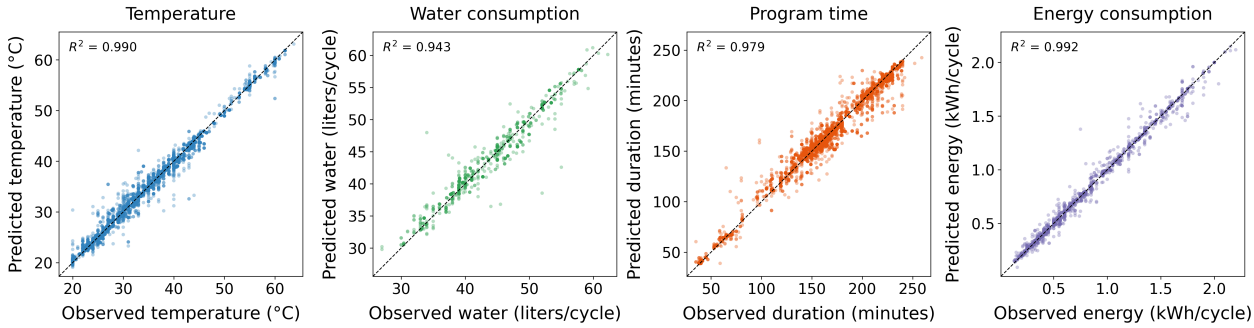



Figure B15: Program inputs out-of-sample fit

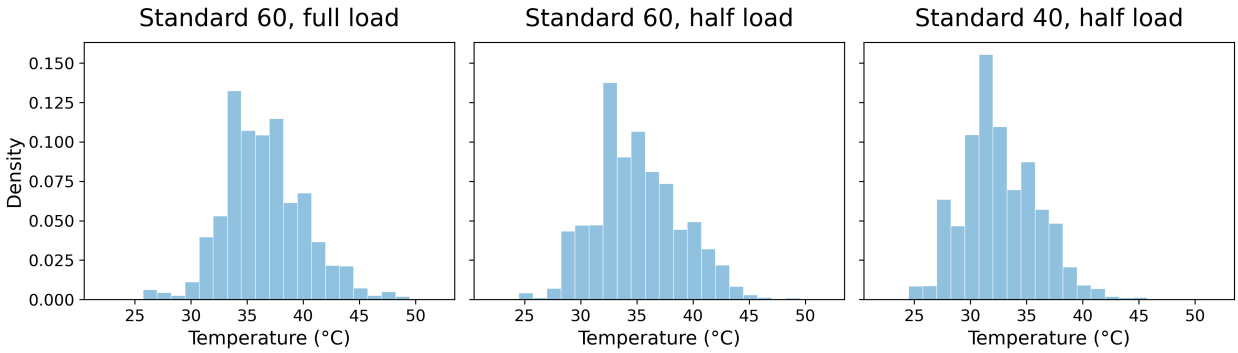



Figure B16: Distribution of standard program temperature

## C Laundry-habits survey

Dear Sir or Madam,

This questionnaire has been designed by the University of Wisconsin–Madison (USA). It is intended for: (1) consumers who own a washing machine with the post-2021 A–G energy label, and (2) the primary person responsible for doing the laundry in the household. Completing the questionnaire will take approximately 5–10 minutes.

1. Who is mainly responsible for doing the laundry in your household? Please choose one option.

   - ☐ I am mainly responsible.
   - ☐ I share responsibility with someone else.

- ☐ Someone else in the household is mainly responsible.

2. In which federal state (Bundesland) do you live? Please choose one option.

   - ☐ Baden-Württemberg
   - ☐ Bavaria (Bayern)
   - ☐ Berlin
   - ☐ Brandenburg
   - ☐ Bremen
   - ☐ Hamburg
   - ☐ Hesse (Hessen)
   - ☐ Lower Saxony (Niedersachsen)
   - ☐ Mecklenburg-Vorpommern
   - ☐ North Rhine-Westphalia (Nordrhein-Westfalen)
   - ☐ Rhineland-Palatinate (Rheinland-Pfalz)
   - ☐ Saarland
   - ☐ Saxony (Sachsen)
   - ☐ Saxony-Anhalt (Sachsen-Anhalt)
   - ☐ Schleswig-Holstein
   - ☐ Thuringia (Thüringen)

3. How many people are living in your household, including yourself? Please choose one option.

   - ☐ 1 person
   - ☐ 2 people
   - ☐ 3 people
   - ☐ 4 people
   - ☐ More than 4 people

4. Approximately what is your household's monthly income?

---

5. What is the **brand and model number** of your washing machine?
   (*You can usually find the model number on the energy label or on the* Typenschild */ rating plate—often inside the door opening/door frame, on the back of the machine, or in the user manual/receipt. For Bosch/Siemens, it may be listed as the "E-Nr."; for AEG/Electrolux, it may be the "PNC".*)

   _____

6. What is the energy efficiency class of your washing machine (*A*–*G* label)? Please choose one option.

   ☐ A
   ☐ B
   ☐ C
   ☐ D
   ☐ E
   ☐ I'm not sure.

7. What is the rated load capacity of your washing machine (in kg)? Please choose one option.

   ☐ 6 kg or less
   ☐ 7 kg
   ☐ 8 kg
   ☐ 9 kg
   ☐ 10 kg or more
   ☐ I'm not sure.

8. When did you purchase your washing machine? Please choose one option.

   | | | | | |
   |---|---|---|---|---|
   | 2021: | ☐Q1 | ☐Q2 | ☐Q3 | ☐Q4 |
   | 2022: | ☐Q1 | ☐Q2 | ☐Q3 | ☐Q4 |
   | 2023: | ☐Q1 | ☐Q2 | ☐Q3 | ☐Q4 |
   | 2024: | ☐Q1 | ☐Q2 | ☐Q3 | ☐Q4 |
   | 2025: | ☐Q1 | ☐Q2 | ☐Q3 | ☐Q4 |

9. When purchasing a washing machine, which information do you use to judge energy efficiency? Please choose the option that best describes you.

☐ I use the energy efficiency class (*A*–*G*) as a quick filter.

☐ I mainly compare the energy consumption values (kWh per 100 cycles) on the energy label and choose the lower value.

☐ I look at both the class (*A*–*G*) and the kWh value, but I do not calculate electricity costs.

☐ I convert the kWh value into an electricity cost estimate.

☐ I rarely or never use the energy efficiency class or the kWh value.

☐ I'm not sure.

10. *Only if you selected in the previous question that you convert the kWh value into an electricity cost estimate:*
    Approximately how much do you think your washing machine costs in electricity per year? Please choose one option.

    ☐ Less than €10

    ☐ €10–€20

    ☐ €21–€30

    ☐ €31–€40

    ☐ €41–€60

    ☐ €61–€80

    ☐ More than €80

    ☐ I'm not sure.

11. What would you expect a washing machine with the highest energy efficiency class to save on? Please choose one option.

    ☐ Energy in all programmes

    ☐ Energy and water in all programmes

    ☐ Energy in some selected programmes

    ☐ Energy and water in some selected programmes

12. How often do your laundry loads have the following stain levels? *"Stains" include visible marks (e.g., food, grease, mud) and/or strong odours (e.g., sweat). Please choose one option per row.*

| | **Not at all** | **Once per 4 weeks** | **Once per 3 weeks** | **Once per 2 weeks** | **1–2x per week** | **3–4x per week** | **5–6x per week** | **> 6x per week** |
|---|---|---|---|---|---|---|---|---|
| **Low** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| **Medium** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| **High** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |

**Low stains:** No visible stains; lightly worn; no strong odour.

**Medium stains:** Some visible dirt/marks or noticeable odour (typical everyday laundry).

**High stains:** Multiple items with heavy/visible stains (e.g., mud/grease/food) and/or strong odour.

13. How often do you use the following washing programmes for washing your clothes? Please choose one option per row.

| | Not at all | Once per 4 weeks | Once per 3 weeks | Once per 2 weeks | 1–2x per week | 3–4x per week | 5–6x per week | >6x per week |
|---|---|---|---|---|---|---|---|---|
| Cotton 20 °C / Cold wash | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| Cotton 30 °C | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| Cotton 40 °C | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| Cotton 60 °C | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| Cotton 90 °C | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| Synthetic / Easy care 30 °C / 40 °C | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| Eco 40-60 (Eco) / Energy saving | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| Silk/Wool | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| Quick wash / Short | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| Mix (all fabrics) | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| Other | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |

## D Proofs and Extensions for Regulation-Induced Energy Shifting

### D.1 Proof of Proposition 1

Under logit program choice, expected utility is

$$\mathcal{I}(d_E, d_C) = \log\left(e^{\beta d_E} + e^{\beta d_C}\right).$$

Let $s_r$ denote the corresponding program-choice probability. Differentiating yields

$$\frac{\partial \mathcal{I}}{\partial d_E} = \beta s_E, \qquad \frac{\partial \mathcal{I}}{\partial d_C} = \beta s_C, \qquad \frac{\partial^2 \mathcal{I}}{\partial d_C \partial d_E} = -\beta^2 s_E s_C.$$

Product demand is $Q(z)$, where

$$z = P + \kappa e_E(d_E) - \theta \mathcal{I}(d_E, d_C),$$

with $Q' < 0$, $\kappa > 0$, and $\theta > 0$. Throughout, partial derivatives of $Q$ with respect to program inputs refer to the composite demand function $Q(z)$. Applying the chain rule gives

$$\frac{\partial^2 Q}{\partial d_C \partial d_E} = \theta\beta s_C \left[\beta s_E \left(\theta Q''(z) + Q'(z)\right) - \kappa e_E'(d_E) Q''(z)\right].$$

The sign depends on the curvature of product demand and the direct contribution of Eco energy consumption. Proposition 1 imposes $\frac{\partial^2 Q}{\partial d_C \partial d_E} < 0$ as a sufficient condition for decreasing differences in demand.

With the Eco-duration cap binding, $d_E^* = \bar{d}_E$. The Cotton first-order condition is

$$\frac{\partial Q}{\partial d_C} = C_C'(d_C).$$

Implicit differentiation with respect to $\bar{d}_E$ yields

$$\frac{\partial d_C^*}{\partial \bar{d}_E} = \frac{\frac{\partial^2 Q}{\partial d_C \partial d_E}}{C_C'' - \frac{\partial^2 Q}{\partial d_C^2}} < 0,$$

where the inequality follows from decreasing differences in demand and the strict second-order condition $C_C'' - \frac{\partial^2 Q}{\partial d_C^2} > 0$.

Because Cotton energy consumption satisfies $e_C'(d_C) > 0$,

$$\frac{\partial e_C(d_C^*)}{\partial \bar{d}_E} = e_C'(d_C^*) \frac{\partial d_C^*}{\partial \bar{d}_E} < 0.$$

Thus, tightening the Eco-duration cap increases optimal Cotton duration and energy consumption, establishing Proposition 1. □

### D.2 Extension: Joint Eco Constraints

I extend the stylized model to allow firms to choose a second Eco input $x_E$, such as temperature, while Cotton duration remains its only design input. Eco energy consumption is $e_E(d_E, x_E)$, with $\frac{\partial e_E}{\partial d_E} > 0$ and $\frac{\partial e_E}{\partial x_E} > 0$. Consumer utility from Eco becomes $v_E = \beta d_E + \beta_x x_E + \varepsilon_E$, while Cotton utility remains $v_C = \beta d_C + \varepsilon_C$. Expected program utility is

$$\mathcal{I}(d_E, x_E, d_C) = \log\left(e^{\beta d_E + \beta_x x_E} + e^{\beta d_C}\right).$$

Taking price as given and normalizing the markup to one, the firm solves

$$\begin{aligned} \max_{d_E, x_E, d_C} \quad & Q\left(P + \kappa e_E(d_E, x_E) - \theta \mathcal{I}(d_E, x_E, d_C)\right) - C_E(d_E, x_E) - C_C(d_C) \\ \text{s.t.} \quad & d_E \le \bar{d}_E, \qquad e_E(d_E, x_E) \le \bar{e}. \end{aligned}$$

**Proposition 2** (Energy Shifting under Joint Eco Constraints)**.** *Suppose both Eco constraints bind and Cotton duration has an interior local optimum satisfying the second-order condition. If* $\frac{\partial^2 Q}{\partial d_C \partial d_E} < 0$ *and* $\frac{\partial^2 Q}{\partial d_C \partial x_E} < 0$*, tightening the Eco-energy cap increases optimal Cotton duration and energy consumption, holding the duration cap fixed. Jointly tightening both caps also increases Cotton duration and energy consumption if the second Eco input does not increase.*

*Proof.* Under logit program choice, the cross-partials of expected utility are

$$\frac{\partial^2 \mathcal{I}}{\partial d_C \partial d_E} = -\beta^2 s_E s_C, \qquad \frac{\partial^2 \mathcal{I}}{\partial d_C \partial x_E} = -\beta \beta_x s_E s_C.$$

As in Proposition 1, the corresponding demand cross-partials also depend on the curvature of $Q$ and the direct contribution of Eco energy consumption. The proposition assumes both are negative.

Because both constraints bind,

$$d_E^* = \bar{d}_E, \qquad e_E(\bar{d}_E, x_E^*) = \bar{e}.$$

Differentiating the energy constraint gives

$$\mathrm{d}x_E^* = \frac{\mathrm{d}\bar{e} - \frac{\partial e_E}{\partial d_E}\mathrm{d}\bar{d}_E}{\frac{\partial e_E}{\partial x_E}}. \tag{D.1}$$

The Cotton first-order condition is

$$\frac{\partial Q}{\partial d_C} = C_C'(d_C).$$

Implicit differentiation yields

$$\left(C_C'' - \frac{\partial^2 Q}{\partial d_C^2}\right) \mathrm{d}d_C^* = \frac{\partial^2 Q}{\partial d_C \partial d_E} \mathrm{d}\bar{d}_E + \frac{\partial^2 Q}{\partial d_C \partial x_E} \mathrm{d}x_E^*.$$

Substituting equation (D.1) gives

$$\mathrm{d}d_C^* = \frac{1}{C_C'' - \frac{\partial^2 Q}{\partial d_C^2}} \left[ \left( \frac{\partial^2 Q}{\partial d_C \partial d_E} - \frac{\frac{\partial e_E}{\partial d_E}}{\frac{\partial e_E}{\partial x_E}} \frac{\partial^2 Q}{\partial d_C \partial x_E} \right) \mathrm{d}\bar{d}_E + \frac{\frac{\partial^2 Q}{\partial d_C \partial x_E}}{\frac{\partial e_E}{\partial x_E}} \mathrm{d}\bar{e} \right]. \tag{D.2}$$

Holding the duration cap fixed,

$$\frac{\partial d_C^*}{\partial \bar{e}} = \frac{\frac{\partial^2 Q}{\partial d_C \partial x_E}}{\frac{\partial e_E}{\partial x_E}\left(C_C'' - \frac{\partial^2 Q}{\partial d_C^2}\right)} < 0.$$

The inequality follows from decreasing differences between Cotton duration and the second Eco input, increasing Eco energy consumption in $x_E$, and the strict second-order condition. Thus, tightening the energy cap reduces $x_E$ and increases optimal Cotton duration.

Now consider jointly tightening both caps, with $\mathrm{d}\bar{d}_E < 0$ and $\mathrm{d}\bar{e} < 0$. Equation (D.1) implies that the second Eco input does not increase if

$$\mathrm{d}\bar{e} \leq \frac{\partial e_E}{\partial d_E} \mathrm{d}\bar{d}_E.$$

Under this condition, $\mathrm{d}d_E^* < 0$ and $\mathrm{d}x_E^* \leq 0$. Because both demand cross-partials are negative, the Cotton first-order condition implies $\mathrm{d}d_C^* > 0$.

Finally, since $e_C'(d_C) > 0$, any increase in Cotton duration also increases Cotton energy consumption. This establishes both claims. □

Holding the energy cap fixed instead gives

$$\left.\frac{\partial d_C^*}{\partial \bar{d}_E}\right|_{\bar{e}} = \frac{\frac{\partial^2 Q}{\partial d_C \partial d_E} - \frac{\frac{\partial e_E}{\partial d_E}}{\frac{\partial e_E}{\partial x_E}} \frac{\partial^2 Q}{\partial d_C \partial x_E}}{C_C'' - \frac{\partial^2 Q}{\partial d_C^2}}.$$

The sign is ambiguous because tightening the duration cap permits an offsetting increase in the second Eco input. Joint constraints strengthen energy shifting when they prevent such compensation within Eco.

### D.3 Socially Efficient Program Design

I extend the stylized model to compare private and socially efficient program designs. Consumers perceive Eco energy consumption as representative of both programs. To isolate the resulting information distortion, suppose the social cost of electricity equals its valuation at purchase. Since $\theta$ converts program utility into price-equivalent units, define $\eta = \kappa/\theta$.

Under consumers' beliefs, expected program utility net of perceived electricity costs is

$$\mathcal{I}^P(d_E, d_C) = \mathcal{I}(d_E, d_C) - \eta e_E(d_E).$$

A social planner instead accounts for actual program-specific electricity consumption:

$$\mathcal{I}^S(d_E, d_C) = \log\left(e^{\beta d_E - \eta e_E(d_E)} + e^{\beta d_C - \eta e_C(d_C)}\right).$$

The socially efficient Cotton usage share is

$$s_C^S = \frac{e^{\beta d_C - \eta e_C(d_C)}}{e^{\beta d_E - \eta e_E(d_E)} + e^{\beta d_C - \eta e_C(d_C)}}.$$

Thus, private and socially efficient program choices differ by

$$\log\frac{s_C}{s_E} - \log\frac{s_C^S}{s_E^S} = \eta\left[e_C(d_C) - e_E(d_E)\right].$$

When Cotton consumes more electricity than Eco, private choices allocate too many cycles to Cotton relative to the social benchmark.

To characterize socially efficient design, let

$$z^P = P + \kappa e_E(d_E) - \theta\mathcal{I}(d_E, d_C), \qquad z^S = P - \theta\mathcal{I}^S(d_E, d_C).$$

Let consumer surplus under full information be

$$CS(z^S) = \int_{z^S}^{\infty} Q(u)\, du.$$

Maintaining the unit-markup normalization and abstracting from other distortions, the social planner chooses program durations to maximize total surplus:

$$\max_{d_E, d_C} \quad CS(z^S) + Q(z^S) - C_E(d_E) - C_C(d_C) \qquad \text{s.t.} \qquad d_E \le \bar{d}_E.$$

This objective includes consumer surplus, operating profits, and design costs, with actual electricity costs incorporated into $\mathcal{I}^S$.

The socially optimal Cotton duration satisfies

$$\theta \left[ Q(z^S) - Q'(z^S) \right] s_C^S \left[ \beta - \eta \frac{\partial e_C}{\partial d_C} \right] = C_C'(d_C^S). \tag{D.3}$$

By contrast, the firm's private optimum satisfies

$$-\theta Q'(z^P) \beta s_C = C_C'(d_C^P). \tag{D.4}$$

The social planner accounts for the additional electricity consumed when Cotton duration increases. The firm does not face this direct energy penalty because consumers perceive Cotton energy consumption as equal to Eco energy consumption. However, the two objectives also differ in their valuation of product demand and consumer surplus.

To compare the optimal designs, hold Eco duration fixed at the binding regulatory cap. Suppose the social objective is strictly concave in Cotton duration. Evaluated at the private optimum $d_C^P$, a sufficient condition for excessive Cotton duration is

$$\left[ Q(z^S) - Q'(z^S) \right] s_C^S \left[ \beta - \eta \frac{\partial e_C}{\partial d_C} \right] < -Q'(z^P) \beta s_C. \tag{D.5}$$

Under this condition, the social marginal return to extending Cotton is smaller than the private marginal return at $d_C^P$. Strict concavity then implies

$$d_C^S < d_C^P, \qquad e_C(d_C^S) < e_C(d_C^P).$$

For example, when $\beta > 0$, the condition is satisfied if $\eta \frac{\partial e_C}{\partial d_C} \geq \beta$ at the private optimum: the social marginal electricity cost of extending Cotton is at least as large as its marginal duration benefit. More generally, whether private designs are excessively energy intensive depends on the relative marginal benefits in equation (D.5).

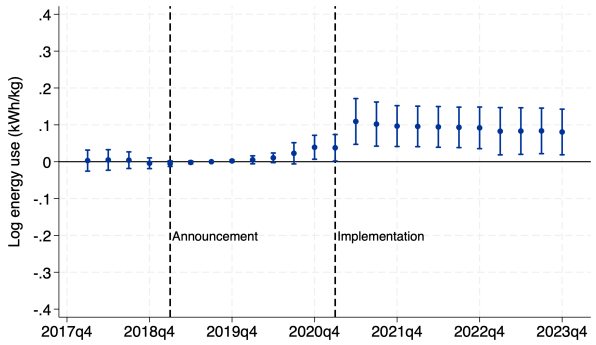
# E Descriptive Evidence

# F Estimation and Computational Implementation

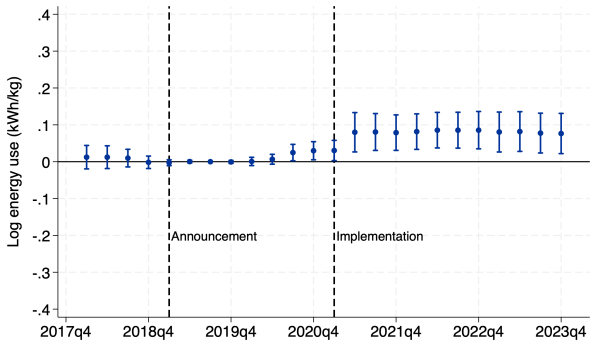
## F.1 Parameter Snapshots and Profit Approximation

The latest engineering regression pools Eco/Standard and nonimputed Cotton observations with common slopes, as described in Appendix **??**. The completed supply and counterfactual calculations instead retain the frozen engineering runtime used to construct their profit labels. Its Eco/Standard elasticities for load, temperature, duration, and water are 0.1984, 1.6492, 0.1625, and 0.3439. Cotton energy is evaluated with the separately saved Cotton relation. Consequently, the completed

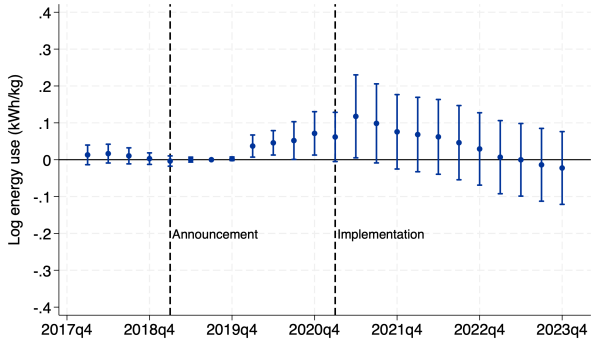
Figure E1: Changes in Energy Consumption: Additional Programs and Robustness



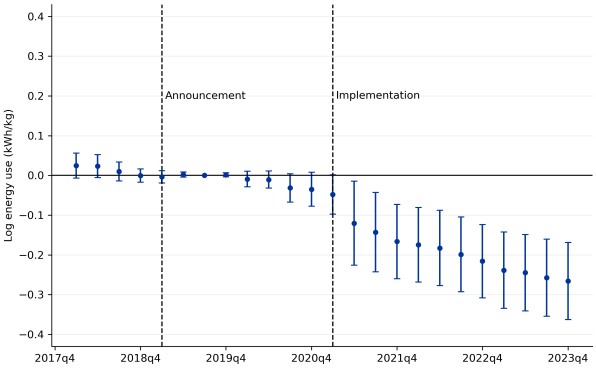

(a) Cotton 40°C




(b) Easy Care




(c) Eco: alternative old-program definition




(d) Eco: label-reported consumption

*Notes:* Panels (a) and (b) report changes in log energy consumption per kilogram for Cotton 40°C and Easy Care, respectively, relative to Wool. Panels (c) and (d) report Eco robustness checks using alternative consumption measures. Panel (c) combines Standard 60°C at full and half load for the old program, retaining the normalized weights and new-program definition from the main specification. Panel (d) uses label-reported consumption under the respective old and new testing protocols. Dots denote coefficients from equation (1), normalized to zero in 2019Q3. The sample contains 31 product lines (firm × capacity × spin speed), observed in every quarter from 2018Q1 to 2023Q4, with products weighted equally within each line. Bars denote 95% confidence intervals based on standard errors clustered by product line. Dashed vertical lines mark the reform's announcement in March 2019 and implementation in March 2021.

Table A1: Energy Equation Estimates

| | Estimate | Std. error |
|---|---|---|
| Log load | 0.167 | 0.008 |
| Log temperature | 1.580 | 0.007 |
| Log duration | 0.401 | 0.013 |
| Log water | 0.321 | 0.015 |
| Product fixed effects | Yes | |
| Program-group fixed effects | Yes | |
| Observations | 8,990 | |
| Products | 1,786 | |
| Adjusted $R^2$ | 0.936 | |

*Notes:* The dependent variable is log electricity consumption per cycle (kWh). Unweighted OLS includes product (brand–model) fixed effects and program-group effects for Eco/Standard, Cotton, and Easy Care, with Eco/Standard omitted. Cotton 20°C, Cotton 40°C, and Cotton 60°C share one group effect. Eco/Standard observations may include imputed values; Cotton and Easy Care observations must be nonimputed for energy and all four inputs. Wool is excluded. Standard errors are conventional homoskedastic OLS standard errors, accounting for absorbed product effects.

policy results are conditional on this earlier engineering mapping, not a recomputation under the pooled slopes in Table A1. The demand and marginal-cost parameters used downstream agree with Tables **??** and **??**.

Demand integration uses five nodes for each of income, stain share, and the two random program tastes, giving the 625 agents per market used in estimation. Profit labels integrate over 32 joint demand and cost shock draws and re-solve Bertrand prices. Own-entrant input derivatives account for the price response through implicit differentiation of the price first-order conditions. Incumbents affect profits but are not treated as redesign opportunities.

The operating-profit approximation is a permutation-invariant set network with shared product encoders and pooled firm and market features (Zaheer et al., 2017). Smooth activations allow its own-input derivatives to enter the training loss alongside profit levels. The saved network has width 96 and 63 input features. Its 32,000 labeled profiles are split into 22,400 training, 4,800 validation, and 4,800 reserved test profiles. Observed profiles are split within quarter; simulated profiles are split by entire path to avoid placing quarters from one path in both training and validation. Training uses standardized profit and derivative losses, float64 arithmetic, AdamW, and an initial learning rate of $3 \times 10^{-4}$. Figure F2 reports the saved validation-loss sequence. The checkpoint's training-complete and acceptance flags remain false; counterfactual-support accuracy has not been established.

Table A2: Changes in Program Inputs and Their Contributions to Energy Consumption

| | Energy/kg | Load | | Duration | | Temperature | | Water | | Technology | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Quarter | Δ% | Δ% | Contrib. (%) | Δ% | Contrib. (%) | Δ% | Contrib. (%) | Δ% | Contrib. (%) | Δ% | Contrib. (%) |
| *Panel A: Eco program* | | | | | | | | | | | |
| 2020Q2 | 1.6 | 0.0 | -0.8 | -1.0 | -14.7 | 0.9 | 71.3 | 0.8 | 11.1 | 0.0 | 2.1 |
| 2020Q4 | 5.8 | 0.6 | -5.1 | -3.2 | -12.2 | 3.6 | 67.6 | 3.3 | 12.1 | -0.2 | -2.9 |
| 2021Q2 | 16.1 | 1.7 | -5.0 | -9.1 | -11.6 | 10.4 | 64.1 | 10.1 | 12.2 | -1.8 | -7.2 |
| 2021Q4 | 10.0 | 2.6 | -9.0 | -9.0 | -12.6 | 7.9 | 54.1 | 9.3 | 12.5 | -2.6 | -11.7 |
| 2022Q2 | 5.8 | 3.7 | -13.7 | -8.4 | -12.5 | 5.9 | 42.8 | 10.7 | 14.6 | -3.4 | -16.4 |
| 2022Q4 | 1.7 | 2.9 | -13.8 | -8.5 | -16.7 | 3.1 | 28.4 | 10.4 | 17.9 | -3.7 | -23.2 |
| 2023Q2 | -2.8 | 3.8 | -23.5 | -8.1 | -20.4 | 0.3 | -0.5 | 11.6 | 25.3 | -3.7 | -30.3 |
| 2023Q4 | -4.5 | 3.9 | -20.4 | -8.0 | -17.1 | -0.7 | -12.2 | 12.2 | 22.5 | -4.0 | -27.8 |
| *Panel B: Cotton 60° C* | | | | | | | | | | | |
| 2020Q2 | 0.9 | 0.0 | -2.2 | 1.1 | 34.9 | -0.2 | -30.3 | 0.9 | 28.3 | 0.0 | 4.3 |
| 2020Q4 | 3.4 | 0.6 | -11.8 | 4.9 | 37.5 | -0.4 | -15.2 | 3.8 | 29.4 | -0.2 | -6.0 |
| 2021Q2 | 12.4 | 1.9 | -10.8 | 14.0 | 33.0 | -0.8 | -11.1 | 12.9 | 31.3 | -1.8 | -13.8 |
| 2021Q4 | 14.3 | 2.9 | -14.7 | 16.0 | 31.4 | -1.0 | -11.0 | 12.4 | 25.7 | -2.6 | -17.2 |
| 2022Q2 | 14.1 | 4.1 | -18.9 | 17.5 | 30.7 | -0.4 | -4.5 | 13.8 | 25.6 | -3.4 | -20.4 |
| 2022Q4 | 16.9 | 3.2 | -15.3 | 18.3 | 33.7 | 0.1 | 0.1 | 14.1 | 27.6 | -3.7 | -23.3 |
| 2023Q2 | 16.2 | 4.3 | -18.2 | 18.5 | 30.5 | 0.6 | 4.6 | 14.7 | 25.6 | -3.7 | -21.1 |
| 2023Q4 | 15.9 | 4.3 | -17.9 | 18.2 | 29.2 | 0.8 | 5.7 | 15.0 | 25.3 | -4.0 | -21.9 |

*Notes:* Changes are relative to 2020Q1. Input changes use arithmetic means; technology reports $100[\exp(\bar{z}_t - \bar{z}_0) - 1]$. Contribution shares equal $100C_k / \sum_h |C_h|$, using changes in mean log inputs and estimated coefficients (load uses $\widehat{\theta}_L - 1$). Absolute shares sum to 100 before rounding; residual and aggregation effects are excluded. Technology is shared within firm–energy-class–entry-quarter cells. The energy regression includes Eco/Standard, Cotton, and Easy Care group effects. Both panels use the same equally weighted product portfolio, excluding Samsung, Beko, Bauknecht, and LG. Eco combines Standard and Eco using common normalized full- and half-load weights; quarter load is excluded. Product turnover and imputed inputs are included.

A second set network predicts optimized inputs and profits net of design costs from pre-choice portfolios. Adjustable entrant inputs are prediction targets, not observed features. Entry estimation uses the saved epoch-90 checkpoint trained on design-cost evaluation 44. The reported design estimates and realized counterfactual design refinement use evaluation 47. Eco and Cotton duration coefficients are identical across these two evaluations, but technology curvature is $1.816 \times 10^8$ in evaluation 44 versus $5.448 \times 10^8$ in evaluation 47, and the Eco-temperature linear term is 45.27 versus 45.92. Realized-design refinement does not update the entry payoffs from the earlier checkpoint; validation of both checkpoints remains incomplete.

Table A3: Changes in Program Inputs and Energy Contributions: Cotton 40°C and Easy Care

| | Energy/kg | Load | | Duration | | Temperature | | Water | | Technology | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Quarter | Δ% | Δ% | Contrib. (%) | Δ% | Contrib. (%) | Δ% | Contrib. (%) | Δ% | Contrib. (%) | Δ% | Contrib. (%) |
| *Panel A: Cotton 40° C* | | | | | | | | | | | |
| 2020Q2 | 0.9 | 0.2 | -6.9 | 1.4 | 34.5 | 0.3 | 23.5 | 0.9 | 17.5 | 0.3 | 17.6 |
| 2020Q4 | 4.0 | 0.6 | -8.1 | 5.7 | 36.7 | 1.0 | 25.0 | 3.9 | 20.5 | 0.6 | 9.7 |
| 2021Q2 | 11.1 | 1.3 | -6.0 | 16.4 | 39.6 | 3.0 | 29.6 | 11.9 | 23.2 | 0.3 | 1.6 |
| 2021Q4 | 8.8 | 2.4 | -11.0 | 16.4 | 36.8 | 3.2 | 29.2 | 11.0 | 19.9 | -0.5 | -3.0 |
| 2022Q2 | 9.0 | 3.4 | -13.3 | 18.1 | 34.2 | 3.5 | 26.7 | 12.6 | 19.3 | -1.3 | -6.6 |
| 2022Q4 | 9.8 | 3.0 | -11.3 | 18.1 | 33.4 | 3.6 | 27.3 | 12.8 | 18.9 | -1.9 | -9.0 |
| 2023Q2 | 9.5 | 4.0 | -14.2 | 18.2 | 31.9 | 3.6 | 25.8 | 13.4 | 18.8 | -2.0 | -9.3 |
| 2023Q4 | 9.2 | 4.3 | -15.3 | 17.9 | 31.0 | 3.5 | 24.8 | 13.5 | 18.7 | -2.2 | -10.2 |
| *Panel B: Easy Care* | | | | | | | | | | | |
| 2020Q2 | 1.0 | -0.1 | 11.6 | 1.1 | 37.5 | -0.1 | -12.3 | 0.4 | 11.7 | 0.3 | 26.8 |
| 2020Q4 | 4.6 | -0.1 | 1.7 | 4.8 | 36.1 | 1.2 | 40.3 | 1.6 | 10.3 | 0.6 | 11.6 |
| 2021Q2 | 13.5 | -0.1 | 1.1 | 14.0 | 34.2 | 4.6 | 52.8 | 5.0 | 10.2 | 0.3 | 1.7 |
| 2021Q4 | 12.2 | -0.2 | 2.6 | 12.8 | 34.0 | 3.9 | 49.7 | 4.4 | 9.9 | -0.5 | -3.8 |
| 2022Q2 | 12.9 | 1.0 | -3.7 | 14.6 | 32.3 | 4.5 | 47.0 | 4.7 | 8.9 | -1.3 | -8.1 |
| 2022Q4 | 14.0 | 0.8 | -2.5 | 15.0 | 30.4 | 5.0 | 47.9 | 5.1 | 8.7 | -1.9 | -10.4 |
| 2023Q2 | 13.2 | 2.4 | -8.5 | 15.9 | 28.4 | 5.4 | 45.5 | 5.0 | 7.5 | -2.0 | -10.0 |
| 2023Q4 | 12.7 | 2.9 | -10.5 | 15.6 | 27.2 | 5.3 | 43.9 | 5.1 | 7.5 | -2.2 | -10.8 |

*Notes:* Changes are relative to 2019Q4. Δ% reports changes in mean energy/kg and inputs; technology reports $100[\exp(\bar{z}_t - \bar{z}_{2019Q4}) - 1]$. Contrib. (%) equals $100C_k/\sum_h |C_h|$, where $C_k$ is the estimated log-energy contribution, using $\widehat{\theta}_L - 1$ for load. Positive shares raise energy/kg; negative shares lower it. Absolute shares sum to 100 percent before rounding and exclude residual and aggregation effects. Estimates and quarterly product portfolios match Table A2. Products are equally weighted, Samsung is excluded, and technology is shared across programs within each product. Product turnover and imputed inputs are included.

### F.2 Reference Designs, Samples, and Moments

Reference designs follow the model: duration remains anchored to the matched $A^{+++}$ benchmark, including fitted baseline Cotton-duration references where supplied by the baseline optimization. Simultaneous technology and temperature donors are averaged. Estimation uses fixed reference data; a simulated path carries its own optimized donor history forward.

The criterion contains five mean marginal profits and three mean optimized durations for $A^{+++}$ products. It also includes four destination-class means for each input group and 416 individual input changes. Independent first-class introductions enter the design targets. Repeated-class copies, fixed coordinates, Samsung Cotton 60°C duration, and first-transformation Eco temperature are excluded from their corresponding targets. Temperature-change moments exclude $A^{+++}$ transitions. Technology-change moments include those transitions but require observed technology below its reference; this selection is based on observed changes, not fitted improvements. Full- and half-load Eco coordinates are pooled, and technology enters once per product. The input-change counts are

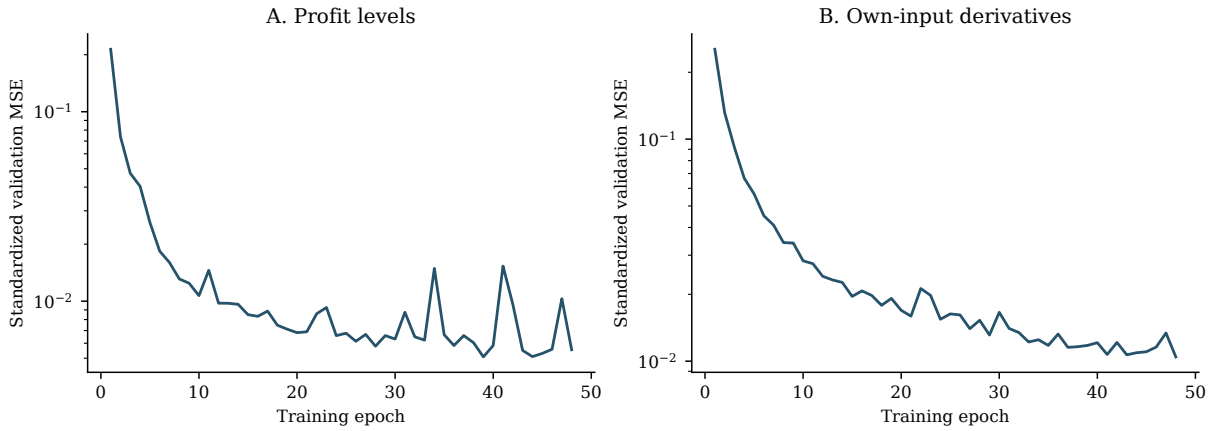



Figure F2: Operating-Profit Approximation: Saved Validation Loss

*Notes:* The figure reports standardized mean squared errors on the saved validation split across 48 recorded epochs. The vertical scales are logarithmic and differ across panels. Loss includes firm-profit levels and own-entrant derivatives of the same network. The saved checkpoint is provisional; this figure is not a final reserved-test or counterfactual-support validation.

77 for Cotton 40°C duration, 67 for Cotton 60°C, 154 for Eco duration, 70 for Eco temperature, and 48 for technology.

The baseline-gradient sample retains the archived trimming of the seven largest Cotton 60°C gradients, Cotton 40°C products 741 and 774, and seven flagged Eco-duration coordinates. These are features of the completed estimator, rather than exclusions inferred from the current fit. Within each input group, the baseline weight is 1/3 and the design weight 2/3. Duration baseline weight is split equally between marginal-profit and optimized-duration restrictions. Design weight is split equally between class means and individual changes, with equal weight within each block. Each group receives one-fifth of the total criterion. Fixed scales are the empirical input standard deviations for design moments and the stored marginal-profit scales for baseline gradient moments.

Table A4 reports selected levels and energy diagnostics; Table **??** summarizes the estimation blocks. Energy rows were computed after the selected fit and did not enter its 444-moment criterion. Figure F3 uses saved eligible input predictions and eligible $A^{+++}$ baseline predictions, retaining fitted errors and boundary observations.

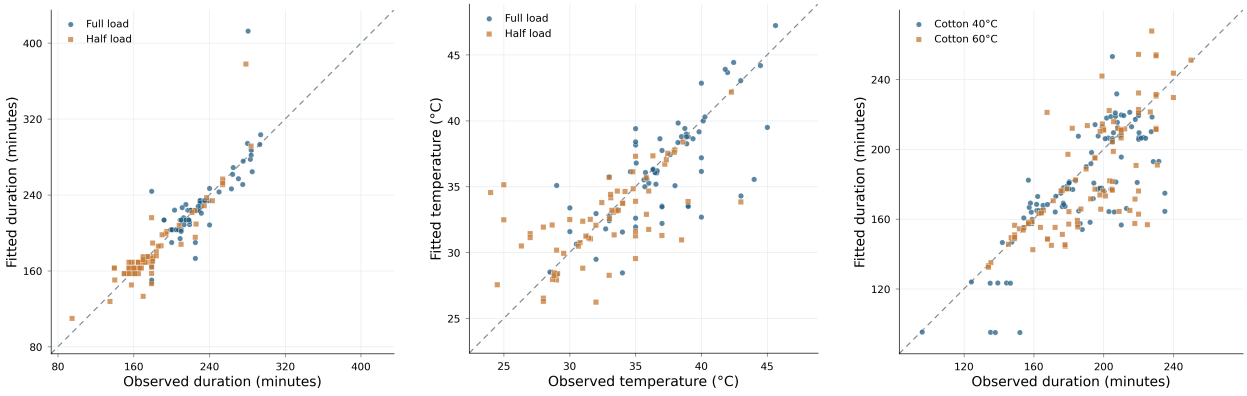
### F.3 Constrained Design Solver

For each candidate cost vector, the inner loop optimizes free first-introduction inputs conditional on rival designs, updates firms in turn, and checks unilateral improvements. It uses at most 20 passes and 300 SLSQP iterations, with solver tolerance $10^{-8}$ and fit tolerance $10^{-4}$. Positive curvatures are searched in logs. The outer search completed 50 evaluations and selected evaluation 47,

Table A4: Selected Design Moments: Observed and Fitted

| Moment | | $A^{+++}$ | A | B | C | D |
|---|---|---|---|---|---|---|
| Cotton 40°C duration (min) | Observed | 176.46 | 191.20 | 194.75 | 199.48 | 183.94 |
| | Fitted | 178.76 | 198.48 | 195.59 | 196.81 | 171.96 |
| Cotton 60°C duration (min) | Observed | 177.11 | 183.48 | 190.39 | 199.29 | 190.47 |
| | Fitted | 179.33 | 186.26 | 183.62 | 195.03 | 184.26 |
| Eco duration (min) | Observed | 219.80 | 191.20 | 190.05 | 190.21 | 188.74 |
| | Fitted | 218.91 | 190.45 | 188.61 | 191.45 | 187.76 |
| Eco temperature (°C) | Observed | 35.75 | 31.46 | 35.43 | 36.40 | 38.50 |
| | Fitted | 35.84 | 32.09 | 34.19 | 34.75 | 34.47 |
| Eco energy (kWh/cycle) | Observed | 0.741 | 0.482 | 0.546 | 0.630 | 0.723 |
| | Fitted | 0.742 | 0.472 | 0.523 | 0.581 | 0.642 |
| Cotton 40°C energy (kWh/cycle) | Observed | 0.991 | 1.120 | 1.059 | 1.133 | 0.998 |
| | Fitted | 0.953 | 1.097 | 1.023 | 1.032 | 0.917 |
| Cotton 60°C energy (kWh/cycle) | Observed | 1.421 | 1.674 | 1.571 | 1.603 | 1.554 |
| | Fitted | 1.527 | 1.738 | 1.584 | 1.653 | 1.517 |

*Notes:* Technology is fixed; the shared Cotton40/Eco curvature is halved. $A^{+++}$ uses baseline products; new-class means weight first introductions equally, including fixed Bauknecht/Beko Cotton60 and Samsung. New-class temperature excludes initial $A^{+++}$ transformations. Eco inputs average full and half loads. Eco energy averages full and half loads for $A^{+++}$ and uses regulatory three-load weights for new classes. Energy uses available observations; Cotton energy is a diagnostic, while Eco energy is targeted by capacity, class, and load.




(a) Eco duration (b) Eco temperature (c) Cotton duration

Figure F3: Observed and Fitted Program Inputs

*Notes:* Points compare observed and fitted inputs for product introductions and $A^{+++}$ baseline products, excluding coordinates at binding regulatory constraints or numerical input bounds. Colors distinguish full and half loads in the Eco panels and Cotton 40°C and Cotton 60°C in the Cotton panel. Fixed quarter-load inputs are omitted. Dashed lines denote equality. The filtered Eco-duration sample consists primarily of $A^{+++}$ baseline products.

with objective 0.3268: 0.00025 from baseline restrictions, 0.05876 from class means, and 0.26776 from introduction-level changes. The inner solution is locally verified; outer convergence, global optimality, and uniqueness remain unestablished.

The implementation retains calibrated constraint offsets and margins of five minutes for duration, one EEI unit at the upper class boundary, and one unit of weighted Eco water. Fixed water, quarter-load attributes, and copied designs are audited separately rather than freely adjusted. The selected fit records four bound hits, 182 copied-design policy audit entries, and 61 fixed-water or quarter-load audit entries. Strict nominal feasibility is not assured; Appendix G reports the corresponding counterfactual audits.

### F.4 Portfolio Conditionals, Gibbs Sampling, and NPL

For each complete own portfolio $a_{ft}$, approximate the expected payoff defined in Section 6.2 by

$$\overline{W}_{ft}(a_{ft};\mathcal{P}_{-f,t}) \simeq \frac{1}{B}\sum_{b=1}^{B} W_{ft}(a_{ft}, a_{-f,t}^{(b)}) - C_{ft}^{\text{entry}}(a_{ft}), \tag{F.1}$$

where $a_{-f,t}^{(b)}$ is a complete rival-portfolio draw. The mixed strategy is

$$\mathcal{P}_{ft}(a_{ft}) = \frac{\exp\{\overline{W}_{ft}(a_{ft};\mathcal{P}_{-f,t})/\sigma\}}{\sum_{b_{ft}\in\mathcal{F}_f(J_{f,t-1})}\exp\{\overline{W}_{ft}(b_{ft};\mathcal{P}_{-f,t})/\sigma\}}. \tag{F.2}$$

Conditioning on the firm's other lines cancels the portfolio normalizer and yields equation (24). The product of observed line conditionals defines a composite likelihood, not the joint portfolio likelihood.

Given rival beliefs, each firm's Gibbs chain visits its eligible lines in random order, draws an action from the line conditional, and records the complete portfolio after each sweep. Payoffs are averaged over rival configurations before the logit transformation. The estimation discards 100 burn-in sweeps, retains 64 portfolios with thinning one, and uses 64 rival draws. Sampling complete rival portfolios preserves dependence within firms; independence across firms is the maintained mixed-strategy assumption. Other own-line actions are fixed at their observed values in the likelihood and at their current sampled values during Gibbs updates.

For fixed beliefs, the inverse-scale and cost-over-scale parameterization makes the logit index affine. The constrained likelihood step uses three optimization starts and enforces nonnegative launch costs on the estimation support. Beliefs and parameters are then updated until relative parameter change is below 0.01 and the largest monitored conditional-probability gap is below 0.02 for two consecutive iterations, with a maximum of 20. Monitoring covers observed and probe own

configurations. An independent-seed update checks monitored probabilities, but does not establish full-chain mixing.

The completed estimation converges in four iterations. Its final independent probability gap is 0.00337. Two launch-cost restrictions bind, while the shock-scale lower bound does not. The likelihood has 469 observations from a broader risk set of 510 records. Table **??** separates entry participation from class choice. Its probabilities condition on observed own actions and should not be interpreted as unconditional simulation frequencies.

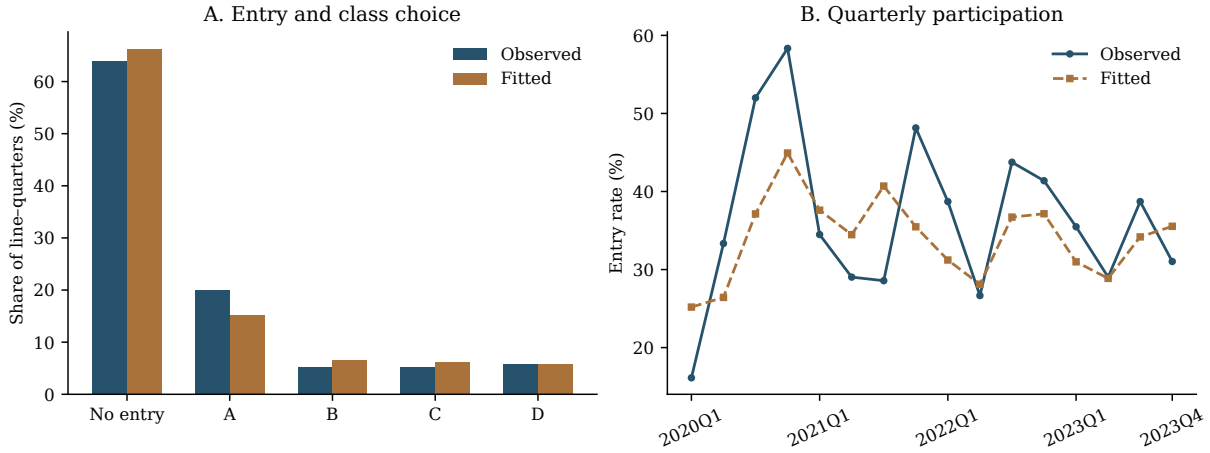



Figure F4: Entry Model Fit

*Notes:* The 469 eligible line–quarters pool transition and post-policy observations. Fitted shares average line-conditional probabilities at observed own other-line actions, integrating over simulated rival portfolios. The same eligible sample supplies each quarterly rate. These probabilities are not unconditional frequencies from simulated entry histories.

### F.5 Conditional Inference

The reported design-cost standard errors use a local numerical Jacobian of the 444 standardized moments with ten free parameters. The stored Jacobian has full column rank. The nonclustered HC1 covariance treats quarter–product–input contributions as independent while retaining their joint contributions to overlapping moments. The firm-clustered alternative aggregates those contributions within each of seven firms. Both condition on demand, engineering, reference construction, the profit approximation, and the selected local solutions. Neither incorporates the incomplete outer search or uncertainty from changing active constraints.

For entry costs, the sandwich uses the conditional composite-likelihood Hessian and score contributions, projected onto the tangent space of the two binding restrictions. The 13-parameter problem therefore has an 11-dimensional local tangent space. The nonclustered version uses individual line–quarter scores; the alternative aggregates scores by firm. A delta-method transformation

yields euro-valued costs and the shock scale. Both condition on fitted beliefs, the selected active set, the intermediate payoff predictor, and upstream estimates. They omit uncertainty from belief feedback and active-set selection. With only seven firm clusters, clustered estimates are a sensitivity comparison rather than a definitive small-sample correction.

## G Counterfactual Computation and Validation

### G.1 Retained-Machine Reference for Nonpurchasers

Nonpurchasers are assumed to retain a machine drawn from the observed 2021Q1 portfolio, denoted $\mathcal{J}^0$. Because household ownership histories are unavailable, I approximate the distribution of retained machines using product sales shares in that quarter:

$$\mu_k^0 = \frac{Q_{k,2021Q1}}{\sum_{h\in\mathcal{J}^0} Q_{h,2021Q1}}, \qquad k \in \mathcal{J}^0, \tag{G.1}$$

where $Q_{k,2021Q1}$ denotes unit sales of product $k$. These weights sum to one and are held fixed across households, periods, and policy scenarios. Sales shares approximate the ownership distribution; they do not measure installed-stock shares.

For each household $i$ and reference machine $k$, I evaluate the program inclusive value $\mathcal{I}_{ik}$ and program-choice probabilities $P_{ikr}$ using the estimated program-use model and the machine's reference attributes. Expected program value and electricity consumption per cycle for the retained machine are

$$\mathcal{I}_{i0} = \sum_{k\in\mathcal{J}^0} \mu_k^0 \mathcal{I}_{ik}, \tag{G.2}$$

$$e_{i0} = \sum_{k\in\mathcal{J}^0} \mu_k^0 \sum_{r\in\mathcal{R}} P_{ikr} e_{kr}. \tag{G.3}$$

Although reference-machine weights are common across households, both quantities vary with household program preferences. I average machine-specific inclusive values rather than taking a logsum across machines: nonpurchasers retain a machine drawn from the reference distribution rather than choose freely from the reference portfolio. The same distribution is used conditional on nonpurchase, assuming no additional relationship between retained-machine ownership and the purchase decision beyond the modeled household characteristics.

The purchase outside option remains normalized to zero in demand estimation. For welfare accounting, its residual purchase-stage value is zero, while the retained machine's services and electricity expenditure enter use-stage welfare. This allocation is an accounting assumption and does not alter estimated purchase probabilities. The purchase outside option is distinct from the

program outside option, whose systematic utility and electricity consumption are normalized to zero.

Purchasers and nonpurchasers are evaluated over the same 12-year horizon and annual wash frequency. Retained-machine services are assumed to remain available throughout this horizon through continued ownership or equivalent replacement; subsequent replacement transactions are not modeled separately. The 2021Q1 portfolio serves as a fixed accounting reference, including for periods preceding that quarter, rather than a literal reconstruction of historical ownership. It is also distinct from the stationary 2020Q1 $A^{+++}$ portfolio used in the no-policy scenario.

Using a common program-utility normalization and horizon for both groups ensures that an additive constant common to all machine-specific expected program utilities cancels in policy differences, provided the represented population and household weights are held fixed. Nonpurchasers' electricity consumption is included explicitly, so a reduction in new-machine purchases does not mechanically eliminate their electricity use.

### G.2 Policy implementation and state transitions

The common simulation starts from the observed 2020Q1 incumbent set and applies the counterfactual rules to eligible introductions throughout 2020Q1–2023Q4, including the transition year. Demand during 2020 is evaluated using the model's old-/new-label regime mixture; later quarters use the post-policy regime. Entry costs use the fixed estimates for launch, quarter, incumbent count, time since the preceding introduction, and class switching. Opportunities and counts are computed from the previous state even after current survivors have been formed. All nonstationary comparisons retain this timing. Legacy availability follows the observed schedule, and the post-2020 compatibility filter remains active. Policy-dependent exit or a counterfactual legacy-survival schedule is not estimated.

The Eco-only index and nominal duration caps follow Appendices A.1.2 and A.2.1. Regulatory test-load weights are distinct from household program-choice probabilities; the modeled entry classes are $A$ through $D$. Removing the cap switches off this policy restriction but retains the supported design bounds.

Optimization retains calibrated offsets, numerical margins, and inherited or copied designs. Fixed water and quarter-load attributes are audited rather than adjusted, so solved portfolios need not satisfy every nominal regulatory constraint. These conventions do not establish regulatory compliance or a grandfathering rule.

Adjustable first-introduction coordinates comprise Eco technology, full- and half-load Eco temperature and duration, and Cotton 40°C/60°C duration, subject to support and brand restrictions.

Quarter-load Eco coordinates, water use, Cotton 20, and EasyCare attributes are fixed; Samsung Cotton 60°C duration is also fixed. Reference and copying rules follow Appendix F.2. For unlabeled entry, the $D$ template supplies physical inputs without an EEI target or explicit class premium. The resulting experiment is broader than restoring pre-reform labels.

Cotton 40°C/60°C electricity uses its separately calibrated engineering relation. In particular, a change in the Eco technology coordinate does not mechanically improve Cotton technology. Household program probabilities determine how engineering outcomes enter realized use. The exported aggregate outcomes are sufficient for the electricity accounting below, but do not identify a causal split between within-product engineering, program switching, purchase substitution, and portfolio availability.

### G.3 Numerical protocol used in the completed runs

The removal batch contains 20 paths for each of current policy, no duration cap, and no EEI/labels, plus one frozen joint-removal solve: 61 completed shards. The additional 2,000 grid paths are documented in Appendix G.5. All nonstationary paths contain 16 quarters and three household-group rows per quarter. Matching uses path IDs 0–19 and the common base seed 20261025; quarter seeds add 10,000 times the path ID and the quarter index. A common seed reduces avoidable variation but does not force realized introductions to match when policy changes payoffs or action sets.

The entry solver uses 16 retained Gibbs draws after 20 burn-in sweeps, with a maximum of 30 belief iterations. It monitors conditional entry probabilities at three fixed own-portfolio configurations per firm: all no-entry, all last available entry action, and one seeded random profile. The maximum probability change must be at most 0.02 for two consecutive iterations. All 960 evolving removal path–quarter solves satisfy that check in three or four iterations. This checks a restricted collection of conditional probabilities; it does not verify the entire joint portfolio distribution, global uniqueness, or complete Gibbs mixing. The subsequently revised protocol with 64 draws, 100 burn-in sweeps, and an independently seeded update has not been applied to these results. No independent-check statistics are therefore reported.

Entry payoffs are supplied by an optimal-outcome predictor. Realized designs use the constrained local solver and tolerances in Appendix F.3. Household outcomes are then evaluated using solved prices and purchase shares. The price routine uses a fixed-point tolerance of $10^{-11}$ (the runtime override), with accelerated iteration and a simple-iteration fallback, allowing 10,000 and 20,000 evaluations respectively. These are numerical tolerances, not bounds on economic approximation error.

Most records fail strict feasibility because fixed-input constraints are audited rather than re-

Table A5: Recorded numerical diagnostics for the removal experiments

| | Current policy | No duration cap | No EEI / labels |
|---|---|---|---|
| Completed path–quarters | 320 | 320 | 320 |
| Passed monitored entry check | 320 | 320 | 320 |
| Mean entry iterations | 3.19 | 3.19 | 3.27 |
| Maximum monitored probability gap | 0.00573 | 0.00307 | 0.00544 |
| Strict-feasibility flag true | 2 | 18 | 4 |
| Fixed-input audit entries | 2,361 | 845 | 1,753 |
| Maximum design passes | 3 | 4 | 3 |
| Median payoff discrepancy (EUR m) | 3.73 | 3.36 | 1.97 |
| 95th percentile discrepancy (EUR m) | 10.68 | 8.95 | 10.75 |
| Maximum payoff discrepancy (EUR m) | 26.82 | 14.15 | 14.59 |

*Notes:* Each nonstationary scenario contributes 320 path–quarter records. All recorded constraint-audit entries are flagged both fixed and audit-only, and no tolerance allowances are recorded. Audit entries can recur for the same product across quarters, so their counts are not counts of unique violations or products. The payoff discrepancy is the maximum across firms within a realized portfolio of the difference between the outcome predictor's net profit and refined net profit under the underlying profit model; the table summarizes these maxima. It is not a comparison with observed or approximation-free profit.

paired. Payoff discrepancies are substantial, and refinement does not correct earlier entry-payoff errors. Results remain conditional on the provisional approximations and outstanding convergence and validation checks; reported Monte Carlo errors exclude approximation and parameter uncertainty.

### G.4 Aggregation, decomposition, and reproducibility

Table A6 reports the accounting levels underlying the removal comparisons. The equality $R = G - D$ is satisfied in every exported path–quarter–group row to within $5.6 \times 10^{-9}$ kWh, and $T = R + B$ to within $7.5 \times 10^{-9}$ kWh. Subgroup totals add to the all-household values to numerical precision. Taxes and rebates are zero in these experiments, so consumer welfare with rebates equals consumer surplus. The reference basket and population are identical across the three nonstationary policies; only the frozen benchmark has a different horizon reference-all total and population.

Table A6: Horizon levels and accounting components for the removal experiments

| | Current policy | No duration cap | No EEI / labels | Frozen joint removal |
|---|---|---|---|---|
| Consumer surplus (EUR m) | 588.351 | 1,191.568 | 689.220 | 1,780.844 |
| Gross purchase use $G$ (GWh) | 200.034 | 165.889 | 142.563 | 256.406 |
| Displaced reference $D$ (GWh) | 184.144 | 197.023 | 138.271 | 256.089 |
| Relative use $R$ (GWh) | 15.890 | -31.134 | 4.293 | 0.317 |
| Reference-all use $B$ (GWh) | 443.824 | 443.824 | 443.824 | 440.119 |
| Population total $T$ (GWh) | 459.714 | 412.690 | 448.117 | 440.436 |
| Decision observations (millions) | 9.660 | 9.660 | 9.660 | 9.579 |

*Notes:* Means of sums across 16 quarterly decision cohorts. Electricity is each cohort's quarterly use, aggregated across cohorts. The frozen benchmark repeats one initial-quarter outcome 16 times; it is not an average of 20 independently solved frozen paths. A negative $R$ means savings relative to the purchase outside reference, not negative physical consumption.

For outcome $Y$, the paired path contrast is $d_p = \sum_t (Y^c_{pt} - Y^0_{pt})$. I report $\bar{d} = 20^{-1} \sum_p d_p$ and $MCSE = \mathrm{sd}(d_p)/\sqrt{20}$. For the frozen benchmark, the same deterministic value replaces $Y^c_{pt}$ for every path; its reported MCSE therefore comes entirely from variation in the current-policy paths. Per-observation welfare effects in the main distribution table divide the horizon aggregate change by the common group population sum. They are neither effects conditional on buying a machine nor gains per distinct household.

The electricity bridge is the exact identity $\Delta R = \Delta G - \Delta D$. Since the tariff and lifetime

factor are common, the electricity-expenditure contribution to consumer surplus is $-0.30A\Delta R$. Table A7 separates that contribution from the remaining corrected purchase-logsum component. This accounting does not re-solve any hybrid economy. A causal mechanism decomposition would require household–product–program probabilities, a common support for changing portfolios, and explicit definitions of the hybrid attribute and choice environments.

Table A7: Accounting decomposition of removal-policy surplus changes

| | Total $\Delta CS$ | Electricity term | Remainder |
|---|---|---|---|
| No duration cap | 603.22 | 515.68 | 87.54 |
| No EEI / labels | 100.87 | 127.18 | -26.31 |

*Notes:* EUR million across 16 cohorts. The electricity term equals $-0.30A\Delta R$; the remainder is total consumer-surplus change less this term. The electricity term is 85.5% of the cap-removal gain and exceeds the total gain under EEI/label removal because the remaining component is negative.

Product-design moments are recovered from the saved solved-design histories and matched to active product IDs in the final-quarter portfolio, deduplicating region and program rows. Within-path equal-product means are averaged over the 20 paths. Class shares are defined only for the labeled scenarios. The saved exports do not provide a ready-to-report sales-weighted design distribution, price series, program-specific energy decomposition, binding-constraint share, producer-surplus series, or household losing fraction. No such quantities are inferred from the aggregate welfare totals, and none of the reported tables require rerunning the simulation.

### G.5 Completed Policy Grids: Implementation and Validation

The downloaded grid contains 50 weighted-EEI policies and 50 producer-tax policies, each with 20 complete paths. Every path has 16 quarters and three household groups: 96,000 outcome rows and 32,000 path–quarter diagnostic records in total. All 2,000 completed-path files, diagnostic files, and terminal checkpoints are present. Recalculated policy means agree with the collector exports to numerical precision, and packaged dependencies match their recorded hashes. Tables A9 and A10 enumerate every tested point, including baseline ID 0.

The weighted-EEI index is

$$EEI_j(w) = \frac{100\,[w_E e_{jE} + w_{40} e_{j40} + w_{60} e_{j60}]}{S(k_j)}, \qquad w_E + w_{40} + w_{60} = 1, \quad w_r \geq 0, \tag{G.4}$$

where Eco energy uses regulatory load weights and $S(k_j)$ is the capacity denominator in Appendix A.1.2. The grid is a deterministic selection from a half-percentage-point lattice with total Cotton weight no greater than 10%. Class thresholds are unchanged. Tax policies use equal program

weights of one third in equation (34) and enter production cost as

$$mc_j^\tau = mc_j + \mathcal{T}_j. \tag{G.5}$$

The EEI entry-payoff checkpoint is epoch 302; the carbon checkpoint is epoch 27 and uses its matching carbon operating-profit and Cotton-engineering files. Both retain provisional training status. Design refinement uses cost evaluation 47 and the saved entry-cost estimates. The different Cotton calibrations preclude treating the two family baselines as one common physical economy.

**Accounting verification.** The grid runtime computes electricity as $G = \sum_i W_i \sum_j s_{ij} E_{ij}$ and subtracts $0.30AG$ in its surplus calculation. It does not export displaced-reference use $D$ or total reference-inclusive electricity $T$. Its corrected purchase-logsum component is recovered as $CS^G + 0.30AG$; thus

$$\Delta CW^G = \Delta(CS^G + 0.30AG) - 0.30A\Delta G + \Delta\text{rebates}. \tag{G.6}$$

The coefficients and units are those in Section **??**. Aggregate rebates equal tax revenue to within EUR $7.5 \times 10^{-9}$; welfare equals surplus plus rebates to within EUR $1.9 \times 10^{-8}$. Subgroup sums agree within $2.3 \times 10^{-8}$ in the exported units. Horizon populations are identical across policies: 9.660 million weighted decision observations, of which 2.254 million have positive duration preferences. Equal market-level rebates do not remove differences in pre-transfer incidence.

At baseline ID 0, mean gross electricity is 165.89 GWh for EEI and 128.87 GWh for tax. Corresponding $CS^G$ levels are minus EUR 969.04 million and minus EUR 643.27 million. Negative surplus levels reflect the specified utility normalization and subtraction of gross lifetime electricity expenditure; the reported comparisons are within-family changes. Because $D$ can change with purchasing and product availability, a common aggregate outside-option correction cannot convert these grids to the removal experiments' reference-adjusted accounting. Harmonized comparisons require re-evaluating household purchase allocations and reference electricity.

**Numerical checks and scope.** Both grids use the same path/quarter seeds and 16-draw, 20-burn protocol described above. All recorded belief checks pass: weighted EEI takes three to five iterations, with maximum final monitored gap 0.0158; tax takes three to four, with maximum gap 0.0115. These are the recorded conditional-probability checks, not an independent-seed test of the full portfolio distribution.

Strict-feasibility flags pass in 947 EEI records and no tax records. EEI audits contain 35,629 weighted-water, 5,883 upper-EEI, and 881 lower-EEI entries, all marked fixed and audit-only. Tax audits contain 25,803 weighted-water entries marked audit-only; the tax export does not supply their fixed-coordinate flag. No numerical policy allowances are recorded. As in the removal experiments,

Table A8: Numerical diagnostics for the completed policy grids

| | Weighted EEI | Producer tax |
|---|---|---|
| Path–quarter solves | 16,000 | 16,000 |
| Belief checks passed | 16,000 | 16,000 |
| Strict-feasibility flags passed | 947 | 0 |
| Audit-only constraint entries | 42,393 | 25,803 |
| Median maximum profit error (EUR m) | 2.84 | 1.46 |
| 95th-percentile maximum error (EUR m) | 9.48 | 5.03 |

*Notes:* Each family contains 16,000 path–quarter solves. Profit errors are absolute discrepancies between predicted and refined net profits, taking the maximum across firms within each realized portfolio. Audit entries may repeat across quarters. They are not counts of distinct products, binding constraints, or failed numerical solves.

completed solves do not ensure nominal feasibility, and Monte Carlo errors exclude the substantial payoff discrepancies in Table A8 and parameter uncertainty.

Product moments in Table 7 use the final active portfolio and its saved solved-input history, deduplicated across region/program rows before averaging. They are equal-product means. Neither these means nor aggregate $G$ identify a causal design/use/purchase decomposition, a sales-weighted attribute effect, or the fraction of households losing. The frontier compares means over the tested policies under the recorded accounting; a boundary maximizer does not establish a global optimum or the ranking after an outside-reference correction.

Table A9: Complete weighted-EEI grid

| ID | $w_E$ | $w_{40}$ | $w_{60}$ | $\Delta G$ | MCSE | $\Delta CW^G$ | MCSE |
|---|---|---|---|---|---|---|---|
| 0 | 1.000 | 0.000 | 0.000 | 0.00 | 0.00 | 0.00 | 0.00 |
| 1 | 0.990 | 0.010 | 0.000 | -0.21 | 0.12 | 1.30 | 2.13 |
| 2 | 0.990 | 0.000 | 0.010 | 0.58 | 0.98 | 0.31 | 4.70 |
| 3 | 0.980 | 0.020 | 0.000 | -0.57 | 0.17 | 4.23 | 2.34 |
| 4 | 0.980 | 0.010 | 0.010 | 0.23 | 0.96 | 2.98 | 4.44 |
| 5 | 0.980 | 0.000 | 0.020 | -0.72 | 1.03 | 11.76 | 6.50 |
| 6 | 0.900 | 0.050 | 0.050 | -6.63 | 1.34 | 64.19 | 9.38 |
| 7 | 0.900 | 0.100 | 0.000 | -1.48 | 0.90 | 21.15 | 4.43 |
| 8 | 0.900 | 0.000 | 0.100 | -10.00 | 1.69 | 91.59 | 12.93 |
| 9 | 0.930 | 0.010 | 0.060 | -5.72 | 1.46 | 54.44 | 10.13 |
| 10 | 0.930 | 0.060 | 0.010 | -1.36 | 1.10 | 16.50 | 7.36 |
| 11 | 0.940 | 0.025 | 0.035 | -2.07 | 1.34 | 19.38 | 8.81 |
| 12 | 0.900 | 0.020 | 0.080 | -8.71 | 1.52 | 80.29 | 9.67 |
| 13 | 0.900 | 0.070 | 0.030 | -4.04 | 1.13 | 37.31 | 7.50 |
| 14 | 0.950 | 0.035 | 0.015 | -0.19 | 1.09 | 2.56 | 7.61 |
| 15 | 0.900 | 0.035 | 0.065 | -7.05 | 1.49 | 65.55 | 10.91 |
| 16 | 0.900 | 0.085 | 0.015 | -2.71 | 1.03 | 28.52 | 6.64 |
| 17 | 0.955 | 0.005 | 0.040 | -1.85 | 1.37 | 14.60 | 8.70 |
| 18 | 0.920 | 0.000 | 0.080 | -7.83 | 1.47 | 71.69 | 10.00 |
| 19 | 0.920 | 0.050 | 0.030 | -2.88 | 1.39 | 28.96 | 9.06 |
| 20 | 0.955 | 0.045 | 0.000 | -0.09 | 0.97 | 7.39 | 4.60 |
| 21 | 0.925 | 0.075 | 0.000 | -0.83 | 0.89 | 14.66 | 4.51 |
| 22 | 0.920 | 0.030 | 0.050 | -5.41 | 1.33 | 50.74 | 8.95 |
| 23 | 0.900 | 0.010 | 0.090 | -9.37 | 1.70 | 84.91 | 13.95 |
| 24 | 0.960 | 0.015 | 0.025 | 0.17 | 1.18 | 0.89 | 8.02 |
| 25 | 0.920 | 0.040 | 0.040 | -4.29 | 1.36 | 43.16 | 8.63 |
| 26 | 0.900 | 0.060 | 0.040 | -4.85 | 1.35 | 48.37 | 8.63 |
| 27 | 0.950 | 0.000 | 0.050 | -2.79 | 1.42 | 22.09 | 10.06 |
| 28 | 0.935 | 0.000 | 0.065 | -5.49 | 1.50 | 49.85 | 10.99 |
| 29 | 0.915 | 0.010 | 0.075 | -7.36 | 1.41 | 66.95 | 9.03 |
| 30 | 0.940 | 0.015 | 0.045 | -3.17 | 1.43 | 26.58 | 10.02 |
| 31 | 0.965 | 0.020 | 0.015 | -0.57 | 1.07 | 10.65 | 7.16 |
| 32 | 0.925 | 0.020 | 0.055 | -5.94 | 1.48 | 56.13 | 10.14 |
| 33 | 0.905 | 0.025 | 0.070 | -7.10 | 1.49 | 65.55 | 10.81 |
| 34 | 0.965 | 0.030 | 0.005 | 0.09 | 0.99 | 5.11 | 4.85 |
| 35 | 0.945 | 0.030 | 0.025 | -0.54 | 1.21 | 7.67 | 8.20 |
| 36 | 0.905 | 0.040 | 0.055 | -6.44 | 1.37 | 61.41 | 9.12 |
| 37 | 0.935 | 0.045 | 0.020 | -0.97 | 1.14 | 9.08 | 7.91 |
| 38 | 0.915 | 0.060 | 0.025 | -1.86 | 1.23 | 21.97 | 7.89 |
| 39 | 0.915 | 0.070 | 0.015 | -1.57 | 1.13 | 17.17 | 7.91 |
| 40 | 0.905 | 0.090 | 0.005 | -1.95 | 1.07 | 22.56 | 6.41 |
| 41 | 0.970 | 0.000 | 0.030 | -1.01 | 1.19 | 12.19 | 7.08 |
| 42 | 0.910 | 0.000 | 0.090 | -8.57 | 1.72 | 79.38 | 15.25 |
| 43 | 0.950 | 0.015 | 0.035 | -1.74 | 1.37 | 15.46 | 8.92 |
| 44 | 0.930 | 0.040 | 0.030 | -2.70 | 1.38 | 26.22 | 8.98 |
| 45 | 0.945 | 0.045 | 0.010 | -0.37 | 1.08 | 8.33 | 7.92 |
| 46 | 0.910 | 0.050 | 0.040 | -4.52 | 1.33 | 45.17 | 8.39 |
| 47 | 0.945 | 0.055 | 0.000 | -0.35 | 0.97 | 9.87 | 4.86 |
| 48 | 0.935 | 0.065 | 0.000 | -0.62 | 0.97 | 12.65 | 4.96 |
| 49 | 0.990 | 0.005 | 0.005 | 0.67 | 0.96 | -0.79 | 4.47 |

Each row uses 20 paired paths. $\Delta G$ is gross purchase electricity in GWh; $\Delta CW^G$ is consumer welfare with rebates in EUR million under gross-use accounting. Changes use the family-specific zero-Cotton-weight or zero-tax baseline (ID 0). MCSE is the standard error of the paired simulation mean, conditional on parameters and numerical approximations. IDs identify downloaded shards; all tested policies are shown.

Table A10: Complete producer-tax grid

| ID | $\tau_E$ | $\tau_C$ | $\Delta G$ | MCSE | $\Delta CW^G$ | MCSE |
|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0.00 | 0.00 | 0.00 | 0.00 |
| 1 | 10 | 10 | -4.03 | 0.12 | 40.72 | 1.06 |
| 2 | 25 | 25 | -9.94 | 0.29 | 95.61 | 2.88 |
| 3 | 50 | 50 | -19.02 | 0.40 | 185.29 | 3.37 |
| 4 | 75 | 75 | -28.78 | 1.00 | 265.98 | 5.82 |
| 5 | 100 | 100 | -37.34 | 0.97 | 339.48 | 6.31 |
| 6 | 100 | 0 | -8.97 | 0.39 | 87.18 | 4.36 |
| 7 | 0 | 100 | -29.02 | 0.95 | 274.25 | 6.57 |
| 8 | 25 | 0 | -2.37 | 0.28 | 20.56 | 3.35 |
| 9 | 0 | 25 | -7.61 | 0.21 | 76.73 | 1.49 |
| 10 | 50 | 0 | -4.51 | 0.32 | 40.98 | 3.37 |
| 11 | 0 | 50 | -14.96 | 0.36 | 147.16 | 2.74 |
| 12 | 40 | 95 | -31.42 | 0.97 | 287.50 | 6.18 |
| 13 | 95 | 40 | -20.19 | 0.57 | 191.39 | 4.18 |
| 14 | 15 | 75 | -23.73 | 0.89 | 226.13 | 5.35 |
| 15 | 65 | 25 | -13.28 | 0.33 | 128.66 | 3.51 |
| 16 | 65 | 100 | -34.33 | 1.01 | 314.37 | 6.46 |
| 17 | 100 | 65 | -28.05 | 0.99 | 261.78 | 6.28 |
| 18 | 25 | 50 | -16.96 | 0.34 | 165.45 | 2.88 |
| 19 | 75 | 0 | -6.74 | 0.33 | 64.01 | 3.78 |
| 20 | 40 | 70 | -24.70 | 0.97 | 228.16 | 5.23 |
| 21 | 75 | 50 | -21.42 | 0.56 | 203.03 | 4.02 |
| 22 | 45 | 20 | -10.17 | 0.31 | 99.50 | 3.29 |
| 23 | 85 | 20 | -13.54 | 0.37 | 131.63 | 3.69 |
| 24 | 20 | 95 | -29.71 | 0.93 | 274.35 | 5.58 |
| 25 | 55 | 80 | -28.36 | 0.97 | 263.02 | 5.92 |
| 26 | 85 | 90 | -34.08 | 0.92 | 309.39 | 5.91 |
| 27 | 0 | 70 | -20.35 | 0.45 | 200.09 | 4.11 |
| 28 | 40 | 35 | -14.10 | 0.35 | 136.12 | 3.16 |
| 29 | 60 | 65 | -25.05 | 0.97 | 232.14 | 5.54 |
| 30 | 80 | 35 | -17.69 | 0.54 | 167.94 | 3.71 |
| 31 | 100 | 25 | -16.24 | 0.65 | 154.10 | 4.60 |
| 32 | 0 | 85 | -24.74 | 0.83 | 240.38 | 5.83 |
| 33 | 100 | 80 | -32.52 | 0.96 | 298.59 | 5.67 |
| 34 | 10 | 35 | -11.59 | 0.24 | 115.02 | 1.87 |
| 35 | 10 | 60 | -18.35 | 0.41 | 180.35 | 2.98 |
| 36 | 25 | 65 | -21.32 | 0.49 | 205.73 | 3.21 |
| 37 | 30 | 80 | -26.81 | 0.97 | 247.85 | 5.76 |
| 38 | 35 | 10 | -6.37 | 0.28 | 59.12 | 3.08 |
| 39 | 55 | 35 | -15.27 | 0.34 | 148.63 | 3.15 |
| 40 | 60 | 10 | -8.55 | 0.34 | 82.94 | 3.69 |
| 41 | 85 | 60 | -25.55 | 0.97 | 238.43 | 5.89 |
| 42 | 10 | 90 | -27.36 | 0.92 | 257.44 | 5.85 |
| 43 | 15 | 20 | -7.59 | 0.22 | 75.98 | 1.47 |
| 44 | 15 | 45 | -14.90 | 0.34 | 146.33 | 2.34 |
| 45 | 30 | 40 | -14.64 | 0.31 | 142.57 | 2.68 |
| 46 | 30 | 100 | -31.87 | 0.94 | 295.27 | 7.16 |
| 47 | 35 | 55 | -19.03 | 0.37 | 185.74 | 3.12 |
| 48 | 45 | 60 | -21.49 | 0.54 | 204.71 | 3.52 |
| 49 | 45 | 85 | -29.04 | 0.98 | 267.66 | 5.89 |

Each row uses 20 paired paths. $\Delta G$ is gross purchase electricity in GWh; $\Delta CW^G$ is consumer welfare with rebates in EUR million under gross-use accounting. Changes use the family-specific zero-Cotton-weight or zero-tax baseline (ID 0). MCSE is the standard error of the paired simulation mean, conditional on parameters and numerical approximations. IDs identify downloaded shards; all tested policies are shown.